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(54) **RAMAN SPECTROSCOPY SYSTEM WITH
OPTICAL-FIBER EXTENSION**

(52) **U.S. CL.**

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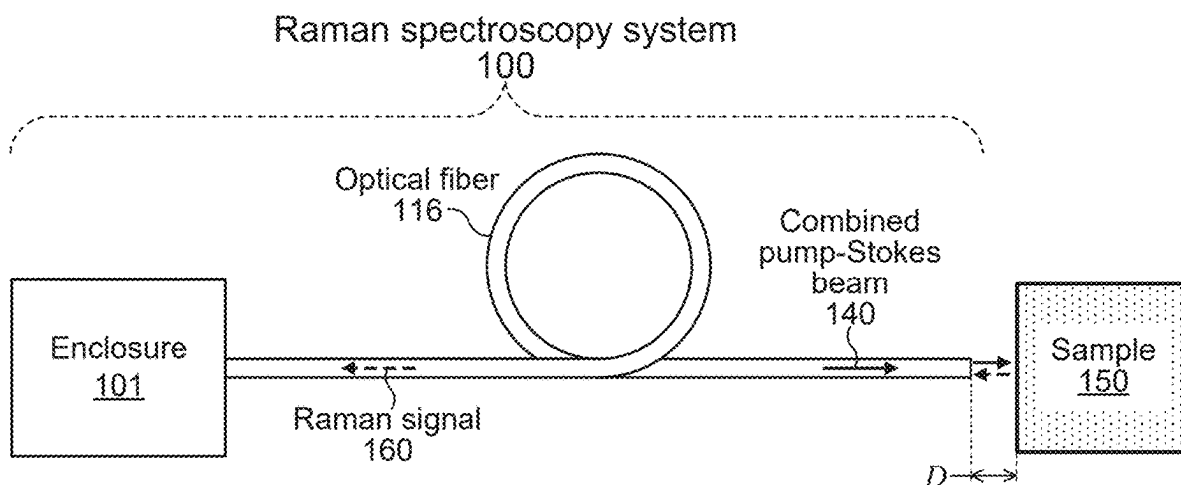
(51) **Int. CL.**

G01N 21/65 (2006.01)

(57)

ABSTRACT

In one embodiment, a Raman spectroscopy system includes a pump light source configured to produce a pump beam of light at a pump frequency and a Stokes light source configured to produce a Stokes beam of light at a Stokes frequency, where the pump and Stokes frequencies are offset by a frequency offset Ω . The system further includes one or more optical fibers configured to (i) direct the pump and Stokes beams to a sample located external to the system and (ii) direct a Raman signal to the system, where the Raman signal is produced by coherent Raman scattering of the pump and Stokes beams of light at the sample. The system also includes an optical receiver configured to detect the Raman signal. The optical receiver includes a probe light source and an optical detector configured to coherently mix the Raman signal with a probe beam of light.



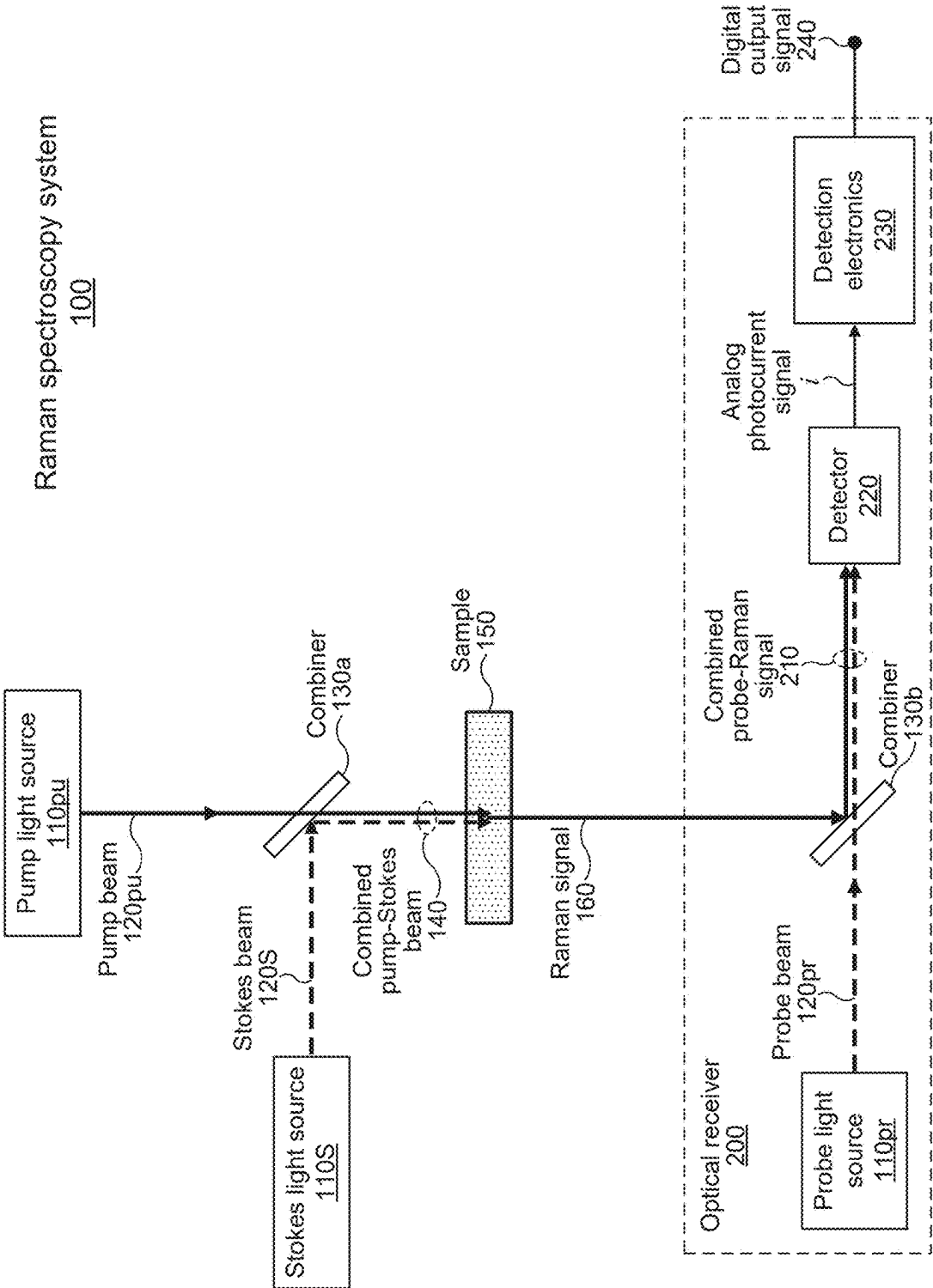


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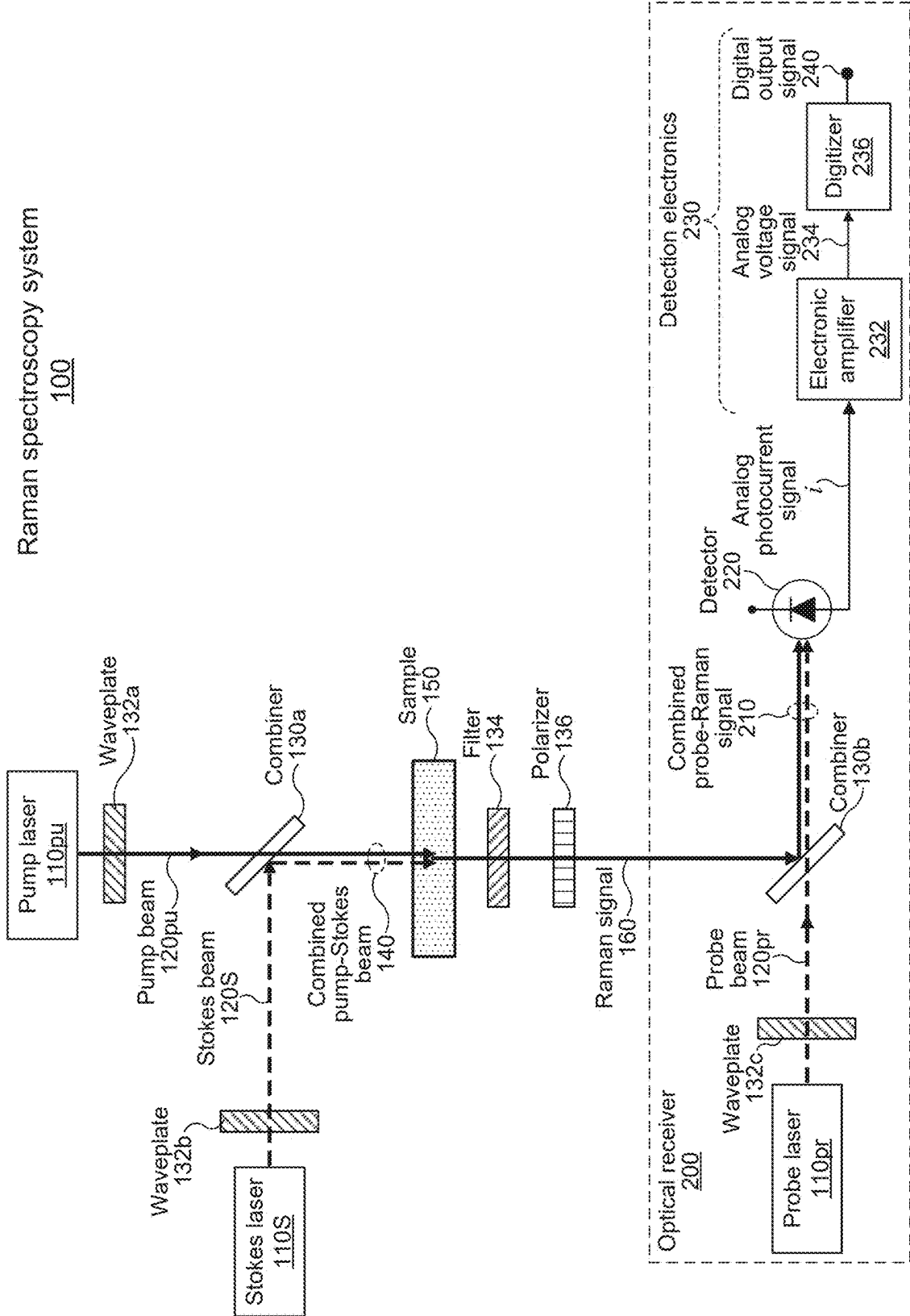


Figure 2

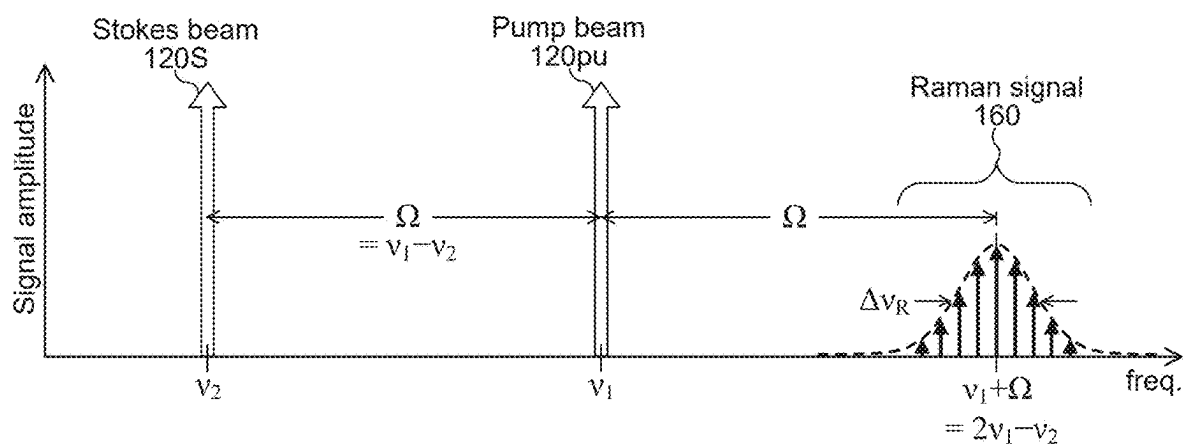


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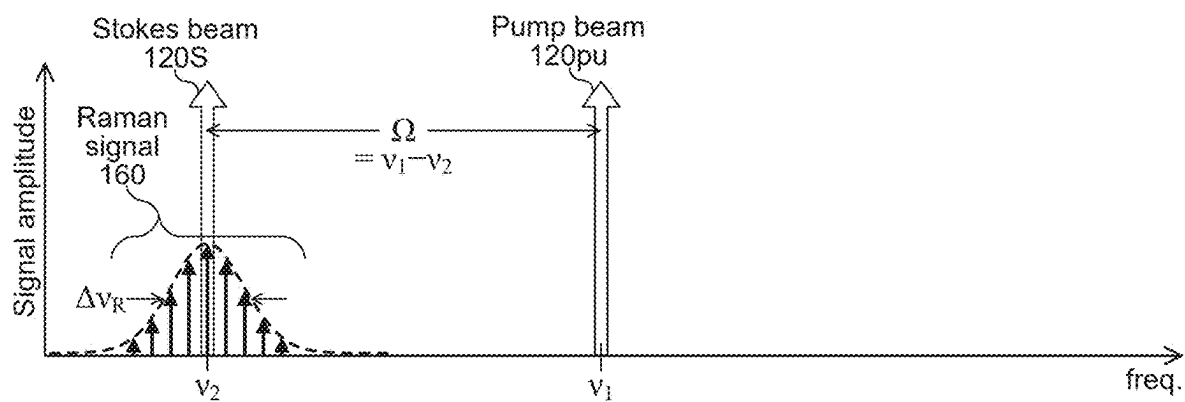


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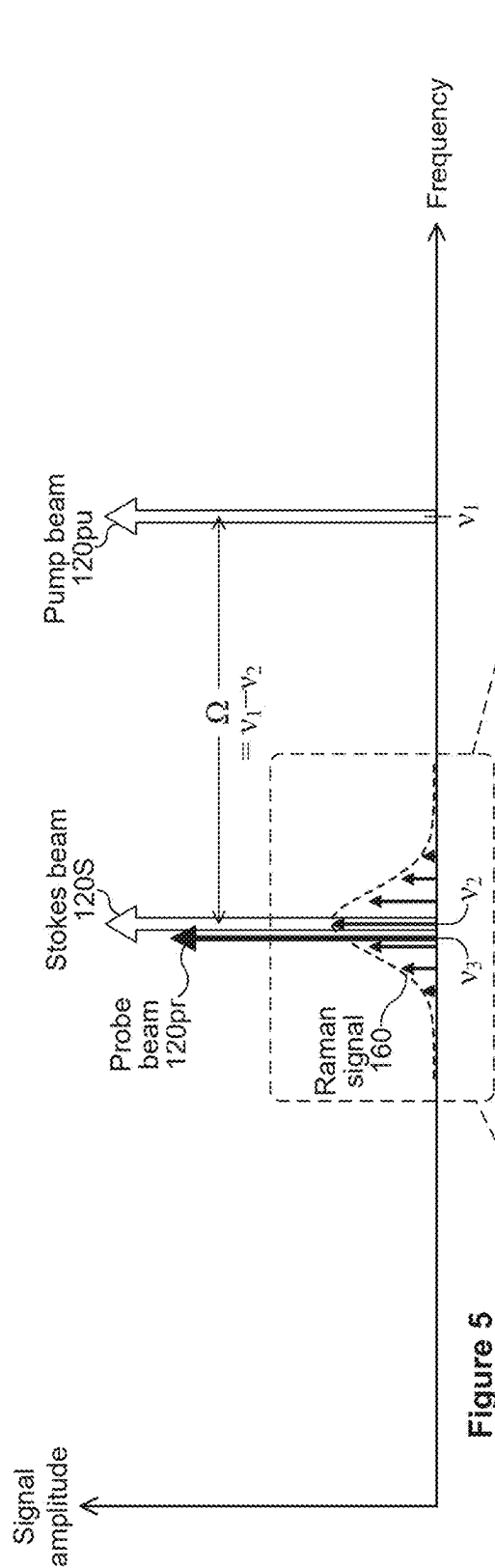


Figure 5

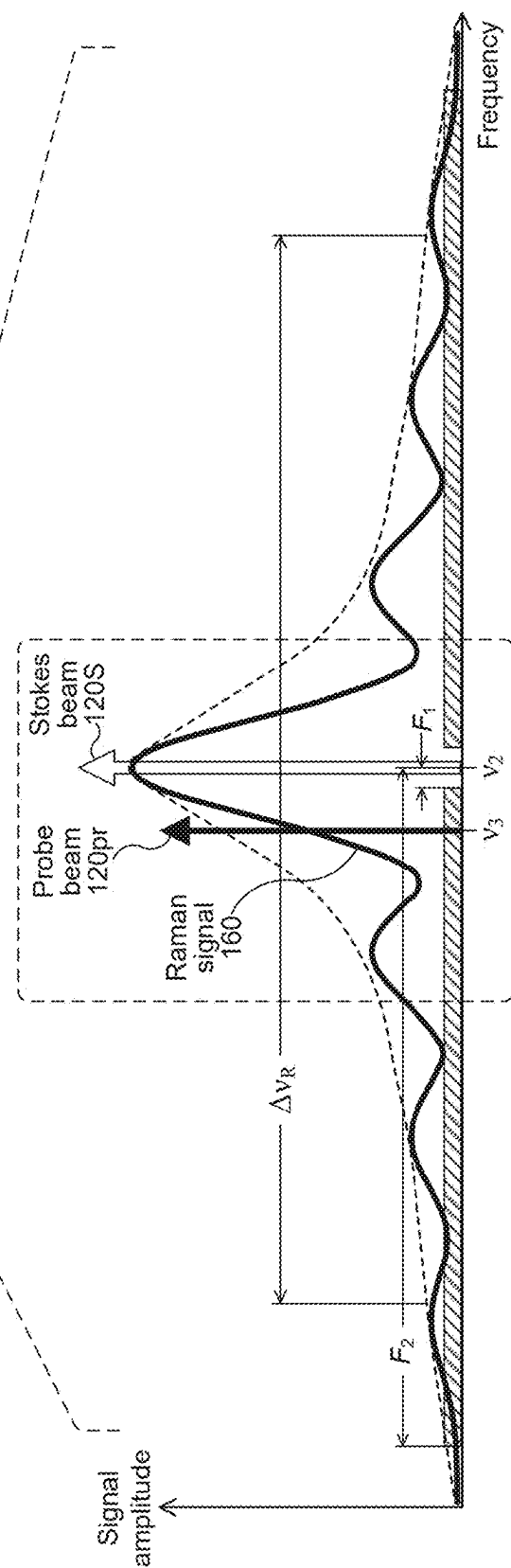


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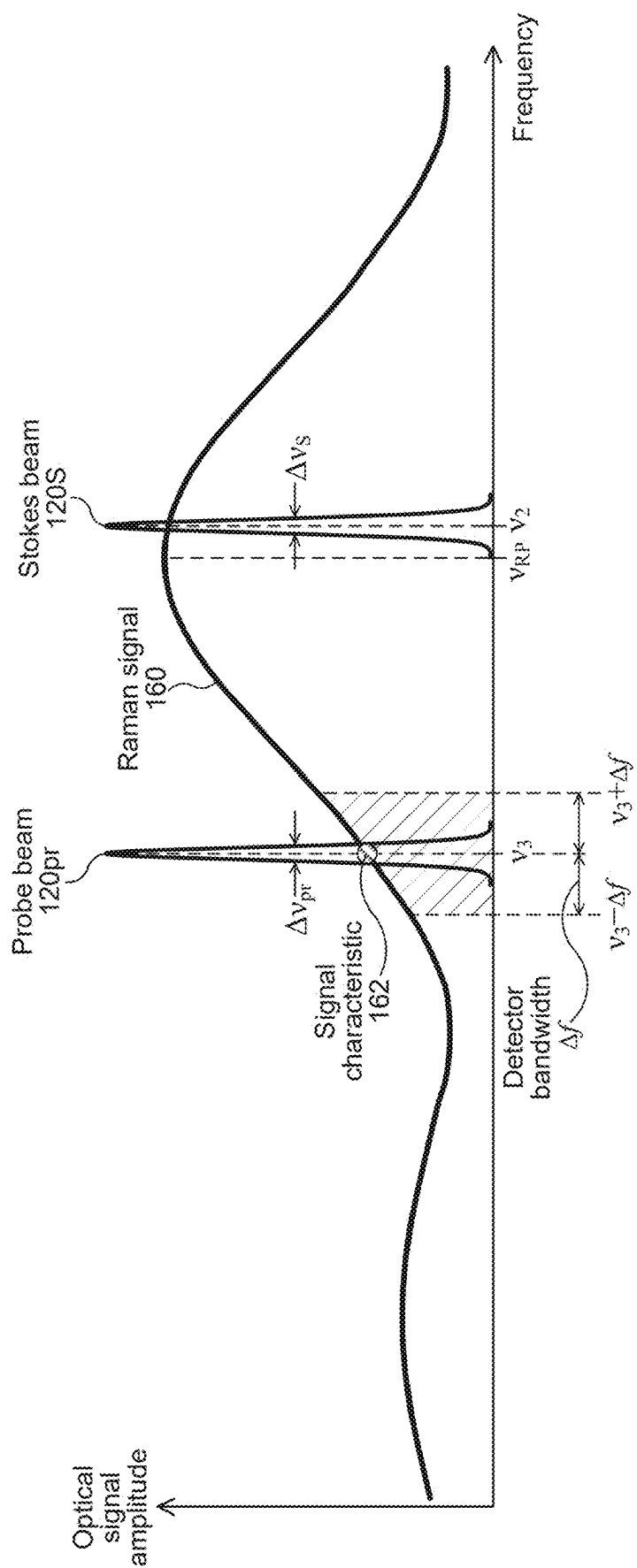


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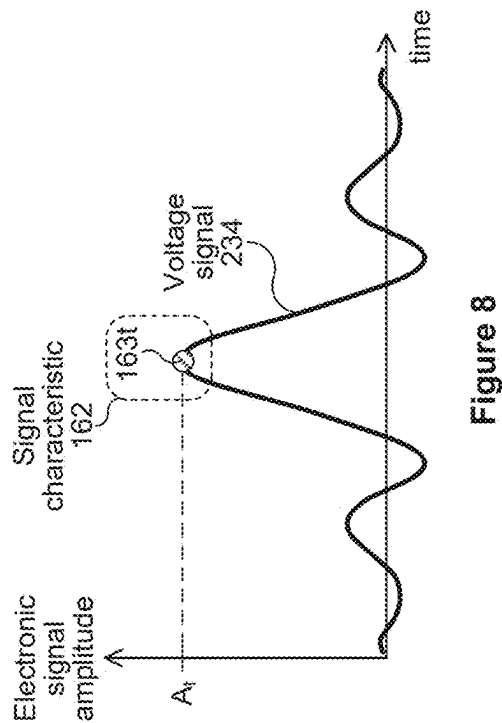


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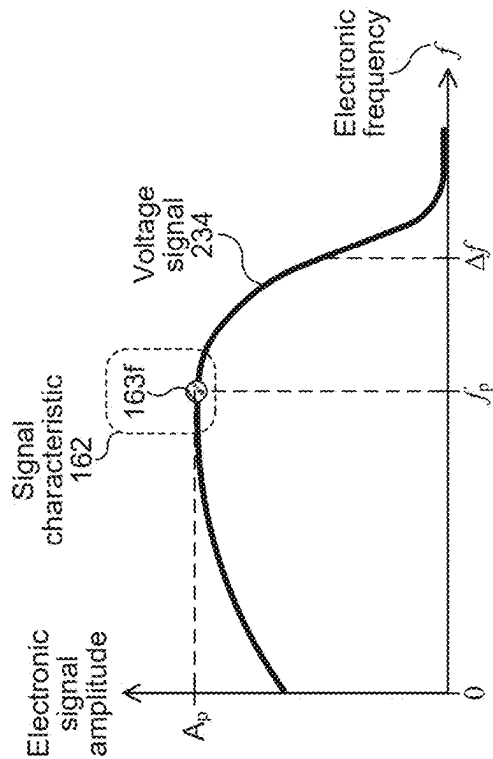


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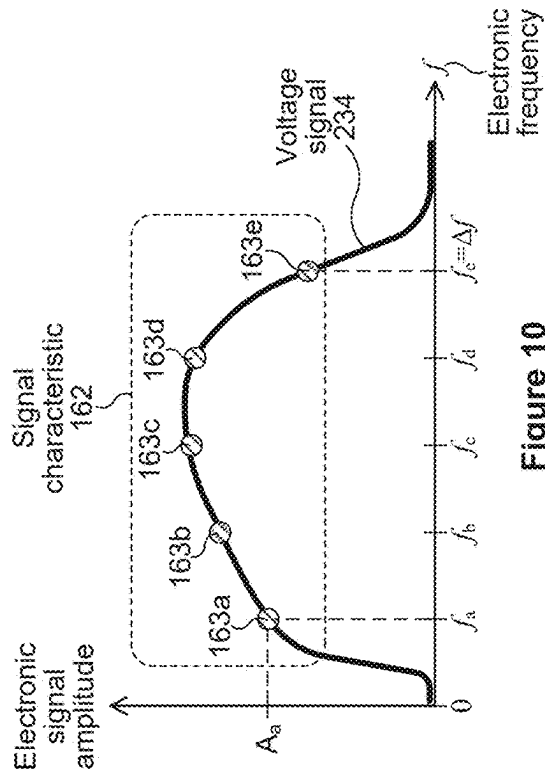


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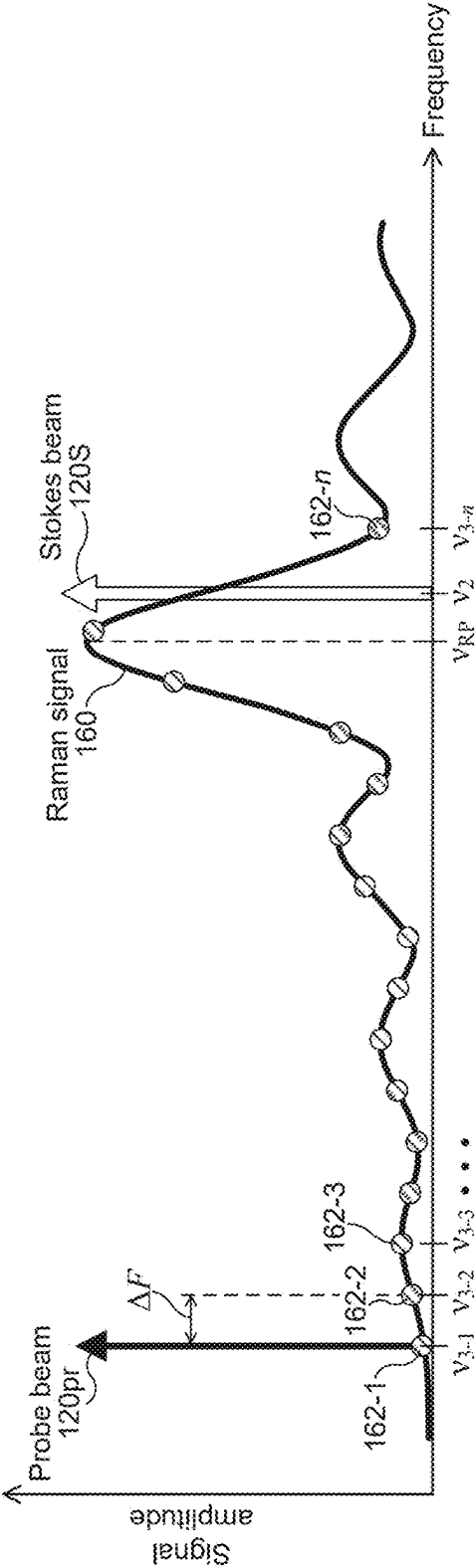


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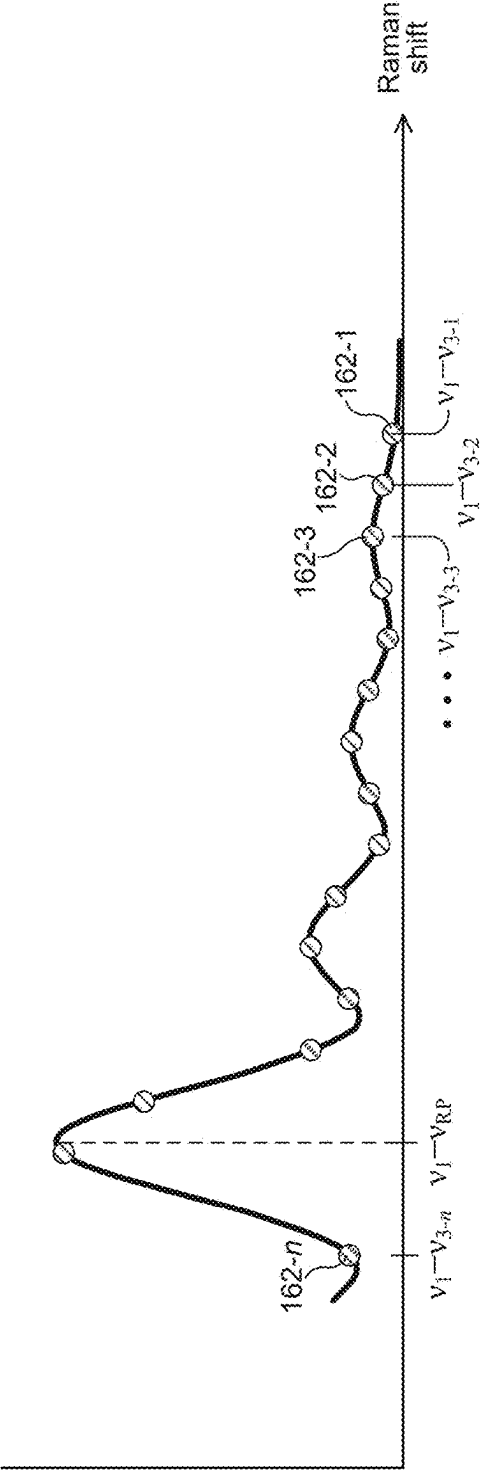


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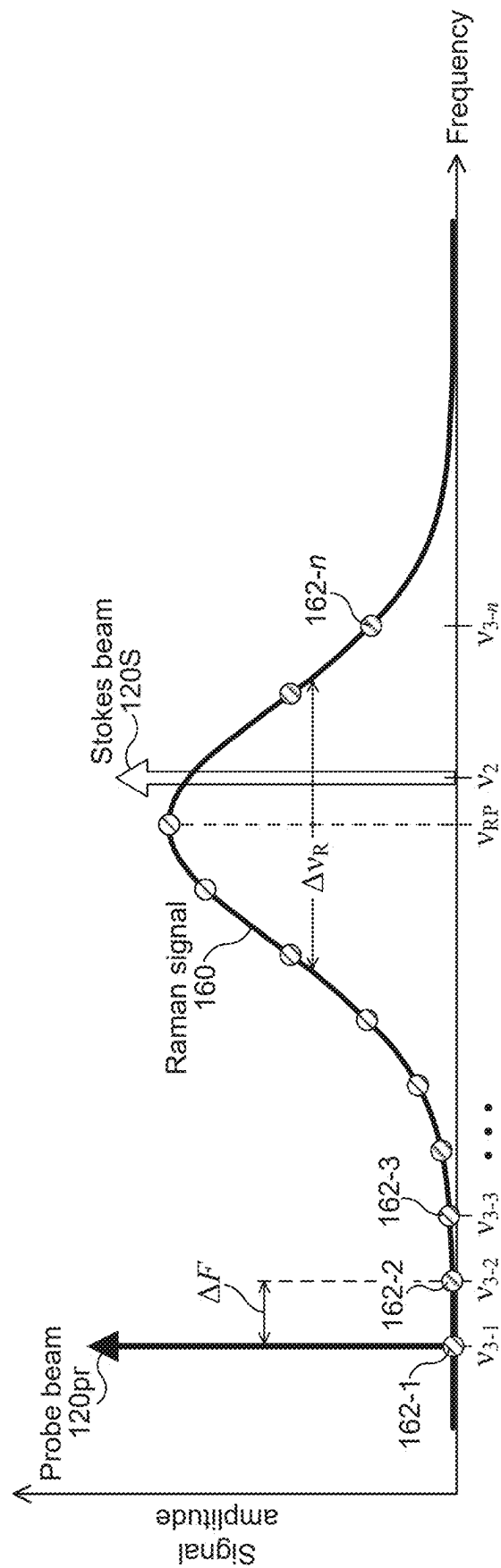


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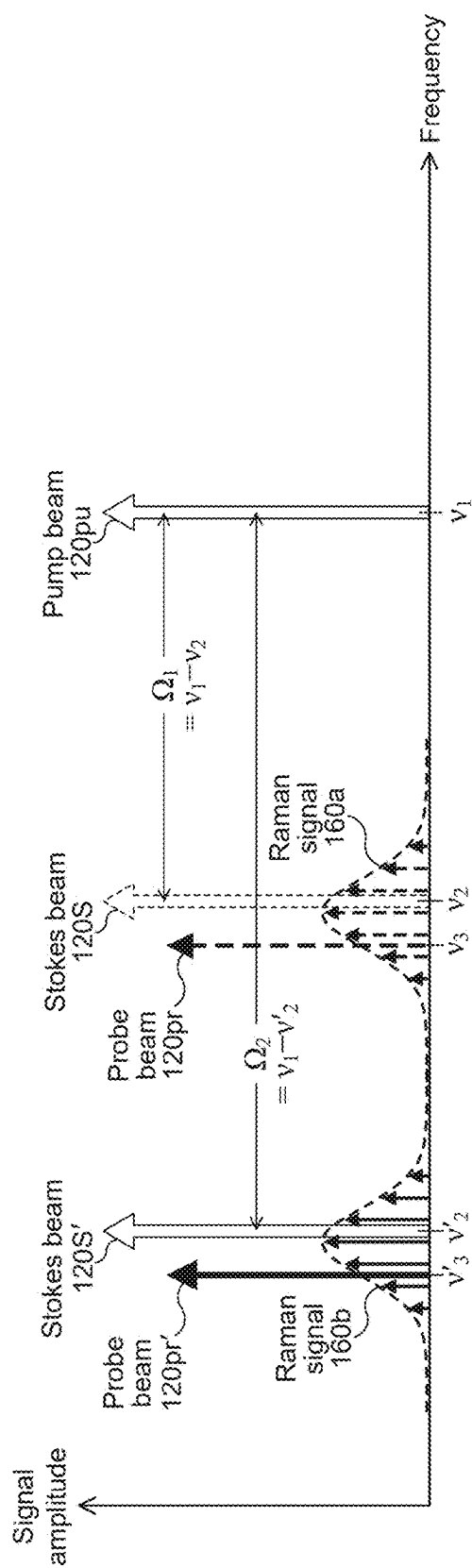


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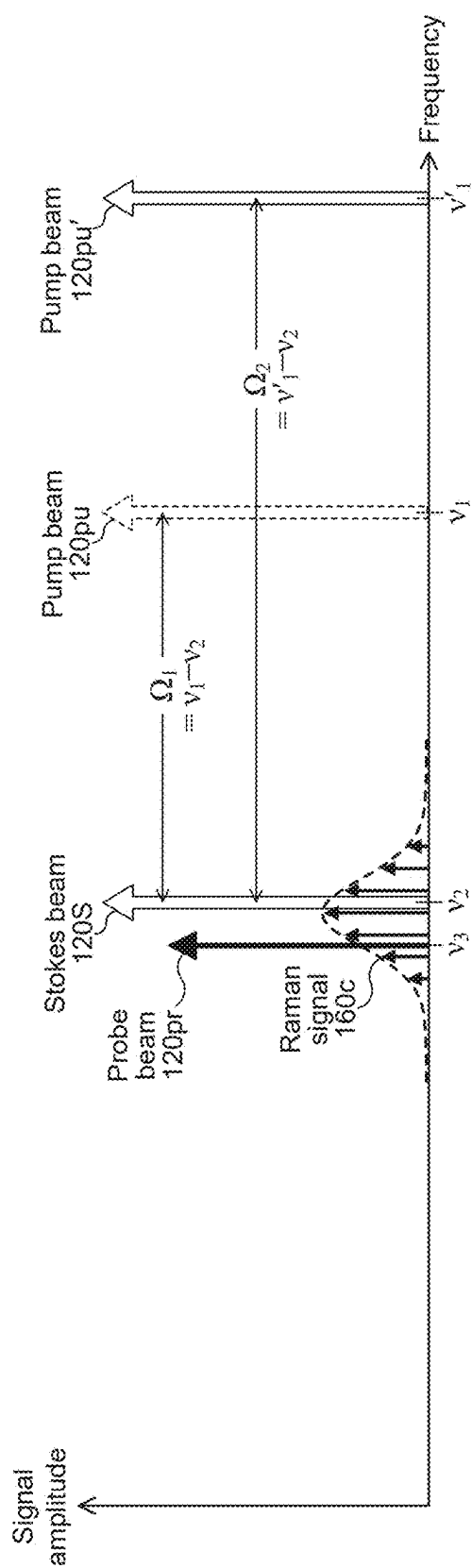


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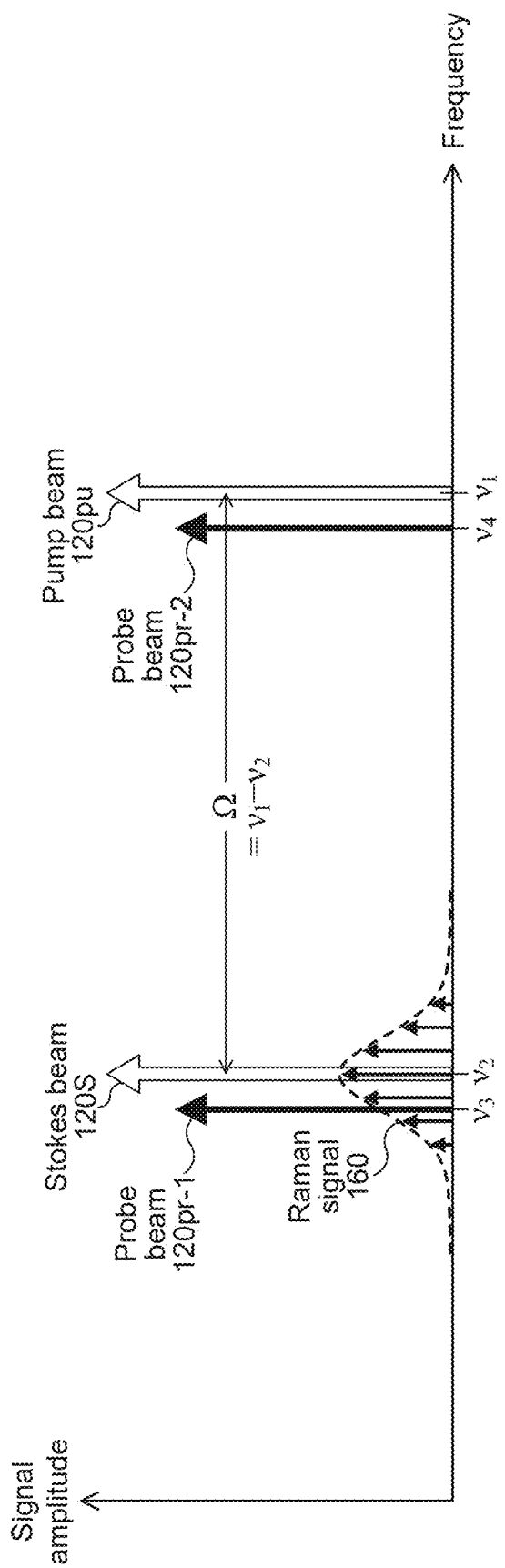


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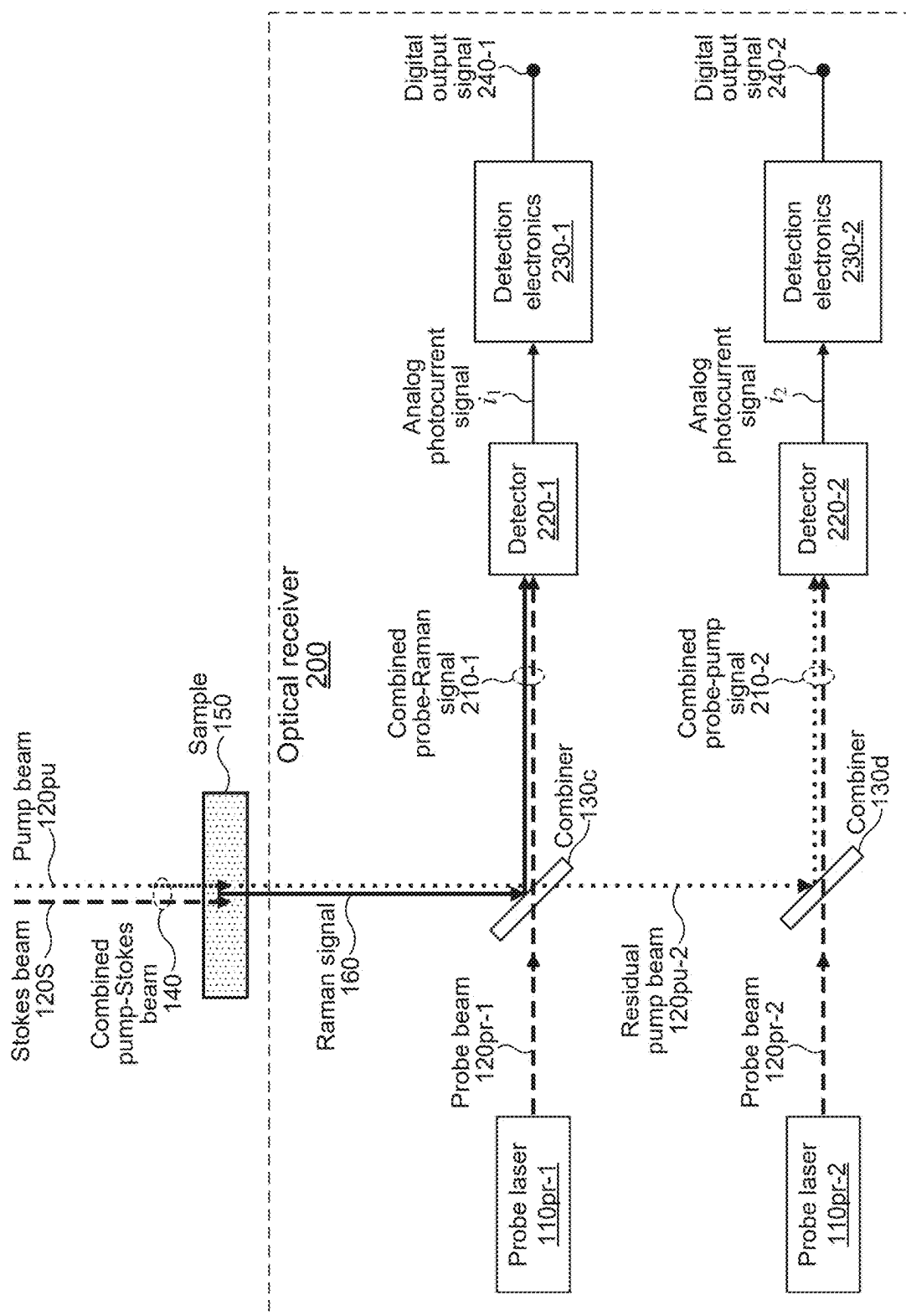


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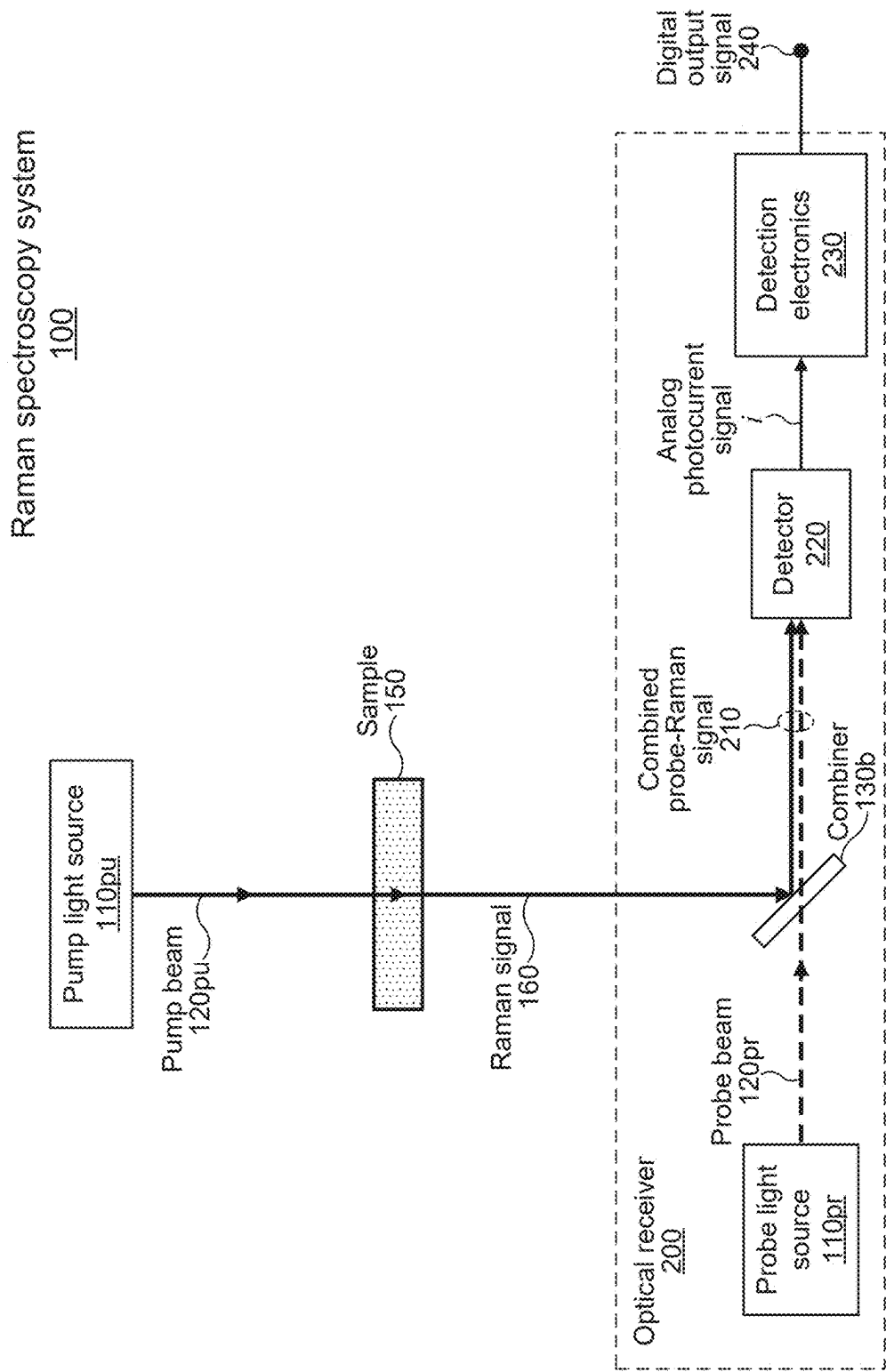


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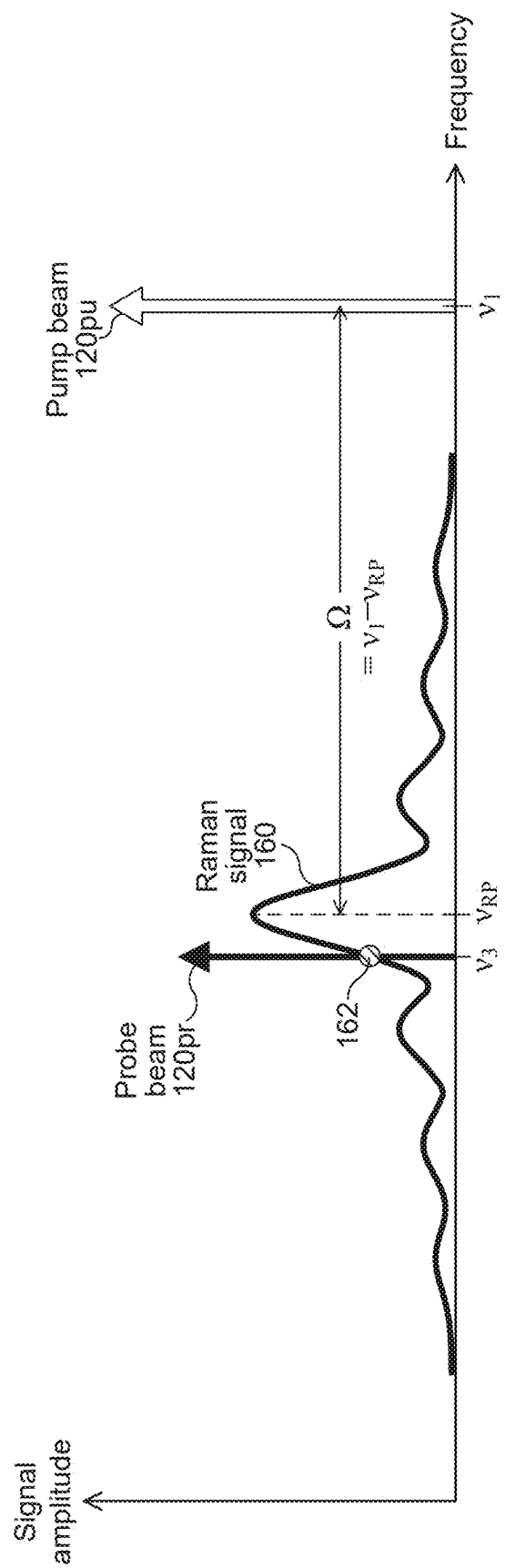


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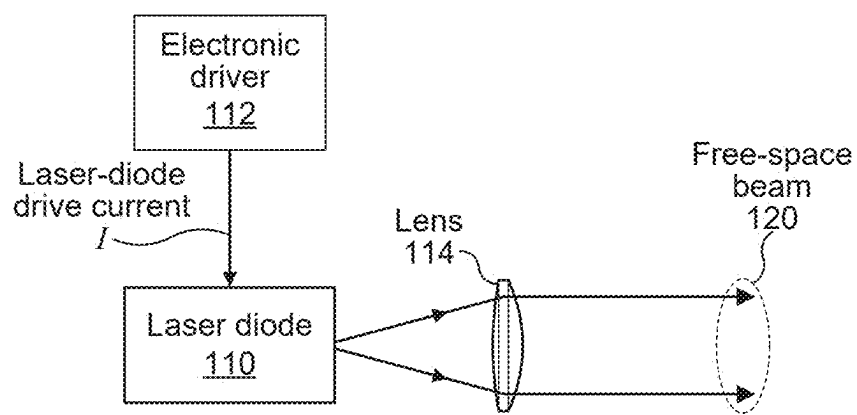


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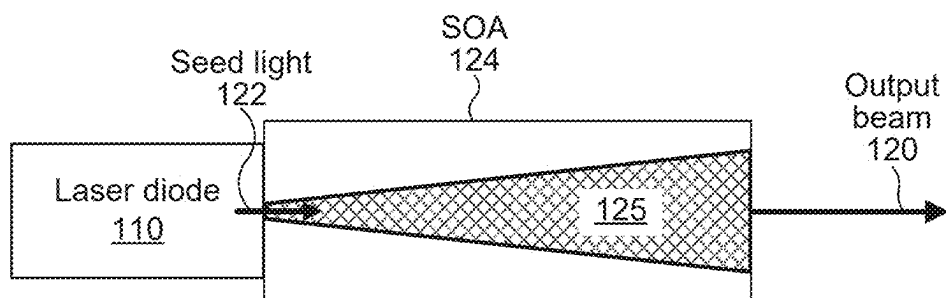


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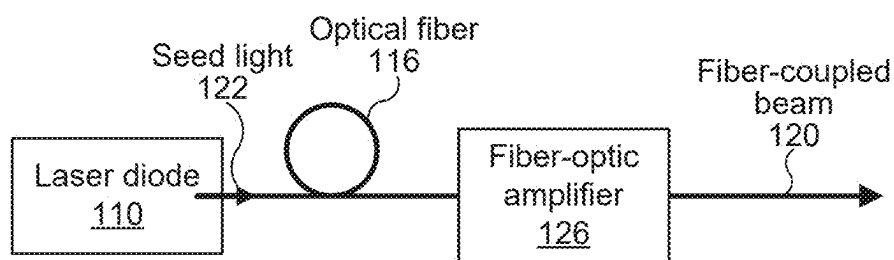


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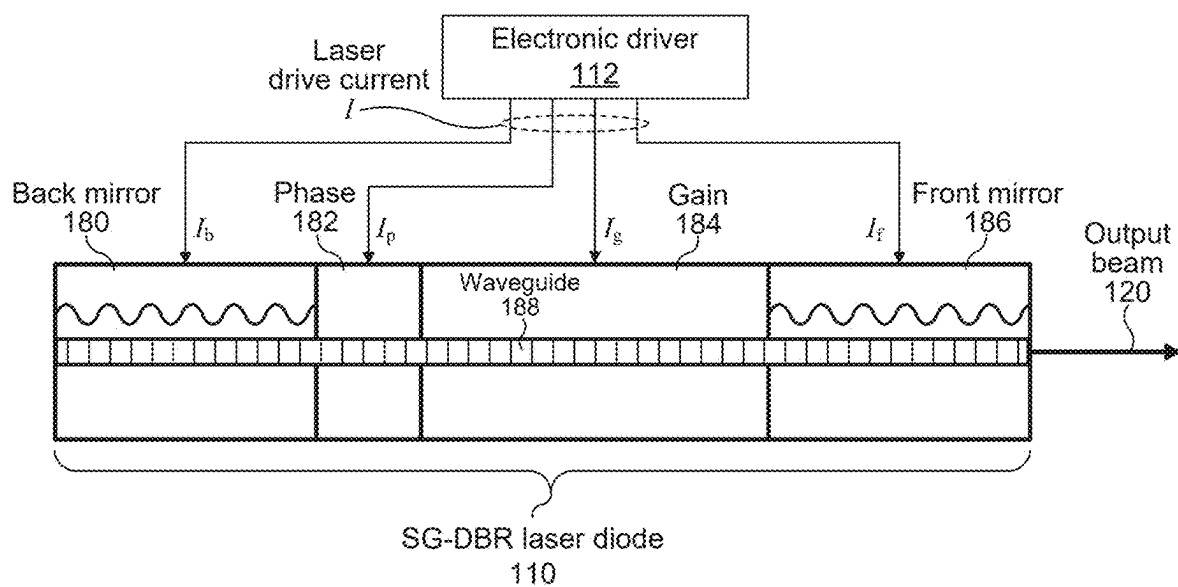


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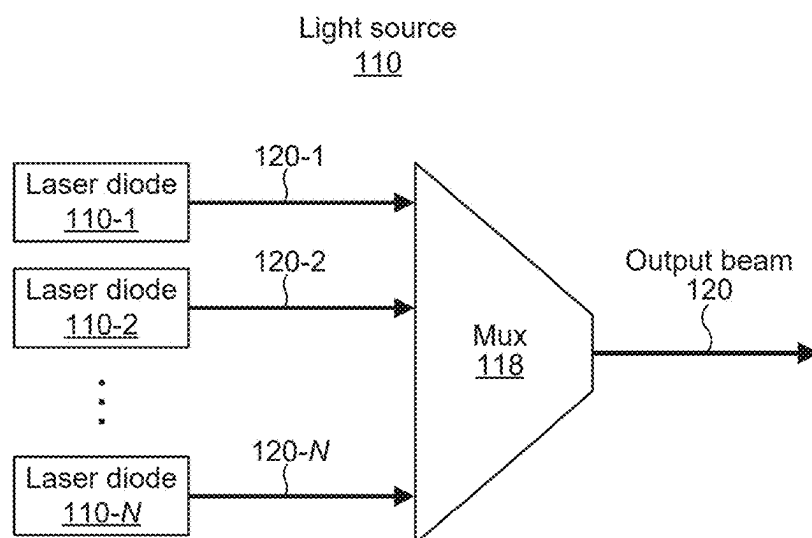


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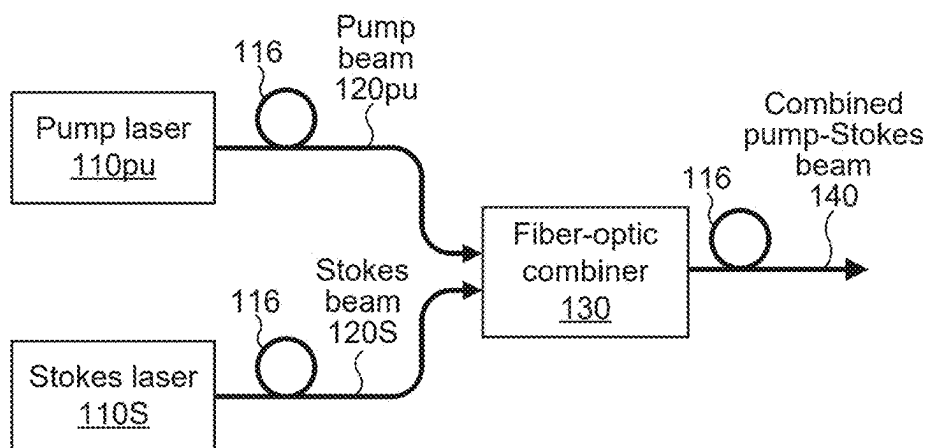


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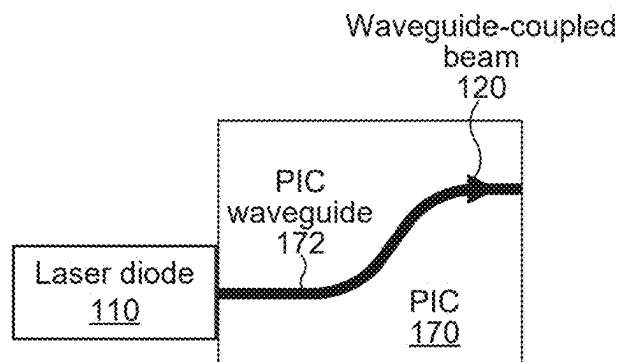


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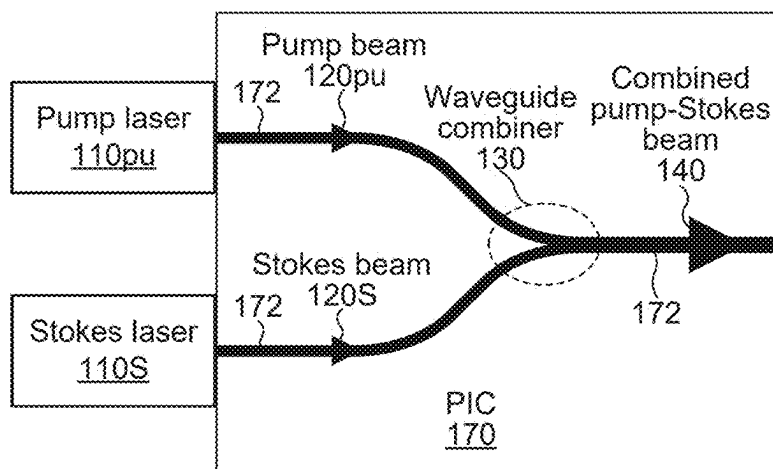


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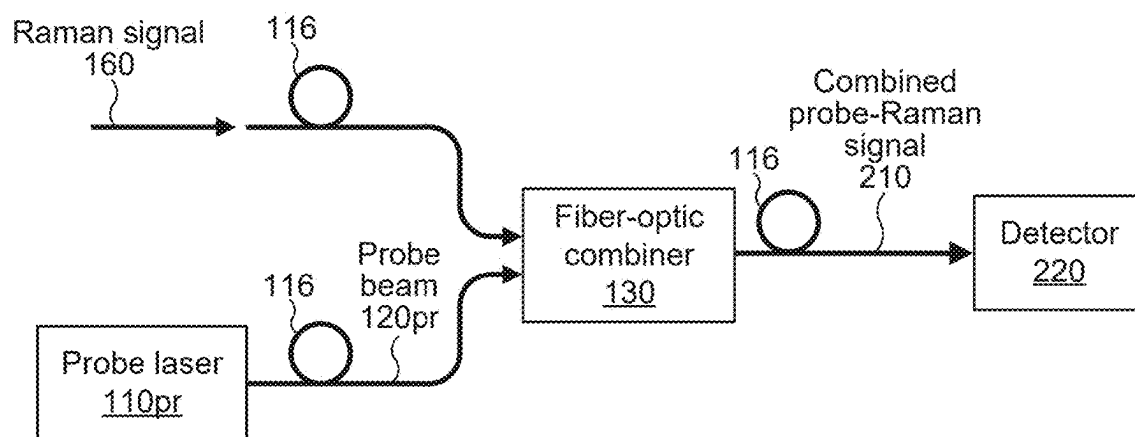


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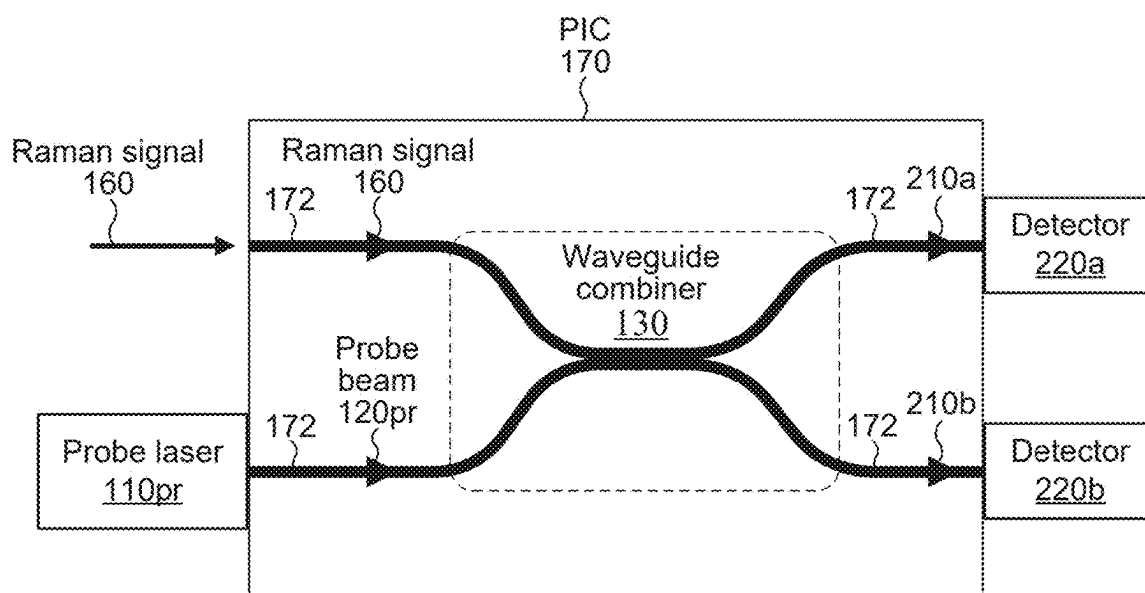


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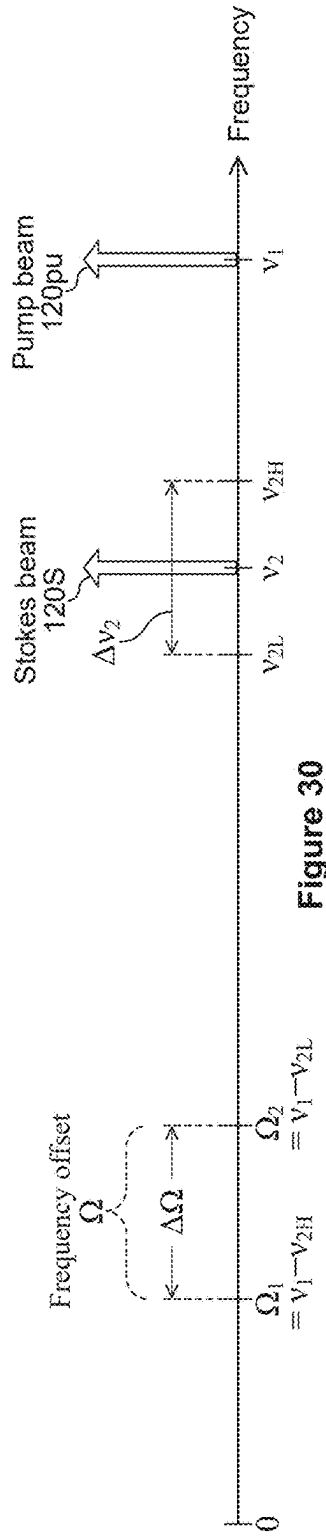


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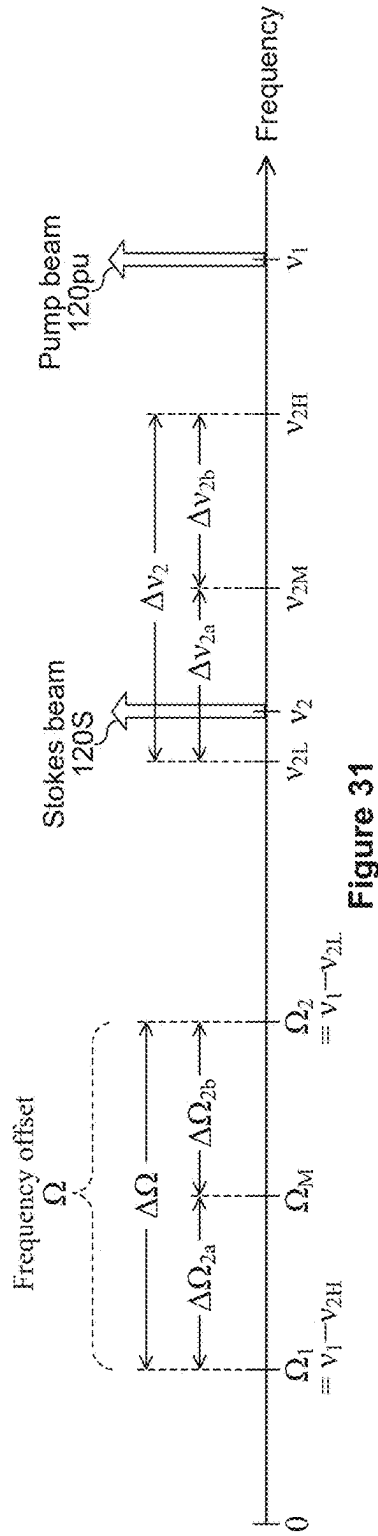


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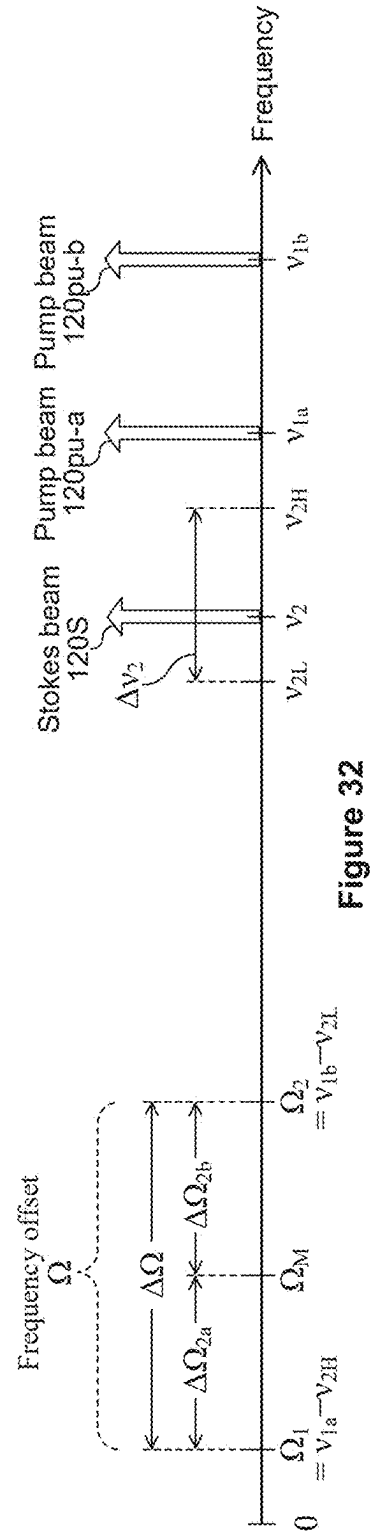


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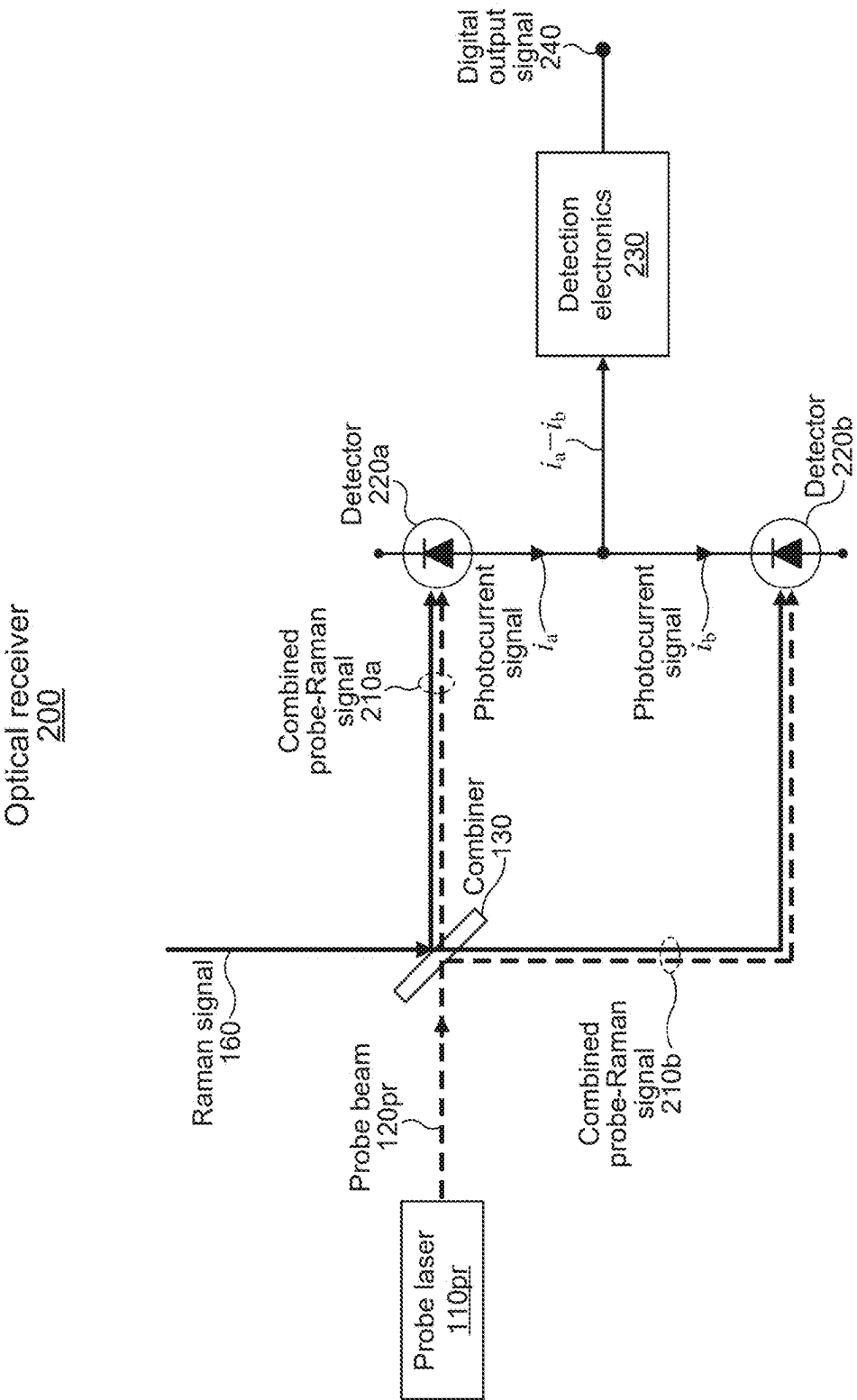


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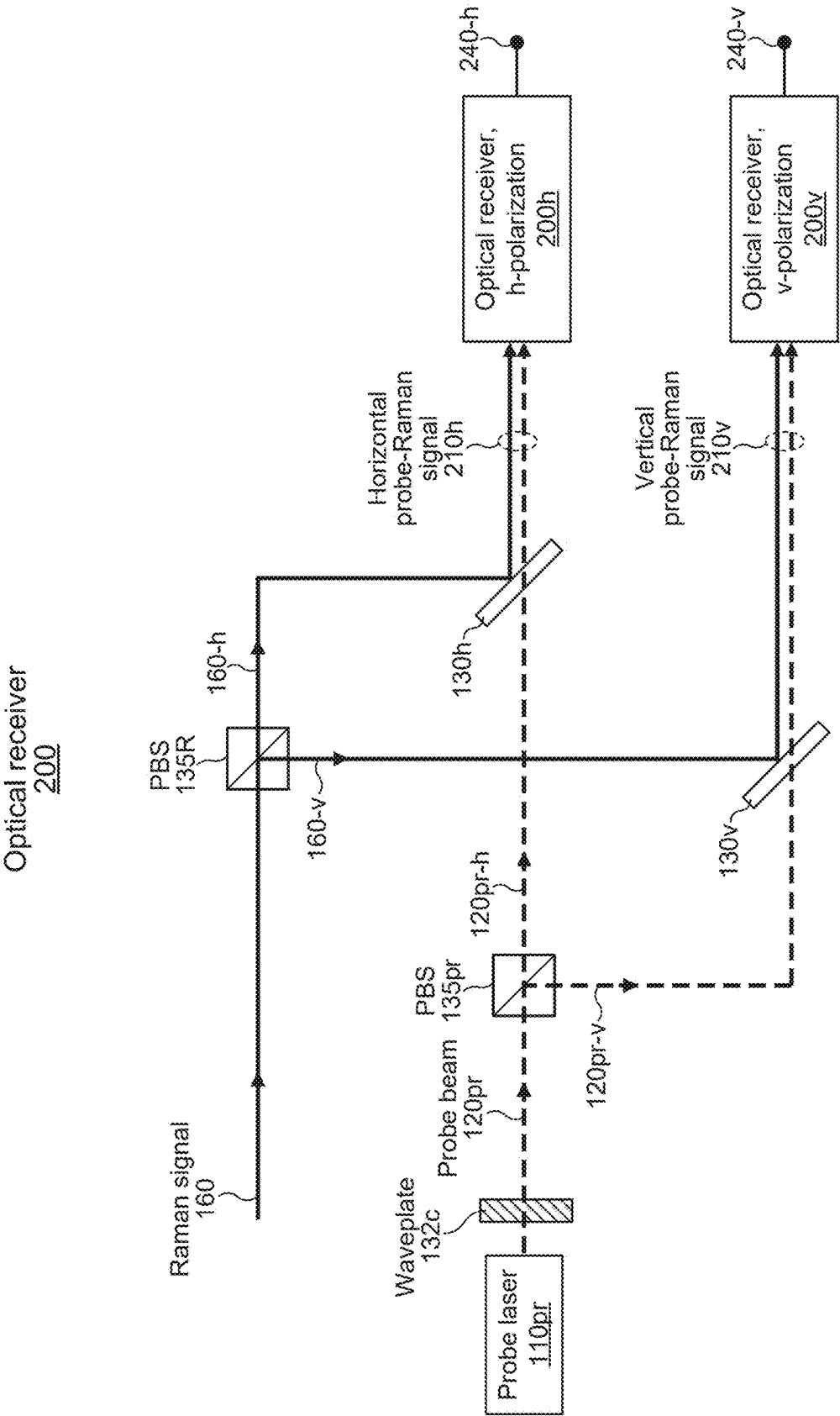


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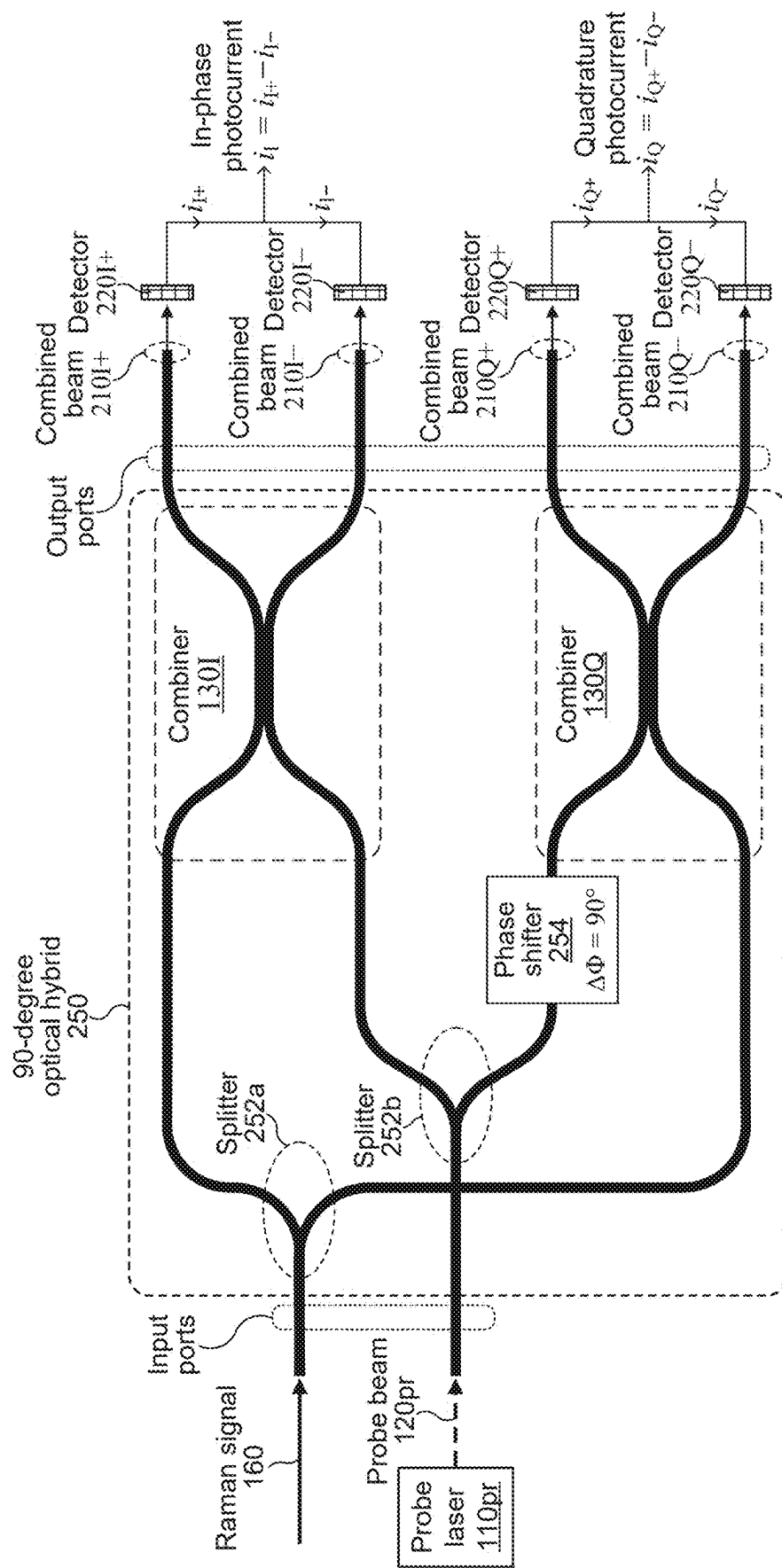


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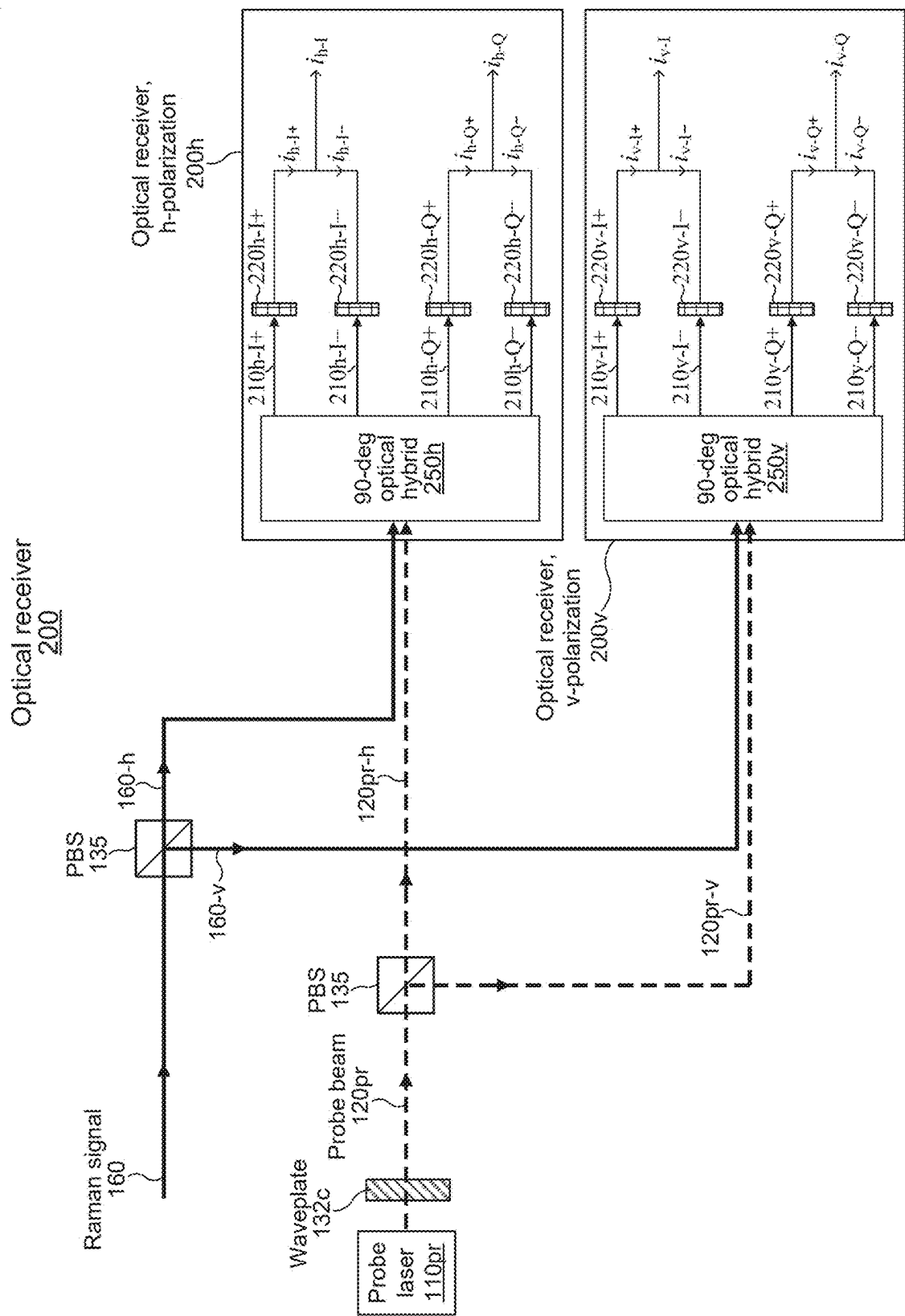


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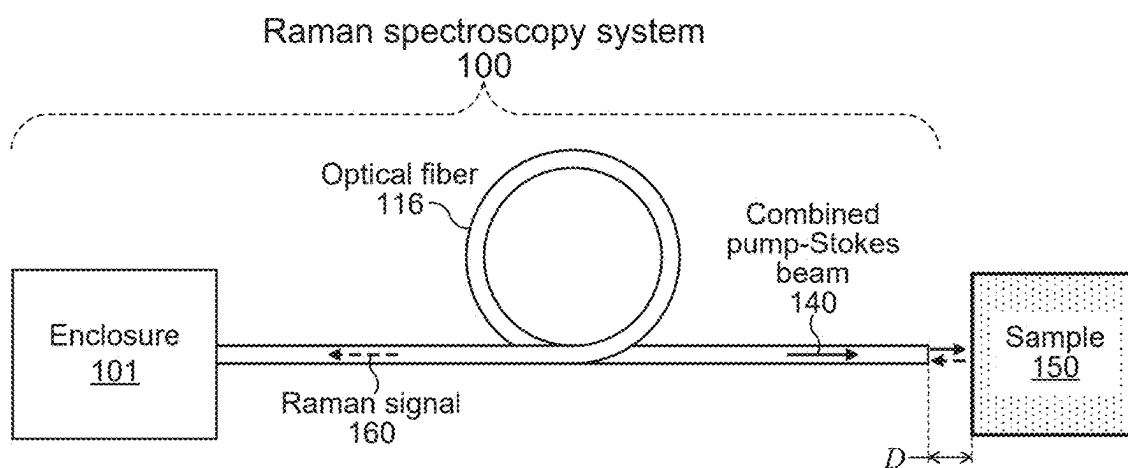


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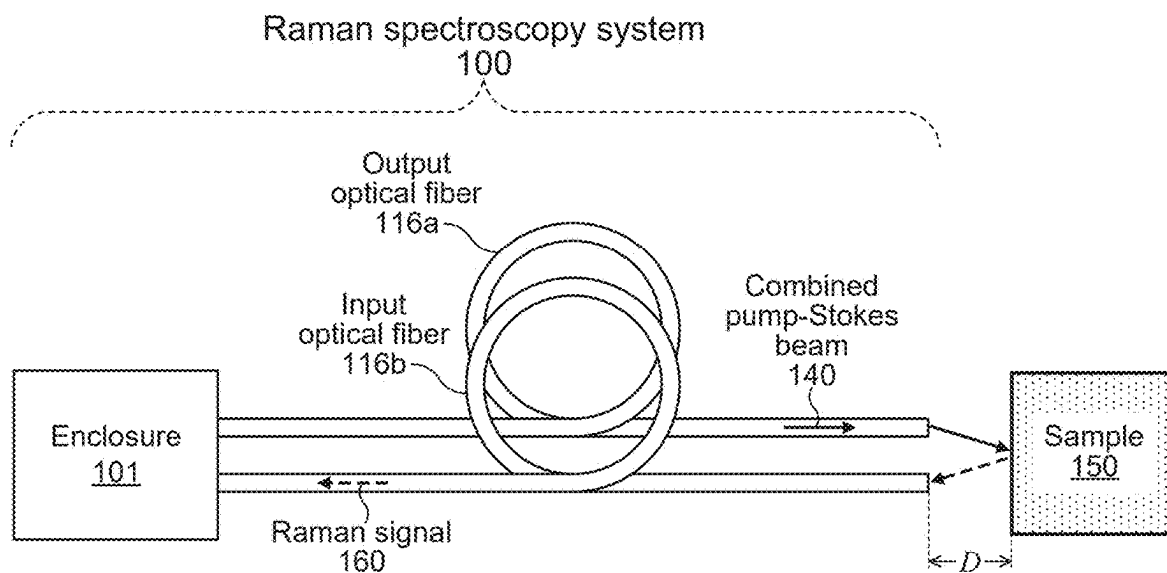


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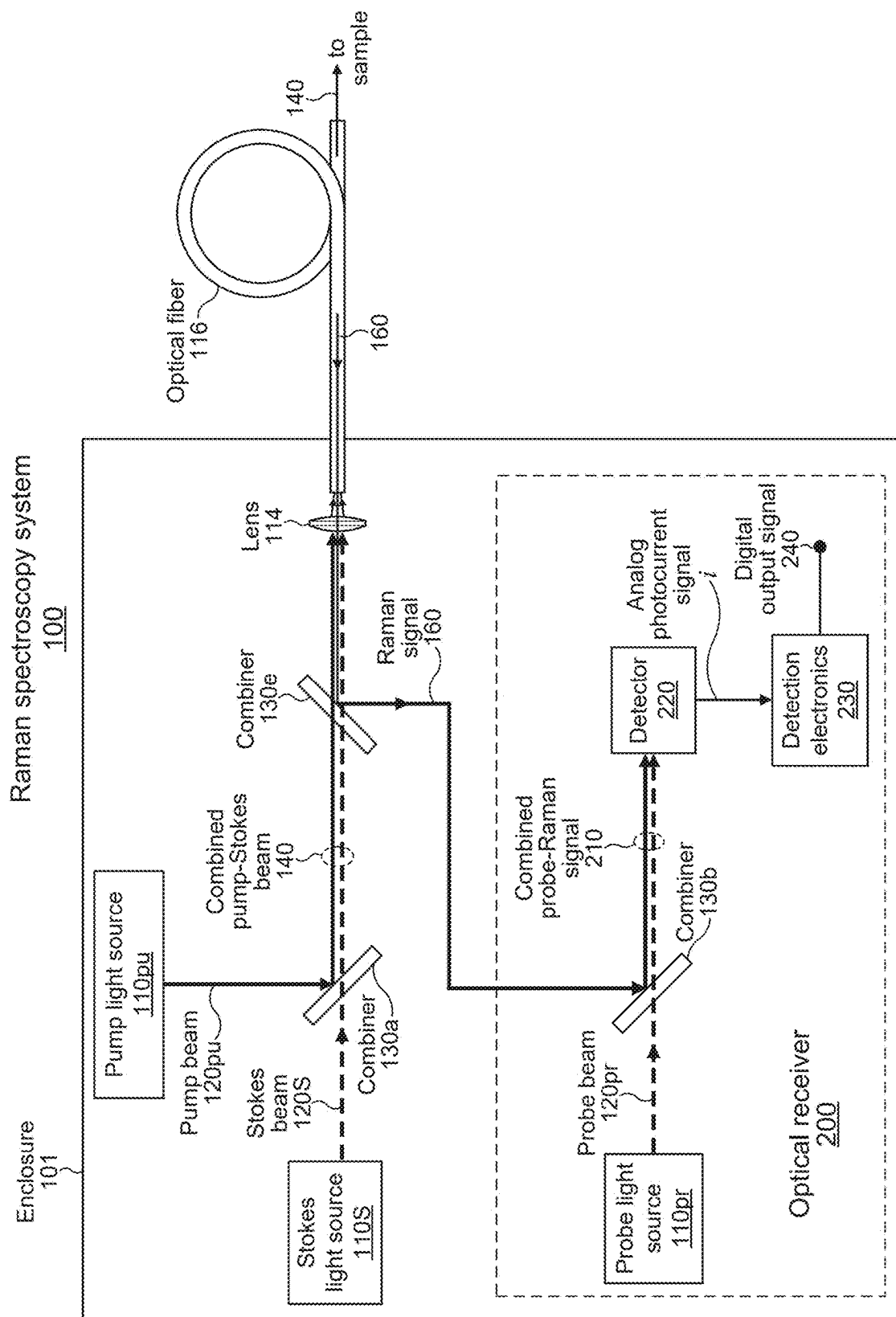


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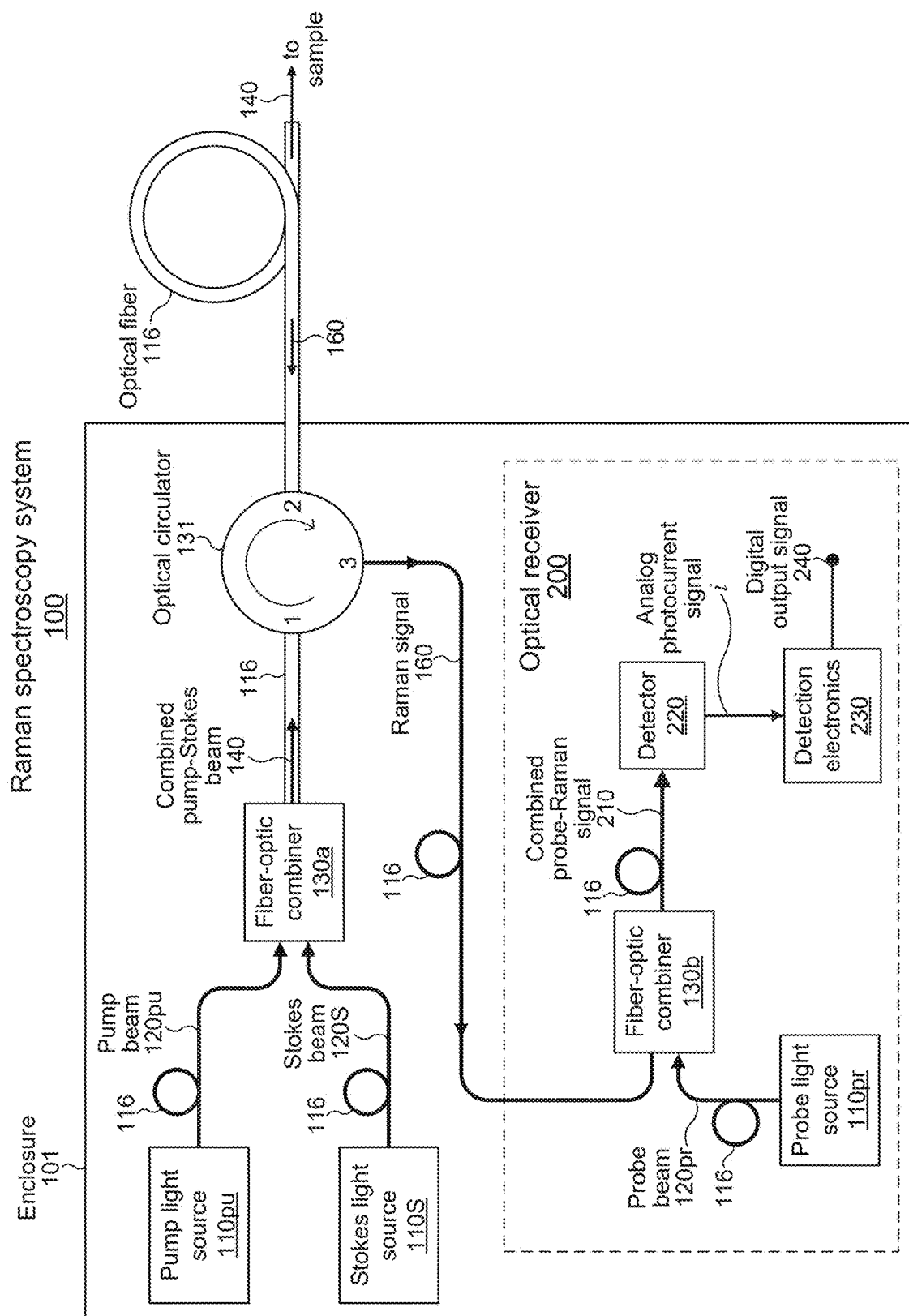


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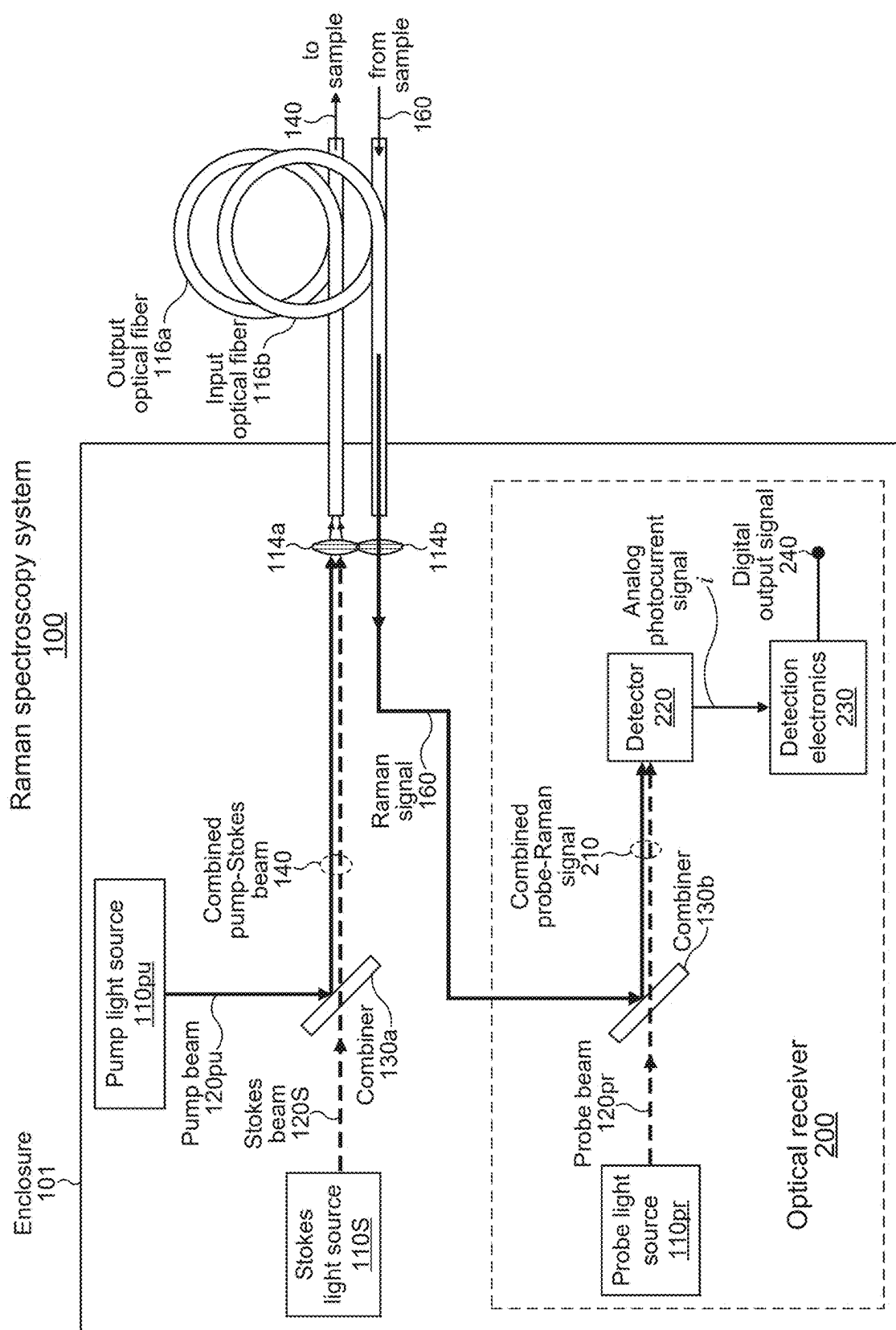


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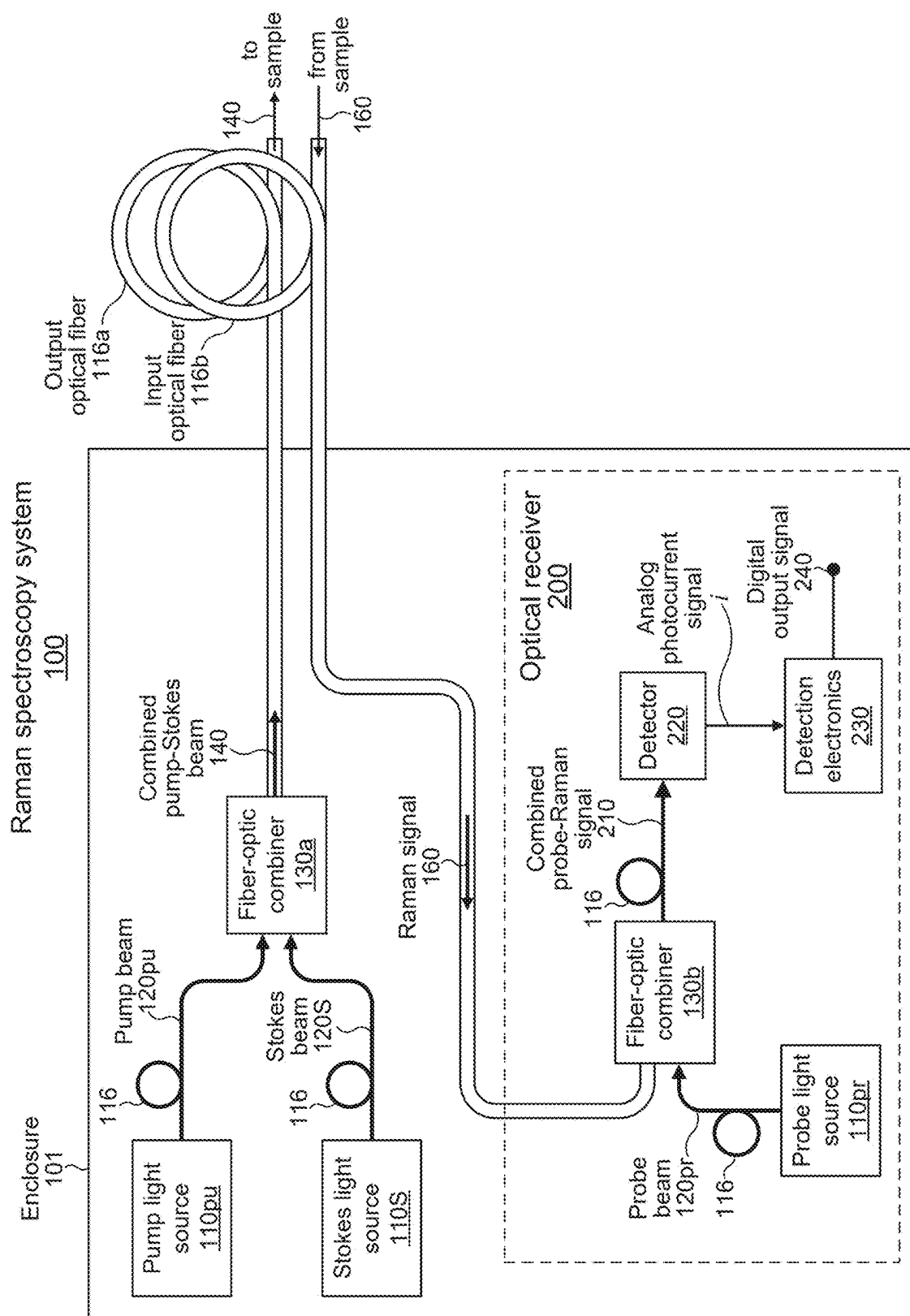


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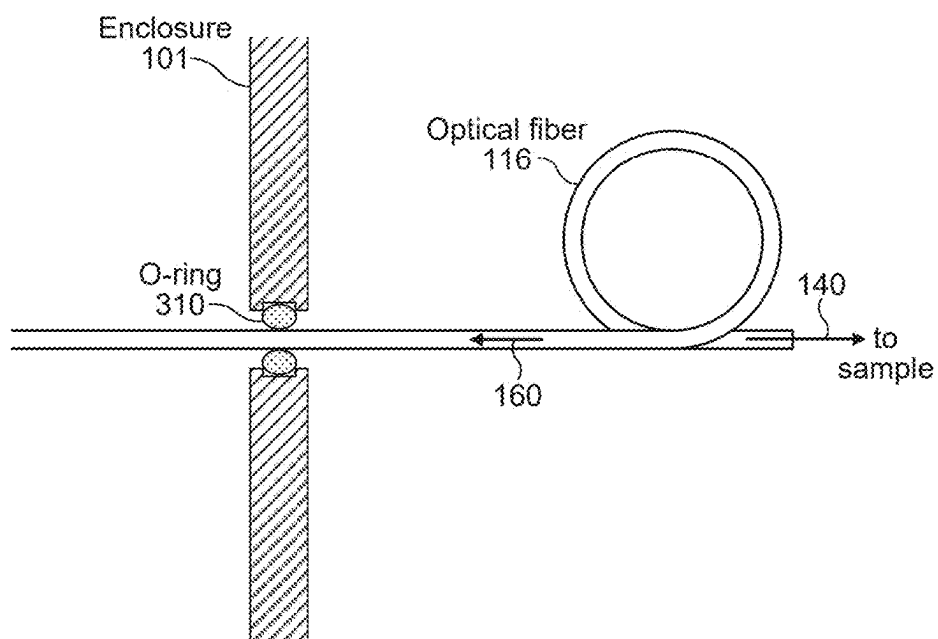


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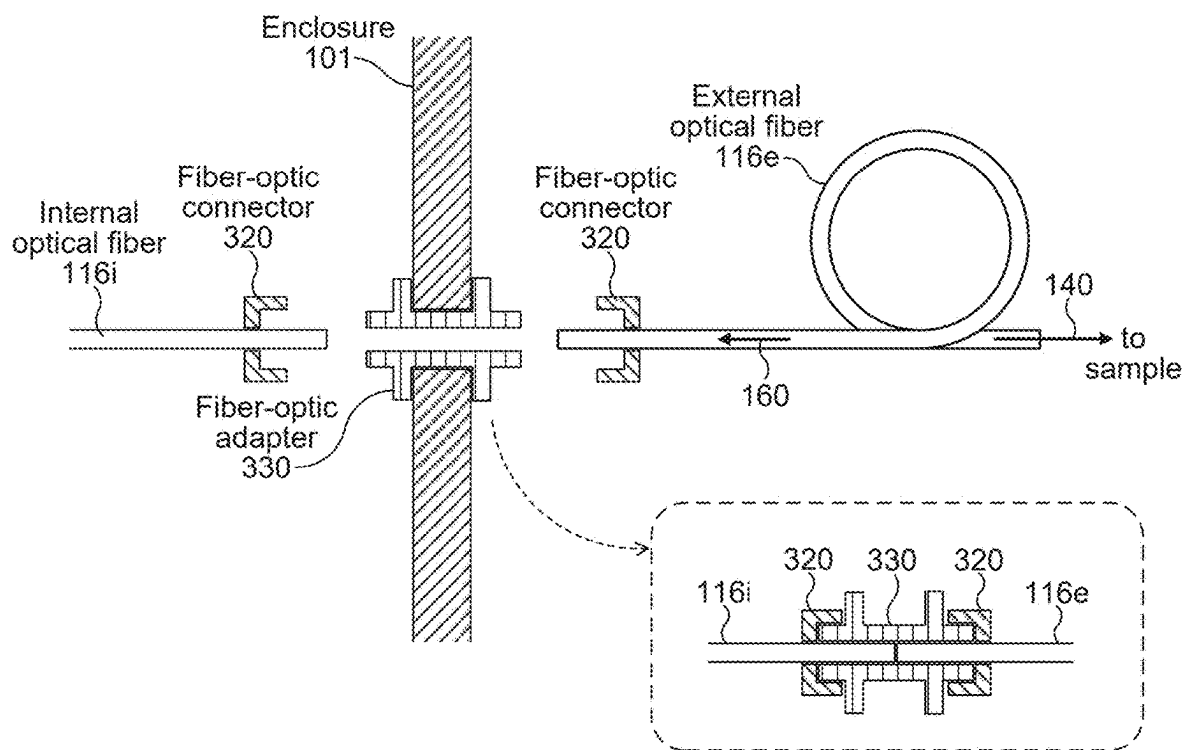


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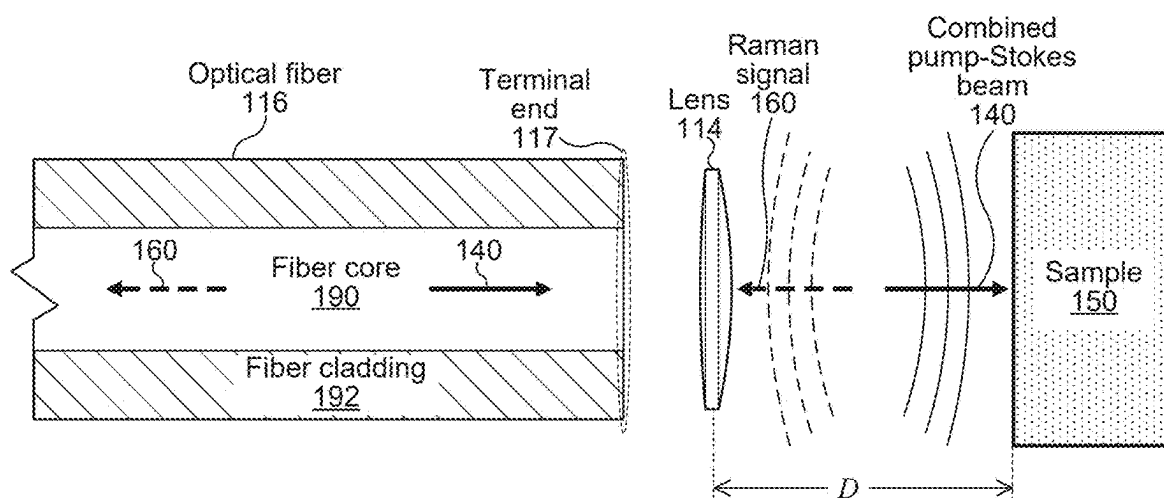


Figure 48

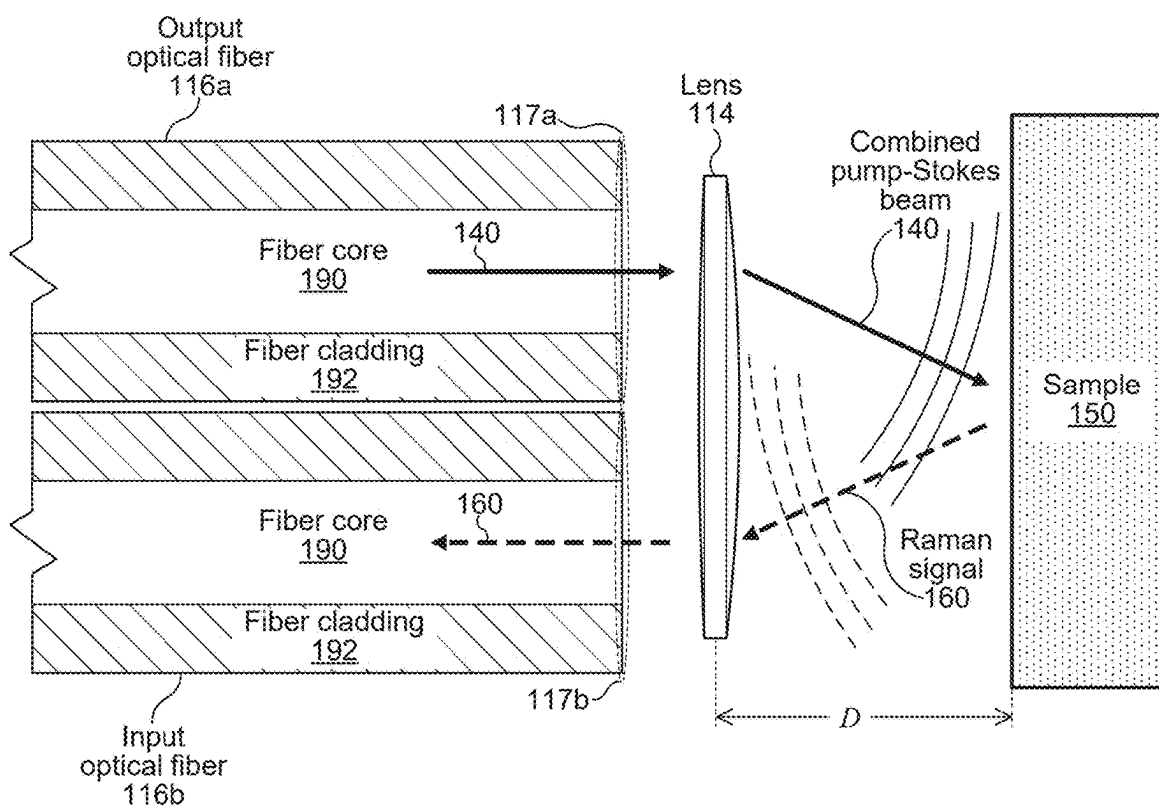
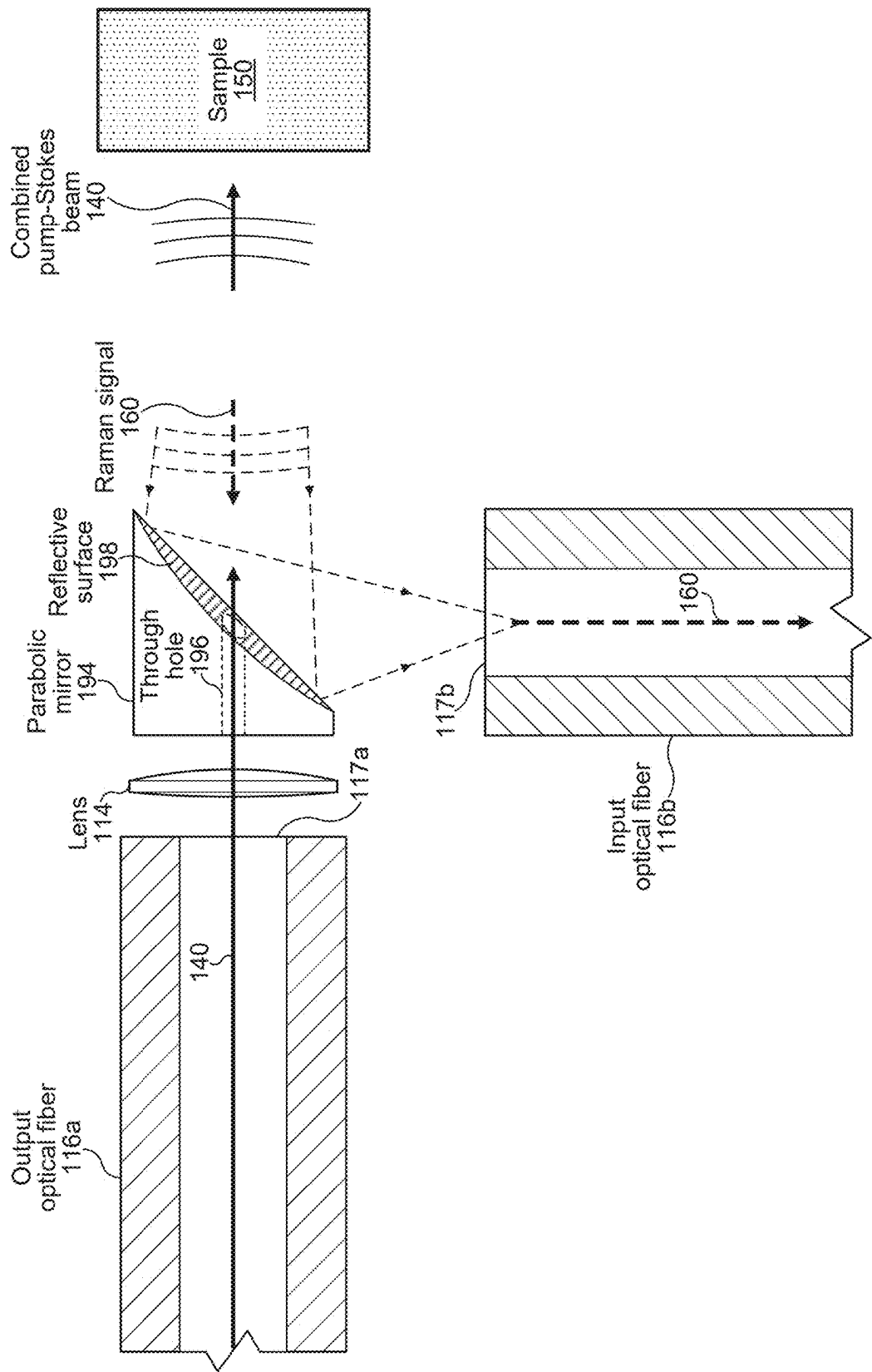


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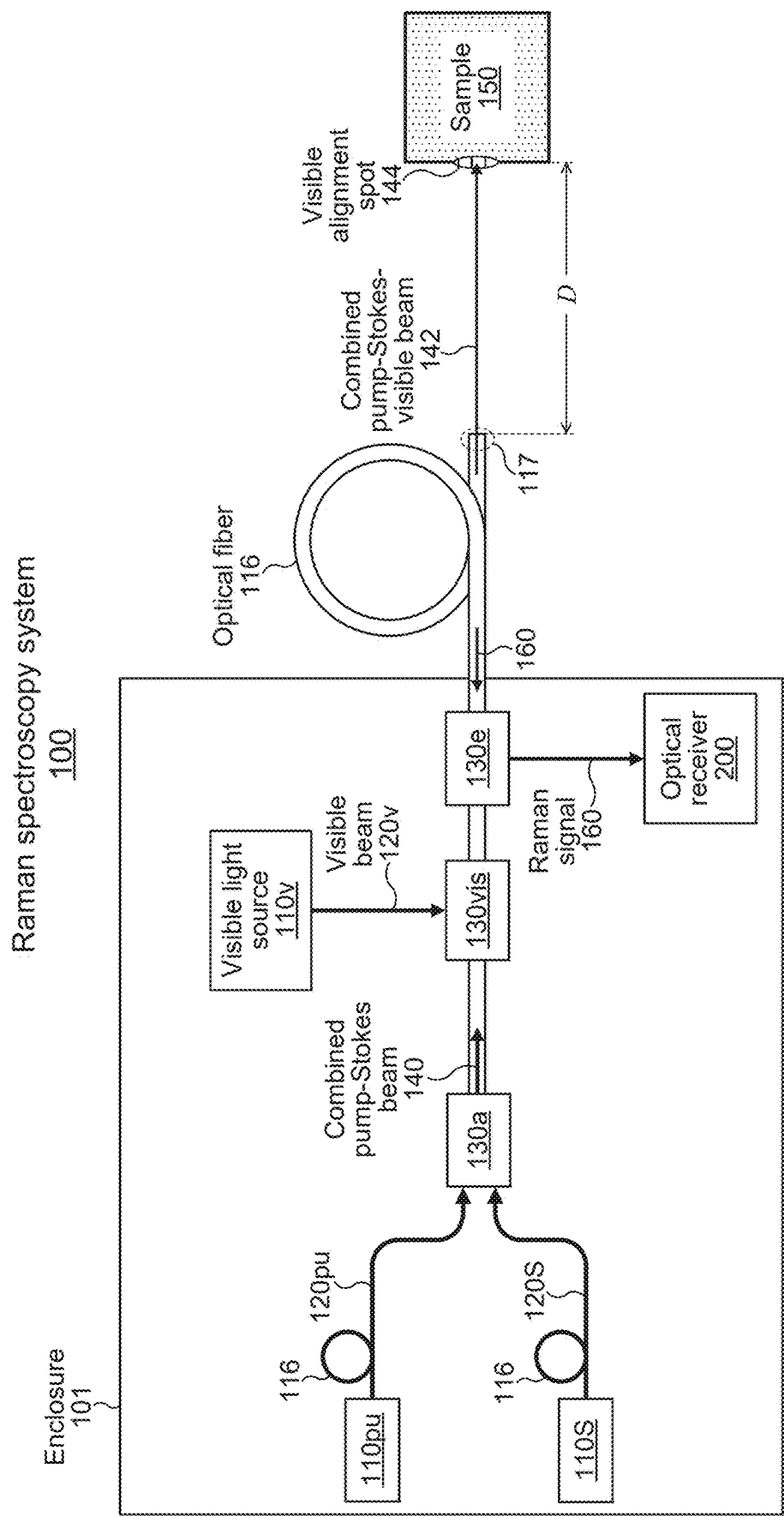


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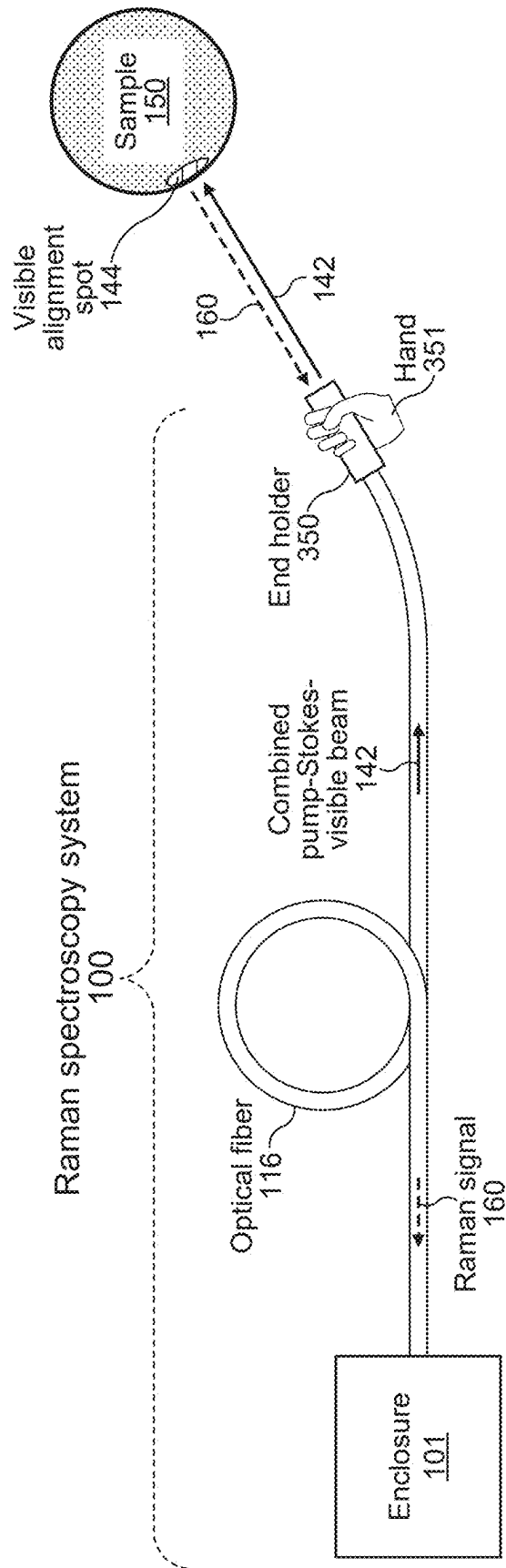


Figure 52

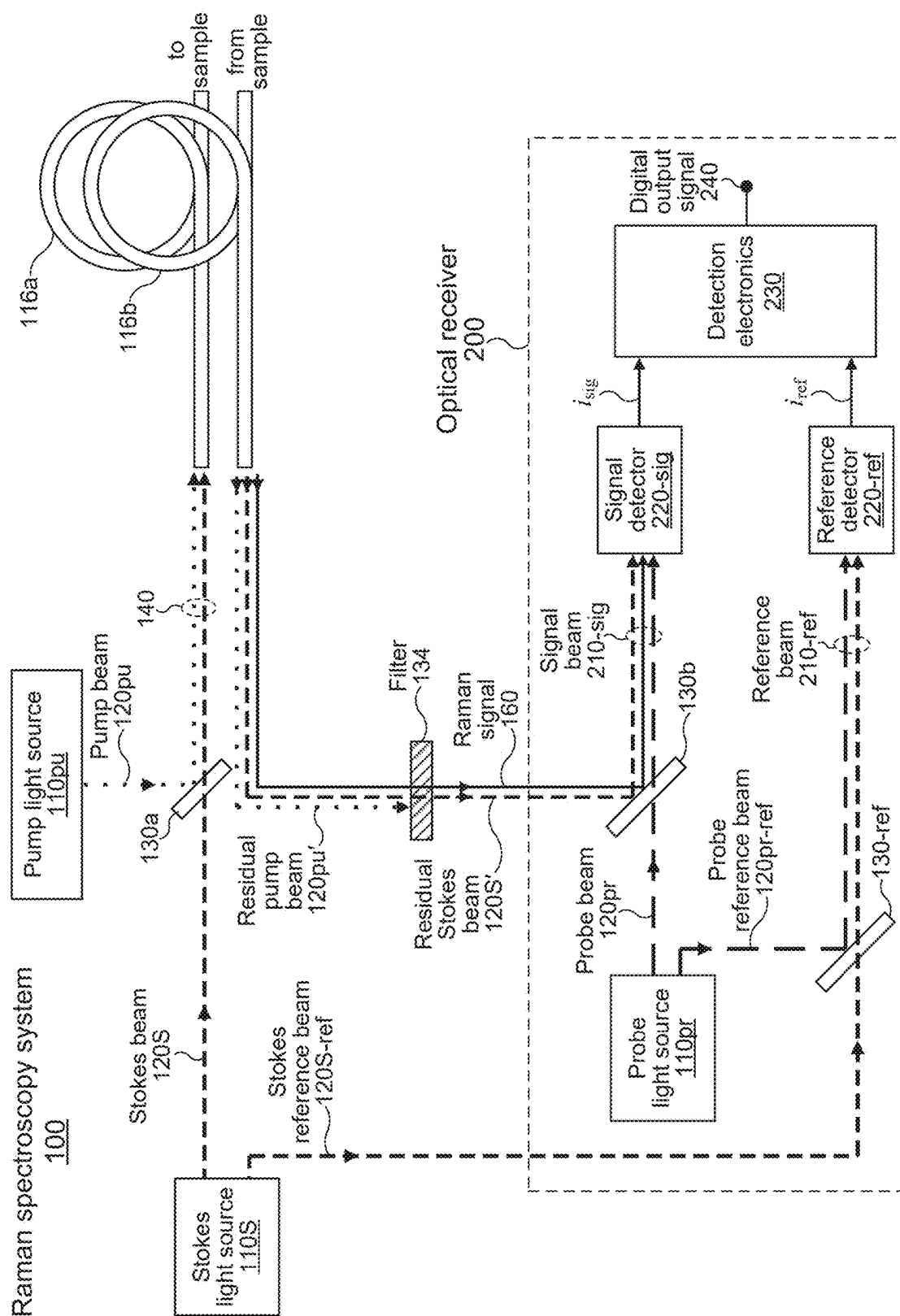
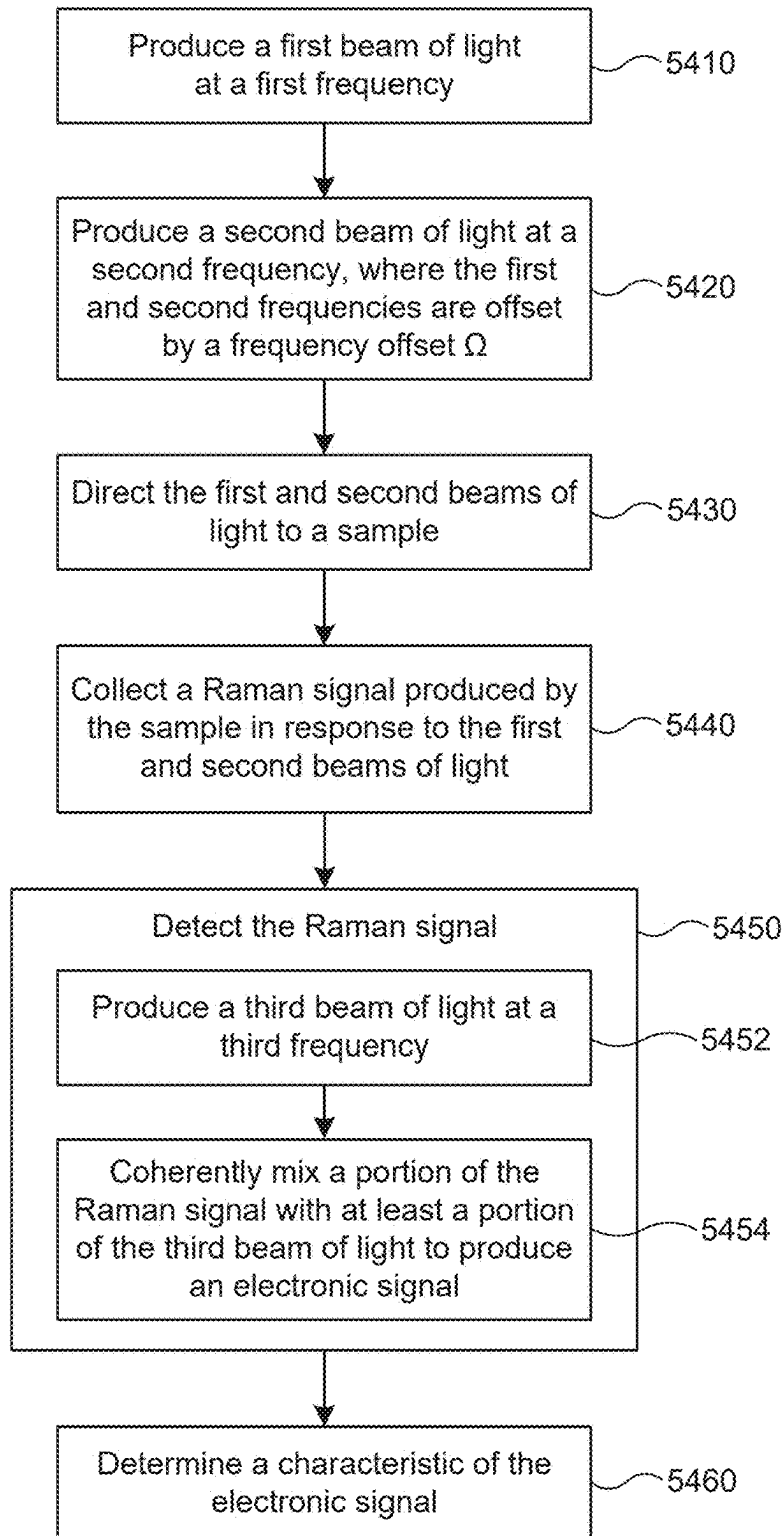


Figure 53

5400**Figure 54**

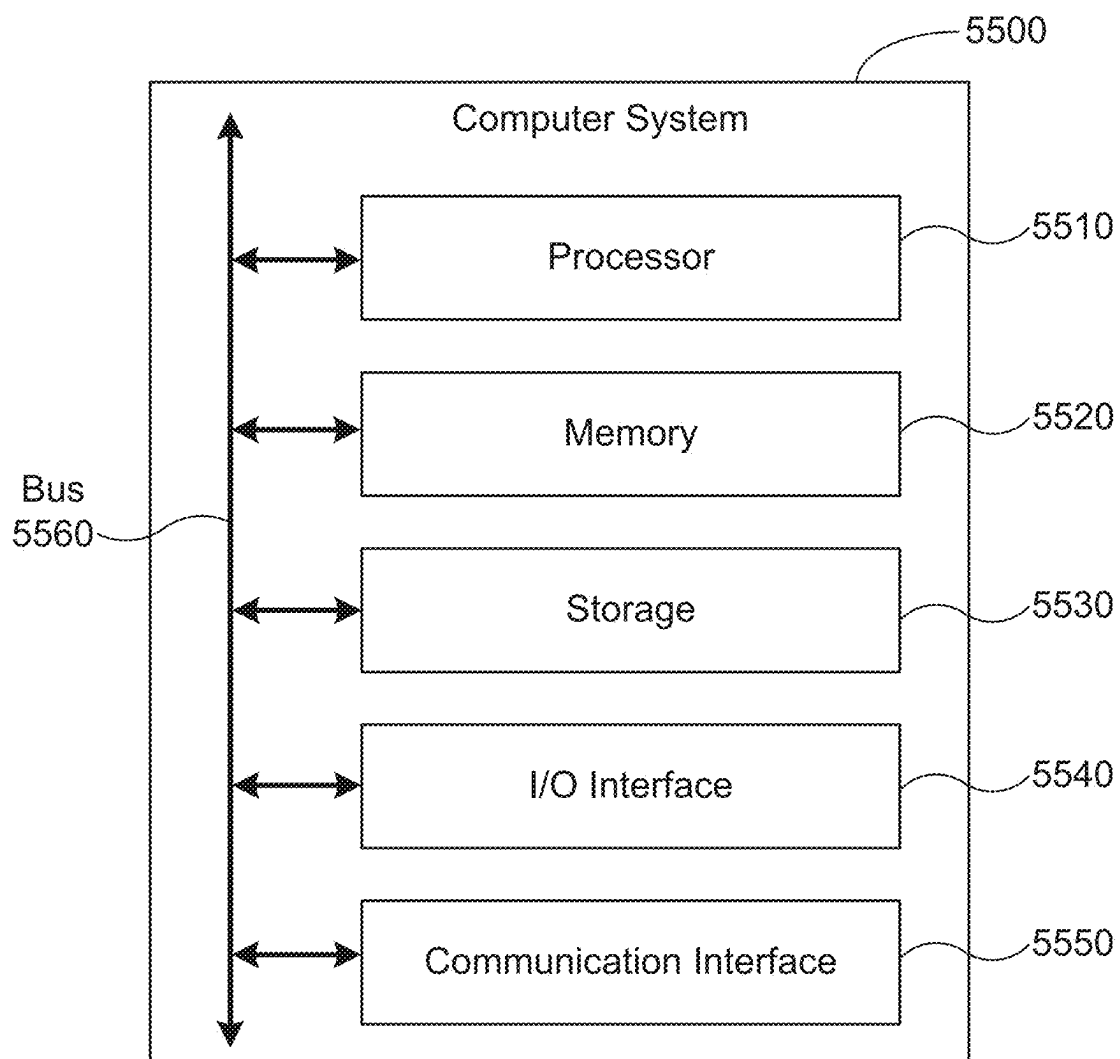


Figure 55

RAMAN SPECTROSCOPY SYSTEM WITH OPTICAL-FIBER EXTENSION

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the benefit, under 35 U.S.C. § 119 (e), of U.S. Provisional Patent Application No. 63/560,521, filed 1 Mar. 2024, which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] This disclosure generally relates to Raman spectroscopy systems and methods that use Raman scattering.

BACKGROUND

[0003] Raman spectroscopy is an optical measurement technique that can be applied to the study of molecular dynamics (e.g., to investigate vibrational and rotational states of molecules). Molecules typically exhibit molecular vibrations with frequencies ranging from less than 10 terahertz (THz) to approximately 100 THz, which corresponds to wavenumbers of approximately 300 cm^{-1} to 3000 cm^{-1} and wavelengths of approximately 30 to 3 micrometers (μm). Raman spectroscopy is based on the inelastic scattering of photons (referred to as Raman scattering) that occurs when light interacts with molecular vibrations or phonons in a sample. Raman scattering causes the energy (or equivalently, the frequency) of scattered light to be shifted, and this shift in energy can provide information about the vibrational modes of molecules in the sample.

[0004] Raman spectroscopy can be used in various chemical sensing applications to identify molecular components in a sample. Since many molecules exhibit a unique Raman scattering spectrum, the spectrum of Raman-scattered light produced when light interacts with a sample can serve as a fingerprint to sense or identify various molecular species within the sample. A sample illuminated with light may produce Raman scattered light at different wavelengths from the illumination light, and measurement of the spectrum of the Raman scattered light is typically performed in the optical domain. For example, the spectrum of Raman scattered light can be measured using an optical spectrometer which separates the Raman scattered light into its optical frequency components using a diffractive element, such as a diffraction grating.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIGS. 1-2 each illustrates an example Raman spectroscopy system.

[0006] FIGS. 3-4 each illustrate an example Raman signal produced by coherent Raman scattering.

[0007] FIG. 5 illustrates the example Raman signal of FIG. 4 along with a probe beam.

[0008] FIG. 6 illustrates an expanded view of a portion of the Raman signal of FIG. 5.

[0009] FIG. 7 illustrates an expanded view of a portion of the Raman signal of FIG. 6.

[0010] FIGS. 8-10 illustrate time-domain and frequency-domain plots of an example electronic signal resulting from coherent mixing of the Raman signal and probe beam of FIG. 7.

[0011] FIG. 11 illustrates an example Raman signal that is measured at multiple probe frequencies.

[0012] FIG. 12 illustrates an example Raman spectrum corresponding to the Raman signal of FIG. 11.

[0013] FIG. 13 illustrates another example Raman signal that is measured at multiple probe frequencies.

[0014] FIGS. 14-15 each illustrate a second example Raman signal obtained by changing the frequency offset between a pump beam and a Stokes beam.

[0015] FIG. 16 illustrates an example Raman signal along with two probe beams of light.

[0016] FIG. 17 illustrates an example optical receiver for measuring the Raman signal and pump beam of light from FIG. 16.

[0017] FIG. 18 illustrates an example Raman spectroscopy system for measuring a Raman signal produced by spontaneous Raman scattering.

[0018] FIG. 19 illustrates an example Raman signal produced by the Raman spectroscopy system of FIG. 18.

[0019] FIG. 20 illustrates an example laser diode that produces a free-space beam of light.

[0020] FIG. 21 illustrates an example laser diode that produces seed light that is amplified by a semiconductor optical amplifier (SOA).

[0021] FIG. 22 illustrates an example laser diode that produces seed light that is amplified by a fiber-optic amplifier.

[0022] FIG. 23 illustrates an example sampled-grating distributed Bragg reflector (SG-DBR) laser.

[0023] FIG. 24 illustrates an example light source with multiple laser diodes and an optical multiplexer that combines light produced by the laser diodes into a single output beam of light.

[0024] FIG. 25 illustrates an example pump laser and Stokes laser with a fiber-optic combiner that produces a combined pump-Stokes beam coupled into an optical fiber.

[0025] FIG. 26 illustrates an example laser diode coupled to a waveguide of a photonic integrated circuit (PIC).

[0026] FIG. 27 illustrates an example pump laser and Stokes laser with a photonic integrated circuit (PIC) that produces a combined pump-Stokes beam coupled into an optical waveguide of the PIC.

[0027] FIG. 28 illustrates an example fiber-optic combiner that combines a Raman signal with a probe beam.

[0028] FIG. 29 illustrates an example photonic integrated circuit (PIC) with a waveguide combiner that combines a Raman signal with a probe beam.

[0029] FIGS. 30-35 each illustrate example frequency ranges of a pump beam and a Stokes beam.

[0030] FIG. 36 illustrates an example optical receiver with two detectors.

[0031] FIG. 37 illustrates an example optical receiver configured for polarization-sensitive detection of a Raman signal.

[0032] FIG. 38 illustrates an example optical receiver configured to detect in-phase and quadrature components of a Raman signal.

[0033] FIG. 39 illustrates an example optical receiver configured to detect polarization as well as in-phase and quadrature components of a Raman signal.

[0034] FIGS. 40-45 each illustrates an example Raman spectroscopy system that includes one or more optical fibers.

[0035] FIGS. 46-47 each illustrates a cross-section of a portion of an enclosure with an example fiber-optic feed-through.

[0036] FIG. 48 illustrates the terminal end of an example optical fiber along with a lens.

[0037] FIG. 49 illustrates example input and output optical fibers along with a lens.

[0038] FIG. 50 illustrates example input and output optical fibers along with a parabolic mirror.

[0039] FIG. 51 illustrates an example Raman spectroscopy system that includes a visible light source.

[0040] FIG. 52 illustrates an example Raman spectroscopy system with a moveable end holder.

[0041] FIG. 53 illustrates an example Raman spectroscopy system with a balanced-detection optical receiver.

[0042] FIG. 54 illustrates an example method for measuring a Raman signal.

[0043] FIG. 55 illustrates an example computer system.

DETAILED DESCRIPTION

[0044] FIGS. 1-2 each illustrates an example Raman spectroscopy system 100. The Raman spectroscopy system 100 in each of FIGS. 1-2 may detect a Raman signal 160 by coherently mixing the Raman signal 160 with a probe beam of light 120_{pr} produced by a probe light source 110_{pr}. The Raman spectroscopy system 100 in each of FIGS. 1-2 may be referred to as a coherent Raman spectroscopy system, a coherent Raman spectroscopy system with heterodyne detection, or a high-resolution coherent Raman spectroscopy system. One or more of the systems or methods described herein may be applied to any suitable form of coherent Raman spectroscopy or coherent Raman scattering (CRS), such as for example, coherent anti-Stokes Raman scattering (CARS), stimulated Raman scattering (SRS), or Raman-induced Kerr effect (RIKE).

[0045] The Raman spectroscopy system 100 in FIG. 1 includes a pump light source 110_{pu} that produces a pump beam of light 120_{pu} and a Stokes light source 110_S that produces a Stokes beam of light 120_S. The pump light source 110_{pu} produces the pump beam of light 120_{pu} at a pump frequency, which may be referred to as a first frequency and may be represented by ν_1 , ν_{pu} , ω_1 , or ω_{pu} . The pump light source 110_{pu} may be referred to as a first light source, and the pump beam of light 120_{pu} may be referred to as a first beam of light. The Stokes light source 110_S produces the Stokes beam of light 120_S at a Stokes frequency, which may be referred to as a second frequency and may be represented by ν_2 , ν_S , ω_2 , or ω_S . The Stokes light source 110_S may be referred to as a second light source, and the Stokes beam of light 120_S may be referred to as a second beam of light. The pump and Stokes frequencies may be offset by a frequency offset Ω , where Ω equals $\nu_{pu} - \nu_S$ (or equivalently, $\Omega = \nu_1 - \nu_2$). Generally, the pump frequency ν_{pu} is greater than the Stokes frequency ν_S , and the frequency offset Ω is a positive value. The pump and Stokes light sources in FIG. 1 may each include a laser. For example, the Raman spectroscopy system 100 in FIG. 2 includes a pump laser 110_{pu} that produces a pump beam of light 120_{pu} and a Stokes laser 110_S that produces a Stokes beam of light 120_S.

[0046] In FIGS. 1-2, the pump beam 120_{pu} and the Stokes beam 120_S are directed to a sample 150, and the sample 150 produces a Raman signal 160 in response to the pump and Stokes beams. For example, the Raman signal 160 may be produced by coherent Raman scattering of the pump and Stokes beams within the sample 150. A Raman spectroscopy system may include one or more optical elements that direct

the pump beam 120_{pu} and the Stokes beam 120_S to a sample 150. Additionally, the optical elements may collect the Raman signal 160 produced by the sample 150 in response to the pump beam 120_{pu} and Stokes beam 120_S and may direct the Raman signal 160 to an optical receiver 200. The optical elements may include free-space optics, fiber-optic components, waveguide-based optics, metamaterials, or any combination thereof. For example, the optical elements may include a mirror, lens, optical combiner (e.g., beamsplitter), optical fiber, photonic integrated circuit (PIC), optical waveguide, or metamaterial-based optic. As another example, the optical combiner 130_a in each of FIGS. 1-2 may be a free-space dichroic beamsplitter that transmits light at the pump-beam wavelength and reflects light at the Stokes-beam wavelength. The combiner 130_a may combine the pump beam 120_{pu} and the Stokes beam 120_S to produce a combined pump-Stokes beam 140 that is directed to the sample 150. The pump and Stokes beams may be combined so that they are substantially overlapped with one another and propagate in the same direction along approximately the same optical axis. Alternatively, a Raman spectroscopy system 100 may not include a pump-Stokes optical combiner 130_a, and the pump beam 120_{pu} and the Stokes beam 120_S may be directed to a sample 150 as separate beams (e.g., the pump and Stokes beams may be overlapped or combined at the sample rather than being combined earlier). In this embodiment, the pump and Stokes beams may enter the sample 150 from opposite sides (e.g., the pump and Stokes beams may propagate to the sample in opposite directions along approximately the same optical axis). As another example, the optical elements may include a lens (not illustrated in FIGS. 1-2) that focuses the pump-Stokes beam 140 onto the sample 150. Additionally, the optical elements may include a lens (not illustrated in FIGS. 1-2) that collects the Raman signal 160 to produce a Raman-signal beam that is directed to the optical receiver 200.

[0047] In FIGS. 1-2 the combined pump-Stokes beam 140 is directed to one side of the sample 150, and the Raman signal 160 is emitted from the opposite side of the sample. A Raman signal 160 that is collected and directed to an optical receiver 200 may be emitted from a sample 150 in any suitable direction. For example, a Raman signal 160 that is sent to an optical receiver 200 may be emitted from a sample 150 in a forward-scattered direction (e.g., as illustrated in FIGS. 1-2), in a backward-scattered direction (e.g., back towards the pump or Stokes beam), or in a sideways-scattered direction (e.g., in a direction approximately orthogonal to the combined pump-Stokes beam 140 in FIGS. 1-2).

[0048] The Raman spectroscopy system 100 in each of FIGS. 1-2 includes an optical receiver 200 that detects the Raman signal 160 produced by the sample 150. The optical receiver 200 (which may be referred to as a heterodyne optical receiver or a high-resolution optical receiver) may detect the Raman signal 160 using an optical heterodyne technique in which the Raman signal 160 is coherently mixed with a probe beam of light 120_{pr}. The optical receiver 200 in FIG. 1 includes a probe light source 110_{pr} that produces a probe beam of light 120_{pr} at a probe frequency. The probe light source 110_{pr} may be referred to as a third light source, and the probe beam of light 120_{pr} may be referred to as a third beam of light. The probe frequency may be referred to as a third frequency and may be represented by ν_3 , ν_{pr} , ω_3 , or ω_{pr} . The probe light source 110_{pr} in FIG.

1 may include a laser. For example, the optical receiver **200** in FIG. **2** includes a probe laser **110_{pr}** that produces a probe beam of light **120_{pr}**.

[0049] The optical combiner **130_b** in each of FIGS. **1-2** may be a dichroic or a non-dichroic beamsplitter that combines the Raman signal **160** and the probe beam **120_{pr}** to produce a combined probe-Raman signal **210** that is directed to a detector **220**. An optical receiver **200** may include one or more optical detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with at least a portion of a probe beam **120_{pr}** to produce an electronic signal. Each of the optical receivers **200** in FIGS. **1-2** includes one detector **220** that receives the combined probe-Raman signal **210**. The Raman signal **160** and the probe beam **120_{pr}** are coherently mixed at the detector **220**, and this heterodyne mixing process produces an electronic signal, which in FIGS. **1-2** is indicated as analog photocurrent signal **i**. In FIGS. **1-2**, the detection electronics **230** receives the photocurrent signal **i** and produces a digital output signal **240** that corresponds to the photocurrent signal. The digital output signal **240** may be sent to a processor, and the processor may determine a characteristic of the analog photocurrent signal **i** based on the digital output signal **240**. For example, the characteristic of the analog photocurrent signal (or the characteristic of an analog voltage signal that corresponds to the photocurrent signal) determined by the processor may include one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization. An analog photocurrent signal may be referred to as a photocurrent, a photocurrent signal, or a current signal, and an analog voltage signal may be referred to as a voltage signal. A processor of a Raman spectroscopy system **100** may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. A processor may be similar to the computer system **5900** illustrated in FIG. **59** and described herein. In some embodiments, a processor or a portion of a processor may be located in the detection electronics **230**. For example, a processor located in the detection electronics **230** may receive a digital output signal **240** and perform some preprocessing of the digital signal or determine a characteristic of a photocurrent signal based on the digital signal.

[0050] In FIGS. **1-2**, the probe beam **120_{pr}** does not travel through the sample **150**, and the probe beam is combined with the Raman signal **160** after the Raman signal has exited the sample **150**. In other embodiments, a Raman spectroscopy system **100** may include a probe laser **110_{pr}** that produces a probe beam **120_{pr}** that is directed through the sample **150**. For example, the probe beam **120_{pr}** may be combined with the pump beam **120_{pu}** and the Stokes beam **120_S**, and all three beams may be directed to the sample **150**. The probe beam **120_{pr}** may travel through the sample **150** and may exit the sample along with the Raman signal **160**. The optical receiver **200** may not include a combiner **130_b**, since the probe beam **120_{pr}** and the Raman signal **160** may already be combined after they exit the sample **150**. The probe beam **120_{pr}** and the Raman signal **160** may be directed to a detector **220** without being transmitted or reflected by an optical combiner **130_b**.

[0051] The detection electronics **230** in FIG. **2** includes an electronic amplifier **232** and a digitizer **236**. The electronic amplifier **232** may include a transimpedance amplifier that amplifies the photocurrent signal **i** to produce an analog voltage signal **234** that corresponds to the photocurrent signal **i** (e.g., the photocurrent signal **i** and the analog voltage signal **234** may have similar temporal shapes or may include similar electronic frequency components). The electronic amplifier **232** may include an additional gain stage that further amplifies an intermediate voltage signal produced by the transimpedance amplifier to produce the voltage signal **234**. Additionally, the electronic amplifier **232** may include an electronic filter (e.g., a low-pass, high-pass, or band-pass filter) that filters the photocurrent signal or voltage signal. For example, the electronic amplifier **232** may include (i) a high-pass filter that removes a DC offset and low-frequency components (e.g., frequency components below 10 MHz) from the photocurrent signal or (ii) a band-pass filter that removes the DC and low-frequency components as well as high-frequency components (e.g., frequency components above 5 GHz). Herein, an electronic signal produced in response to coherent mixing may refer to a current signal (e.g., a photocurrent **i** produced by a detector **220**) or may refer to a corresponding voltage signal (e.g., a voltage signal **234** produced by an electronic amplifier that amplifies a photocurrent produced by a detector to produce the voltage signal). In FIG. **2**, the digitizer **236** receives the voltage signal **234** and produces a digital output signal **240** that includes a digital representation of the voltage signal **234**. The digital output signal **240** may be a time-domain digital representation of the voltage signal **234**. The digital output signal **240** may be referred to as corresponding to or representing the voltage signal **234** or the photocurrent signal **i**. The digitizer **236** may include an analog-to-digital converter (ADC) that produces a digital version of the voltage signal **234**. Additionally or alternatively, the digitizer **236** may include a peak detector that determines a peak value of the voltage signal **234**.

[0052] A Raman spectroscopy system **100** may include one or more optical detectors **220**. An optical detector **220** (which may be referred to as a detector, photodetector, or photodiode) may include a PN photodiode, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), silicon photomultiplier (SiPM), or photomultiplier tube (PMT). A PN photodiode refers to a photodiode structure formed by a p-doped semiconductor and an n-doped semiconductor, where the PN acronym refers to the structure having p-doped and n-doped regions. A PIN photodiode refers to a photodiode structure formed by an undoped intrinsic semiconductor region located between p-doped and n-doped regions, where the PIN acronym refers to the structure having p-doped, intrinsic, and n-doped regions.

[0053] A PN photodiode, PIN photodiode, APD, or SPAD may include any suitable semiconductor material, such as for example: silicon, germanium, gallium arsenide (GaAs), indium phosphide (InP), indium arsenide (InAs), aluminum arsenide (AlAs), indium antimonide (InSb), aluminum antimonide (AlSb), gallium antimonide (GaSb), aluminum gallium arsenide (AlGaAs), indium gallium arsenide (InGaAs), indium aluminum arsenide (InAlAs), indium arsenide antimonide (InAsSb), aluminum arsenide antimonide (AlAsSb), aluminum gallium antimonide (AlGaSb), gallium arsenide antimonide (GaAsSb), aluminum indium arsenide anti-

monide (AlInAsSb), indium gallium arsenide antimonide (InGaAsSb), indium gallium aluminum arsenide (InGaAlAs), aluminum gallium arsenide antimonide (AlGaAsSb), or silicon germanium (SiGe). For example, the Raman signal **160** and the probe beam **120_{pr}** in FIGS. **1-2** may each have a wavelength in the 400-1100 nanometer (nm) range, and the detector **220** may include a silicon PIN photodiode. As another example, the Raman and probe beams in FIGS. **1-2** may each have a wavelength in the 1000-1700 nm range, and the detector **220** may include an InGaAs PIN photodiode. As another example, a detector **220** may include an APD that includes a semiconductor material with antimonide (e.g., InSb, AlSb, GaSb, InAsSb, AlAsSb, AlGaSb, GaAsSb, AlInAsSb, InGaAsSb, or AlGaAsSb).

[0054] In order for a detector **220** to detect an optical signal, the wavelength of the optical signal must be within the detector's wavelength range of responsivity (e.g., approximately 400-1100 nm for a silicon detector, and approximately 1000-1700 nm for an InGaAs detector) and a frequency of an amplitude modulation of the optical signal must be within the electronic bandwidth of the detector. The electronic bandwidth (Δf) of a detector **220** refers to the range of electronic modulation frequencies over which a detector may detect an optical signal, where detection of the optical signal refers to (i) the detector producing a photocurrent signal **i** that corresponds to the optical signal and (ii) an electronic amplifier **232** producing a voltage signal **234** that corresponds to the photocurrent signal. If a silicon detector **220** has an electronic bandwidth that extends from 100 MHz to 5 GHz, then the detector may detect optical signals with (i) wavelengths between approximately 400 nm and 1100 nm and (ii) amplitude modulation between 100 MHz and 5 GHz. For example, a 900-nm optical signal with an amplitude modulation at a frequency between 100 MHz and 5 GHz may be detected by the silicon detector **220**. The silicon detector **220** may not detect a continuous-wave or substantially constant portion of a 900-nm optical signal (e.g., the substantially constant portion of the optical signal may produce a DC current in the detector that may be electronically filtered out), and the silicon detector **220** may not detect a portion of the optical signal with an amplitude modulation greater than approximately 5 GHz. Herein, reference to the electronic bandwidth (Δf) of a detector **220** may refer to (i) the electronic bandwidth of just the detector or (ii) the overall bandwidth of the detector in combination with an electronic amplifier **232**.

[0055] A detector **220** may have an electronic bandwidth Δf between approximately 100 MHz and approximately 50 GHz. For example, the detector **220** in each of FIGS. **1-2** may have an electronic bandwidth between 100 MHz and 10 GHz. As another example, the detector **220** in each of FIGS. **1-2** may have an electronic bandwidth that extends from a low-frequency cutoff to a high-frequency cutoff. The low-frequency cutoff may be approximately DC (i.e., zero hertz), 1 MHz, 10 MHz, 50 MHz, or 100 MHz, and the high-frequency cutoff may be approximately 500 MHz, 1 GHz, 2 GHz, 5 GHz, 10 GHz, 20 GHz, or 50 GHz. The electronic bandwidth of a detector **220** may refer to the bandwidth of only the detector **220**. For example, a detector **220** may have an electronic bandwidth that extends from DC to 10 GHz, and the detector may be referred to as having a 10-GHz bandwidth that extends from DC to 10 GHz. Alternatively, the electronic bandwidth of a detector **220** may refer to the overall bandwidth of the detector in combination with an

electronic amplifier **232** that amplifies the photocurrent signal **i** produced by the detector. An electronic amplifier **232** may have a low-frequency cutoff (e.g., DC, 1 MHz, 10 MHz, 50 MHz, or 100 MHz) and a high-frequency cutoff (e.g., 500 MHz, 1 GHz, 2 GHz, 5 GHz, 10 GHz, 20 GHz, or 50 GHz), and a detector-amplifier combination may be referred to as having an electronic bandwidth that extends from the low-frequency cutoff to the high-frequency cutoff. For example, if the detector bandwidth extends from DC to 10 GHz, and the electronic amplifier bandwidth extends from DC to 5 GHz, then the detector (or, the detector-amplifier combination) may be referred to as having an electronic bandwidth of 5 GHz that extends from DC to 5 GHz. As another example, if the detector bandwidth extends from DC to 10 GHz, and the electronic amplifier bandwidth extends from 100 MHz to 5 GHz, then the detector may be referred to as having an electronic bandwidth of 4.9 GHz that extends from 100 MHz to 5 GHz.

[0056] A Raman spectroscopy system **100** may include one or more optical waveplates that change or rotate the polarization of a beam of light. For example, a half-wave plate may be used to rotate a linearly polarized beam of light to a different polarization orientation (e.g., from vertically polarized to horizontally polarized), and a quarter-wave plate may be used to convert a linearly polarized beam of light to a circular or elliptical polarization. The pump laser **110_{pu}** in FIG. **2** may produce linearly polarized light, and the waveplate **132_a** may be a half-wave plate that rotates the polarization of the pump beam **120_{pu}** prior to the pump beam being directed to the sample **150**. Alternatively, the waveplate **132_a** may be a quarter-wave plate that converts the linearly polarized pump beam **120_{pu}** to a circular or elliptical polarization prior to the pump beam being directed to the sample **150**. Similarly, the Stokes laser **110_S** in FIG. **2** may produce linearly polarized light, and the waveplate **132_b** may be (i) a half-wave plate that rotates the polarization of the Stokes beam **120_S** or (ii) a quarter-wave plate that converts the Stokes beam **120_S** to a circular or elliptical polarization.

[0057] The probe laser **110_{pr}** in FIG. **2** may produce linearly polarized light, and the waveplate **132_c** may be (i) a half-wave plate that rotates the polarization of the probe beam **120_{pr}** or (ii) a quarter-wave plate that converts the probe beam **120_{pr}** to a circular or elliptical polarization prior to the probe beam being combined with the Raman signal **160**. Changing the polarization of the probe beam **120_{pr}** may allow the Raman signal **160** and the probe beam to be coherently mixed. The polarization of the probe beam **120_{pr}** can be changed so that it has both horizontal and vertical polarization components, which ensures that at least a portion of the probe beam **120_{pr}** and the Raman signal **160** have polarizations that are oriented in the same direction so that their electric fields may be added together.

[0058] Each of the optical waveplates **132** in FIG. **2** may be a free-space optical element, a fiber-optic component, a waveguide-based optical element, or a metamaterial-based optic. Additionally, each of the optical waveplates **132** in FIG. **2** may be a fixed waveplate or an adjustable waveplate. A fixed waveplate may have a fixed optical phase difference between the two axes of the waveplate (e.g., a quarter-wave plate may have a one-quarter wavelength phase difference, and a half-wave plate may have a one-half wavelength phase difference). An adjustable waveplate may allow for the phase difference between the two axes of the waveplate to be

dynamically changed. For example, an electronically adjustable waveplate may include a Pockels cell, a liquid crystal device, or a photoelastic modulator that allows the phase difference to be adjusted electronically so that the waveplate can be dynamically configured to act as a waveplate having any suitable phase difference (e.g., a phase difference between zero wavelengths and one-half wavelength). An adjustable waveplate may switch between (i) applying no phase difference to incident light so that the transmitted light has the same polarization as the incident light and (ii) applying a one-quarter wavelength or one-half wavelength phase difference so that linearly polarized incident light is converted to circularly polarized light or is rotated to a different polarization. For example, the pump laser **110_{pu}** in FIG. 2 may produce vertically polarized light, and the waveplate **132a** may be an adjustable waveplate that switches between (i) applying no polarization rotation to the pump beam **120_{pu}** so that the pump beam remains vertically polarized and (ii) applying a 90-degree rotation to the pump-beam polarization so that the pump beam **120_{pu}** after the waveplate **132a** is horizontally polarized. Dynamically changing the polarization of the pump beam **120_{pu}** may allow the Raman spectroscopy system **100** in FIG. 2 to perform measurements at two different pump-beam polarizations, which may produce additional data for characterization of the sample **150**.

[0059] In some embodiments, an optical waveplate **132** may be a metamaterial-based waveplate. A metamaterial refers to an engineered material having features or repeating patterns at scales smaller than the wavelength of light interacting with the metamaterial. A metamaterial may be configured to act as a mirror, lens, waveplate, diffractive optical element, optical combiner, or optical waveguide. A metamaterial-based waveplate may affect the polarization of a beam of light based on wavelength. For example, the Raman spectroscopy system **100** in FIG. 2 may include a metamaterial-based waveplate (not illustrated in FIG. 2) located after the combiner **130a** and before the sample **150**. The metamaterial waveplate may change the polarization of the pump beam **120_{pu}** from linear to circular while preserving the polarization of the Stokes beam **120S** (e.g., the Stokes beam may remain linearly polarized and may not be significantly changed by the waveplate).

[0060] A Raman spectroscopy system **100** may include an optical filter that transmits light at one or more wavelengths and blocks light at one or more other wavelengths. The optical filter **134** in FIG. 2 is located between the sample **150** and the optical receiver **200** and may be configured to substantially transmit one or more optical wavelengths associated with the Raman signal **160** and substantially block one or more wavelengths associated with the pump beam **120_{pu}** or the Stokes beam **120S**. For example, the optical filter **134** may transmit greater than 90% of the Raman signal **160** and may block greater than 90% of both the pump and Stokes beams. As another example, the optical filter **134** may transmit greater than 90% of the Raman signal **160** and the Stokes beam **120S**, and the optical filter **134** may block greater than 98% of the pump beam **120_{pu}**.

[0061] A Raman spectroscopy system **100** may include an optical polarizer that transmits light having a particular polarization (e.g., horizontal) and blocks light having an orthogonal polarization (e.g., vertical). The optical polarizer **136** in FIG. 2 is located between the sample **150** and the optical receiver **200** and may be oriented to transmit light

with a polarization associated with the Raman signal. Additionally, the polarizer **136** may block a polarization associated with the pump or Stokes beams. For example, the Stokes beam **120S** incident on the sample **150** may be vertically polarized, and the Raman signal **160** may be at least partially horizontally polarized. The polarizer **136** may be oriented to block vertically polarized light and transmit horizontally polarized light so that the Stokes beam **120S** is blocked and the Raman signal **160** is at least partially transmitted by the polarizer. A Raman spectroscopy system **100** may include both an optical filter **134** and an optical polarizer **136**. For example, the optical filter **136** in FIG. 2 may be configured to transmit the Raman signal **160** and the Stokes beam **120S** and block the pump beam **120_{pu}**. Additionally, the Raman signal **160** and the Stokes beam **120S** may be orthogonally polarized, and the polarizer **136** may be oriented to transmit the Raman signal **160** and block the Stokes beam **120S**. Using a filter or polarizer located after the sample **150** to block light from the pump beam **120_{pu}** or the Stokes beam **120S** may reduce noise in the system by reducing the amount of unwanted background light that reaches the detector **220**.

[0062] The sample **150** in FIGS. 1-2 may be a solid, liquid, or gas, or any combination thereof. The sample **150** may include a biological material, an organic material, an inorganic material, a crystalline material, an amorphous solid material, or any other suitable material or combination of suitable materials. For example, the sample **150** may include a drug, mineral, food, contaminant, or explosive material that produces a Raman signal **160** in response to excitation by the pump and Stokes beams. As another example, the sample **150** may be a biological material (e.g., blood, urine, saliva, sweat, or cerebrospinal fluid) and a component or molecule (e.g., glucose or cortisol) that is part of the biological material may produce a Raman signal **160**. As another example, the sample **150** may be water or wastewater that may include a contaminant, virus, bacteria, or an indicator of an infectious disease. The Raman signal **160** may be produced by coherent Raman scattering of the pump and Stokes beams within the sample **150**, and the frequency offset Ω between the pump and Stokes beams may be approximately equal to a vibrational frequency or an electronic-transition frequency of a particular material that is part of the sample. The vibrational frequency of the particular material may correspond to a molecular vibration of a molecule, or in the case of a crystalline material, may correspond to a lattice vibration of a crystal. For example, the sample **150** may include glucose, and the frequency offset Ω may be approximately equal to a frequency of a molecular vibration of glucose.

[0063] The Raman spectroscopy systems **100** in FIGS. 1-2 may perform one or more measurements of a sample **150**, and each measurement may include determining a characteristic of an electronic signal that results from the coherent mixing of the Raman signal **160** and the probe beam **120_{pr}**. The electronic signal may include a photocurrent i or a corresponding voltage signal **234**. The frequency offset Ω may be approximately equal to a vibrational frequency or electronic-transition frequency of a particular material, and based on the one or more measurements, a processor may determine whether the particular material is present in the sample **150**. Additionally or alternatively, the processor may determine an amount or a concentration of the particular material in the sample based on the measurements. For

example, the frequency offset Ω may be approximately equal to a vibrational frequency of glucose, and, based on one or more optical heterodyne measurements of a Raman signal **160** produced by a sample **150**, the processor may determine (i) whether glucose is present in the sample or (ii) an amount or a concentration of glucose in the sample. The amount of glucose that is present in the sample **150** may be proportional to the amplitude of a photocurrent signal **i** produced by coherent mixing of the Raman signal **160** and the probe beam **120pr**. Based on the amplitude of the photocurrent signal **i**, the processor may determine the amount or concentration of glucose in the sample **150**.

[0064] A technical advantage of a coherent Raman spectroscopy system **100** as described herein is a higher spectral resolution or a better chemical sensitivity than a conventional Raman spectroscopy system. As such, a Raman spectroscopy system **100** as described herein may be referred to as a high-resolution Raman spectroscopy system or as a high-resolution coherent Raman spectroscopy system. In a conventional Raman spectroscopy system, a Raman signal produced by a sample may be measured in the optical domain using an optical spectrometer. A spectrometer typically uses a dispersive optical element (e.g., a diffraction grating) to separate the Raman signal into its various spectral components. However, this type of measurement performed in the optical domain typically has a spectral resolution on the order of 1 cm^{-1} (or, about 30 GHz). In contrast, the spectral resolution of a coherent Raman spectroscopy system with heterodyne detection, as described herein, is determined primarily by the spectral linewidth $\Delta\nu_{pr}$ of the probe beam **120pr** that is coherently mixed with the Raman signal **160**. The probe laser **110pr** may include a wavelength-tunable laser diode with a linewidth of 200 MHz or less, which corresponds to a spectral resolution of the Raman spectroscopy system **100** of less than 200 MHz (or, less than 0.007 cm^{-1}). This 200-MHz spectral resolution is more than 100 times better than the 30-GHz spectral resolution of a conventional Raman spectroscopy system. In some embodiments, the probe laser **110pr** may have a linewidth of 1 MHz or less, which corresponds to a spectral resolution of the Raman spectroscopy system **100** of less than 1 MHz (or, less than $33 \times 10^{-6} \text{ cm}^{-1}$). A related advantage of a coherent Raman spectroscopy system **100** is that the signal capture and analysis are performed in the electronic domain (e.g., at electronic frequencies between DC and 50 GHz) rather than in the optical domain (e.g., at optical frequencies between 60 THz and 1,000 THz). The coherent mixing of two optical signals (Raman signal **160** and probe beam **120pr**) produces an electronic signal which can be analyzed with relatively high resolution compared to an optical signal. This electronic signal analysis, along with the relatively narrow spectral linewidth of the probe laser **110pr**, provides a coherent Raman spectroscopy system **100** with a high spectral resolution. Additionally, the wavelength tunability of the probe laser **110pr** allows a Raman spectrum of a material to be determined at multiple frequencies with high spectral resolution.

[0065] The higher spectral resolution of a coherent Raman spectroscopy system **100** may provide a corresponding improvement in the ability of the coherent Raman spectroscopy system to sense various chemical species. For example, a high-resolution coherent Raman spectroscopy system **100** may be able to distinguish between different chemical species that have Raman peaks located relatively close

together, whereas a conventional Raman spectroscopy system may not be able to resolve spectral features below about 1 cm^{-1} . Additionally, the higher spectral resolution of a coherent Raman spectroscopy system **100** may allow for lower concentrations of materials to be detected, as compared to a conventional Raman spectroscopy system. For example, a coherent Raman spectroscopy system **100** may be able to detect small deviations in the chemical signature of a biological sample, which may indicate the presence of damage or a mutation, which in turn may be correlated with a disease or pathogen.

[0066] Another technical advantage of a coherent Raman spectroscopy system **100** as described herein is its relatively compact size. A coherent Raman spectroscopy system may be packaged in a relatively compact enclosure as compared to a conventional Raman spectroscopy system. Since the spectral resolution of an optical spectrometer scales inversely with the optical path length of the spectrometer (e.g., a longer path length provides better spectral resolution), an optical spectrometer with a spectral resolution around 1 cm^{-1} can be quite large or bulky. In contrast, since the spectral resolution of a coherent Raman spectroscopy system is determined primarily by the spectral linewidth of the probe laser **110pr**, a coherent Raman spectroscopy system does not require a long optical path length to provide high spectral resolution. Thus, an enclosure for a coherent Raman spectroscopy system may be significantly smaller than that for a conventional Raman spectroscopy system. In some embodiments, a coherent Raman spectroscopy system may be packaged as a compact device that may be referred to as a lab-on-a-chip or a spectrometer on a chip. For example, a coherent Raman spectroscopy system may be packaged as a wearable device that provides ongoing, continual monitoring for a person or an animal.

[0067] FIGS. 3-4 each illustrates an example Raman signal **160** produced by coherent Raman scattering. In a coherent Raman spectroscopy system **100**, the pump beam **120pu** and the Stokes beam **120S** are directed to a sample **150**, which produces a Raman signal **160** by coherent Raman scattering of the pump and Stokes beams within the sample. The frequency of the pump beam **120pu** is ν_1 , and the frequency of the Stokes beam **120S** is ν_2 . The frequency offset Ω between the pump and Stokes beams equals $\nu_1 - \nu_2$. The frequencies of the pump and Stokes beams may be set so that the frequency offset Ω is approximately equal to a vibrational frequency or an electronic-transition frequency of a particular material. A coherent Raman spectroscopy system **100** may measure a Raman signal **160** produced by a sample **150** to determine (i) whether the particular material is present in the sample or (ii) an amount or a concentration of the particular material within the sample.

[0068] The frequency offset Ω between the pump and Stokes beams may be any suitable fixed or adjustable value between approximately 5 terahertz (THz) and approximately 100 THz. Expressed in wavenumbers, this corresponds to the frequency offset Ω being between approximately 167 cm^{-1} and approximately 3336 cm^{-1} . For example, in a coherent Raman spectroscopy system **100**, the pump beam **120pu** may have a wavelength between approximately 1220 nanometers (nm) and approximately 1450 nm (which corresponds to a pump-beam frequency ν_1 between approximately 246 THz and approximately 207 THz), and the Stokes beam **120S** may have a wavelength between approximately 1490 nm and approximately 1570 nm (which corre-

sponds to a Stokes-beam frequency ν_2 between approximately 201 THz and approximately 191 THz). This system may produce pump and Stokes beams having a frequency offset Ω between approximately 5.6 THz and approximately 54.8 THz (or, in wavenumbers, between approximately 185 cm^{-1} and 1827 cm^{-1}). For example, if the pump beam **120pu** has a wavelength of 1330 nm (or equivalently, a frequency ν_1 of 225.4 THz) and the Stokes beam **120S** has a wavelength of 1550 nm (or equivalently, a frequency ν_2 of 193.4 THz), then the frequency offset Ω between the pump and Stokes beams is approximately 32 THz (or, in wavenumbers, 1067 cm^{-1}). The pump and Stokes beams in a Raman spectroscopy system **100** may each have any suitable wavelength between approximately 300 nm and approximately 5,000 nm. For example, if the pump beam **120pu** has a wavelength of approximately 785 nm (or equivalently, a frequency ν_1 of 381.9 THz) and the Stokes beam **120S** has a wavelength of approximately 840 nm (or equivalently, a frequency ν_2 of 356.9 THz), then the frequency offset Ω between the pump and Stokes beams is approximately 25 THz (or, in wavenumbers, 834 cm^{-1}). As another example, the pump beam **120pu** may have a wavelength between approximately 700 nm and approximately 850 nm, between approximately 890 nm and approximately 920 nm, or between approximately 1000 nm and approximately 1100 nm.

[0069] The Raman signal **160** in each of FIGS. 3-4 is an optical signal with a spectral linewidth of $\Delta\nu_R$. In FIG. 3, the Raman signal **160** has a center frequency approximately equal to $2\nu_1 - \nu_2$ (which is equal to $\nu_1 + \Omega$, since $\Omega = \nu_1 - \nu_2$). The center frequency of a Raman signal **160** may refer to the frequency of a central peak or the frequency of an approximate center of the Raman signal. Additionally or alternatively, the center frequency of a Raman signal **160** may correspond to a Raman peak of an associated Raman spectrum. The Raman signal **160** in FIG. 3 may be produced by coherent anti-Stokes Raman scattering (CARS) in which the pump and Stokes beams interact with a sample **150** to produce a Raman signal **160** at or around the frequency $2\nu_1 - \nu_2$. For example, if the pump and Stokes beams have respective wavelengths of 1064 nm and 1550 nm (which corresponds to frequencies of approximately 281.8 THz and 193.4 THz), then the frequency offset Ω is 88.3 THz (or, 2947 cm^{-1}), and the anti-Stokes Raman signal **160** has a center wavelength of approximately 810 nm (which corresponds to a frequency of approximately 370 THz). The Raman signal **160** in FIG. 4, which overlaps the frequency of the Stokes beam **120S**, may be produced by stimulated Raman scattering. The Raman signal **160** in FIG. 4 may be centered at or near the frequency ν_2 of the Stokes beam **120S**. For example, the Raman signal **160** in FIG. 4 may have a center frequency that is within approximately 200 GHz (or, 6.7 cm^{-1}) of the Stokes-beam frequency ν_2 .

[0070] FIG. 5 illustrates the example Raman signal **160** of FIG. 4 along with a probe beam **120pr**. The Raman signal **160** may be produced by stimulated Raman scattering of the pump beam **120pu** and Stokes beam **120S**, and the probe beam **120pr** may be used to measure the Raman signal. The Raman signal **160** is centered at or near the frequency ν_2 of the Stokes beam **120S**, and the frequency ν_3 of the probe beam of light **120pr** overlaps the Raman signal **160** and is relatively close to the frequency ν_2 of the Stokes beam of light **120S** (e.g., the probe-beam frequency ν_3 may be within 200 GHz of the Stokes-beam frequency ν_2). In a coherent

Raman spectroscopy system **100**, the pump frequency ν_1 , Stokes frequency ν_2 , and probe frequency ν_3 may each be between approximately 60 THz and approximately 1,000 THz (which corresponds to a wavelength between approximately 5,000 nm and approximately 300 nm). For example, in FIG. 5 the pump frequency ν_1 may be 291 THz (which corresponds to a wavelength of approximately 1030 nm), the Stokes frequency ν_2 may be 240.03 THz (which corresponds to a wavelength of approximately 1249.00 nm), and the probe frequency ν_3 may be 240.00 THz (which corresponds to a wavelength of approximately 1249.14 nm).

[0071] Each of the light sources **110** of a coherent Raman spectroscopy system **100** may include a wavelength-tunable light source. A wavelength-tunable light source refers to a light source **110** that can produce light at multiple different wavelengths within a range of wavelengths (or equivalently, at multiple different frequencies within a range of frequencies). For example, the probe light source **110pr** in FIG. 1 may be a wavelength-tunable light source where the wavelength of the probe beam **120pr** is adjustable over an 80-nm wavelength range from 1490 nm to 1570 nm (which corresponds to a 10.3 THz frequency range from approximately 201.2 THz to approximately 191.0 THz). At any given time, a wavelength-tunable light source **110** may operate at any one of the different wavelengths within its wavelength-tuning range. For example, during a first period of time, the probe light source **110pr** with a 1490-1570 nm wavelength-tuning range may produce a probe beam **120pr** at 1500 nm, and during a subsequent second period of time, the probe light source may be tuned to operate at 1560 nm. A wavelength-tunable light source may be adjustable over a frequency range that corresponds to a wavelength range having a width or span between approximately 10 nm and approximately 100 nm. For example, the width of the wavelength-tuning range of a wavelength-tunable light source may be approximately 10 nm, 20 nm, 30 nm, 40 nm, 50 nm, 60 nm, 80 nm, or 100 nm. A wavelength-tuning range from 1490 nm to 1570 nm may be referred to as having an 80-nm width or an 80-nm span. Around 1550 nm, a 10-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 1.25 THz (or equivalently, 41.6 cm^{-1}), and a 100-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 12.5 THz (or equivalently, 416 cm^{-1}). Around 1050 nm, a 10-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 2.72 THz (or equivalently, 90.7 cm^{-1}), and a 100-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 27.3 THz (or equivalently, 909 cm^{-1}). A wavelength-tunable light source may be referred to as a frequency-tunable light source or a tunable light source.

[0072] Each of the light sources **110** of a coherent Raman spectroscopy system **100** may include one or more laser diodes, and each of the laser diodes may be a fixed-wavelength laser diode or a wavelength-tunable laser diode. A fixed-wavelength laser diode may operate at a single wavelength or within a relatively narrow wavelength range (e.g., within 0.1 nm of a particular wavelength). A fixed-wavelength laser diode may include a distributed feedback (DFB) laser diode, a distributed Bragg reflector (DBR) laser diode, a fiber-Bragg-grating (FBG) stabilized laser diode, a temperature-stabilized laser diode, or any other suitable fixed-wavelength laser diode. A wavelength-tunable laser diode may produce light at multiple different wavelengths

within a range of wavelengths. For example, a wavelength-tunable laser diode may be configured to produce light at any wavelength within a wavelength range having a width between approximately 10 nm and approximately 100 nm. At any given time, a wavelength-tunable laser diode may operate at any one wavelength of multiple different wavelengths within a range of wavelengths. A wavelength-tunable laser diode may include an external-cavity laser diode, a thermally tuned laser diode, or a sampled-grating distributed Bragg reflector (SG-DBR) laser. For example, a wavelength-tunable SG-DBR laser may have a 40-nm wavelength-tuning range that extends from 1530 nm to 1570 nm, and the SG-DBR laser may be adjustable to operate at any single wavelength within the 40-nm wavelength range. A wavelength-tunable laser diode may be referred to as a frequency-tunable laser diode or a tunable laser diode. For example, a tunable laser with a 40-nm wavelength-tuning range that extends from 1530 nm to 1570 nm may also be referred to as a frequency-tunable laser with a 5.0-THz frequency-tuning range that extends from approximately 191 THz to approximately 196 THz.

[0073] Each of the light sources **110** of a coherent Raman spectroscopy system **100** may include one or more of the following: light-emitting diode (LED), super-luminescent light source, short-pulse laser, broadband light source, fiber laser, solid-state laser, quantum-cascade laser. For example, a light source that produces light over a relatively broad range of wavelengths (e.g., a super-luminescent light source, short-pulse laser, or broadband light source) may be used to investigate a sample over the broad range of wavelengths without having to use a wavelength-tunable light source.

[0074] FIG. 6 illustrates an expanded view of a portion of the Raman signal **160** of FIG. 5. The portion of FIG. 5 enclosed by the dashed-line box is expanded in FIG. 6. The peak of the Raman signal **160** in FIG. 6 is approximately coincident with the frequency ν_2 of the Stokes beam **120S**, and the frequency ν_3 of the probe beam **120pr** overlaps the Raman signal.

[0075] The Raman signal **160** in FIG. 6 is an optical signal with a spectral linewidth of $\Delta\nu_R$. The spectral linewidth of a Raman signal **160** may have a value between approximately 30 GHz and 300 GHz (or, in wavenumbers, between approximately 1 cm^{-1} and approximately 10 cm^{-1}). The spectral linewidth $\Delta\nu_R$ of a Raman signal **160** may represent the spectral width of a peak of the Raman signal (e.g., a full-width-at-half-maximum of the peak) or may represent an approximate extent or width of the full Raman signal. For example, the spectral linewidth $\Delta\nu_R$ of the Raman signal **160** in FIG. 13 corresponds to a full-width-at-half-maximum of a peak of the Raman signal. In FIG. 6, the spectral linewidth $\Delta\nu_R$ of the Raman signal **160** corresponds to an extent or width of the full Raman signal. For example, the spectral linewidth of a Raman signal **160** may equal a spectral width at which an envelope of the Raman signal has decreased to a particular level (e.g., to 50%, 20%, or 10% of a peak value). The envelope may be a curve that decreases monotonically away from a peak of the Raman signal and approximately follows an overall shape of the Raman signal. In FIG. 6, the dashed-line curve that traces the peaks of the Raman signal **160** represents an envelope of the Raman signal, and the spectral linewidth $\Delta\nu_R$ corresponds to a full-width-at-10% of the Raman-signal envelope (i.e., the points at which the envelope has decreased to 10% of its maximum value).

[0076] In a coherent Raman spectroscopy system **100**, the difference between the frequency ν_2 of the Stokes beam **120S** and the frequency ν_3 of the probe beam **120pr** may be greater than a low-frequency limit F_1 and less than a high-frequency limit F_2 (i.e., $F_1 < |\nu_2 - \nu_3| < F_2$). For example, the low-frequency limit F_1 may be approximately 1 MHz, 10 MHz, 50 MHz, 100 MHz, 200 MHz, 500 MHz, or 1 GHz, and the high-frequency limit F_2 may be approximately 10 GHz, 20 GHz, 50 GHz, 100 GHz, 200 GHz, 500 GHz, or 1 THz. As another example, the low-frequency limit F_1 may be related to the spectral linewidth $\Delta\nu_{pr}$ of the probe beam **120pr** and the spectral linewidth $\Delta\nu_S$ of the Stokes beam **120S** (e.g., F_1 may be greater than $\Delta\nu_{pr} + \Delta\nu_S$). As another example, the high-frequency limit F_2 may be related to the spectral linewidth $\Delta\nu_R$ of a Raman signal **160** (e.g., F_2 may be approximately equal $(0.5)\Delta\nu_R$, $\Delta\nu_R$, or $2\Delta\nu_R$). As another example, F_1 may be 100 MHz and F_2 may be 200 GHz, which indicates that the frequency ν_2 of the Stokes beam **120S** and the frequency ν_3 of the probe beam **120pr** may differ by greater than 100 MHz and less than 200 GHz (i.e., $100\text{ MHz} < |\nu_2 - \nu_3| < 200\text{ GHz}$). The frequencies F_1 and F_2 represent the frequency range with respect to the Stokes-beam frequency ν_2 over which the probe-beam frequency ν_3 may be scanned. The two hatched rectangles along the frequency axis in FIG. 6 represent the allowed frequency ranges for the probe beam **120pr**. The frequency of the probe beam **120pr** may be (i) between $\nu_2 - F_2$ and $\nu_2 - F_1$ or (ii) between $\nu_2 + F_1$ and $\nu_2 + F_2$. In an optical receiver **200**, the probe beam **120pr** and the Raman signal **160** may be coherently mixed together at an optical detector **220**. The frequency ν_3 of the probe beam **120pr** may be kept away from the frequency ν_2 of the Stokes beam **120S** by at least the low-frequency limit F_1 (e.g., $|\nu_2 - \nu_3| > F_1$) to avoid mixing between the probe and Stokes beams, which could cause the detector to produce an unwanted electronic signal not related to the mixing between the probe beam and Raman signal. Additionally, the frequency ν_3 of the probe beam **120pr** may be kept within the high-frequency limit F_2 of the frequency ν_2 of the Stokes beam **120S** (e.g., $|\nu_2 - \nu_3| < F_2$), since measurements outside the high-frequency limit may not produce a significant electronic signal.

[0077] FIG. 7 illustrates an expanded view of a portion of the Raman signal **160** of FIG. 6. The portion of FIG. 6 enclosed by the dashed-line box is expanded in FIG. 7. The peak of the Raman signal **160** in FIG. 7 is located at frequency ν_{RP} , and the Stokes beam **120S** is located at the Stokes frequency ν_2 . The Raman-signal peak frequency ν_{RP} may correspond to a Raman peak of a material that is part of a sample **150** being measured by a Raman spectroscopy system **100**. For example, the difference between the pump-beam frequency ν_1 and the Raman-signal peak frequency ν_{RP} may equal a vibrational frequency Q of the material (e.g., $\nu_1 - \nu_{RP} = Q$). In some embodiments, a Stokes light source **110S** may be operated so that the Stokes-beam frequency ν_2 is approximately equal to the Raman-signal peak frequency ν_{RP} , and in other embodiments (e.g., as illustrated in FIG. 7), the Stokes-beam frequency ν_2 may be slightly off-resonance or detuned with respect to the Raman-signal peak frequency ν_{RP} . For example, in FIG. 7, the Stokes-beam frequency ν_2 may differ from the Raman-signal peak frequency ν_{RP} by less than or equal to 30 GHz, 10 GHz, 5 GHz, or 1 GHz.

[0078] The beams of light **120** produced by the pump, Stokes, and probe light sources **110** in a Raman spectroscopy

system 100 may each have a spectral linewidth of less than 200 MHz. Additionally, one or more of the beams of light 120 may have a spectral linewidth of less than 1 MHz. For example, the spectral linewidth of a beam of light 120 may be less than 200 MHz, 100 MHz, 50 MHz, 10 MHz, 1 MHz, or 100 kHz. In FIG. 7, the spectral linewidth $\Delta\nu_{pr}$ of the probe beam 120_{pr} and the spectral linewidth $\Delta\nu_S$ of the Stokes beam 120_S may each be less than 200 MHz. In some embodiments, the probe light source 110_{pr} may be configured to produce a probe beam 120_{pr} having a relatively narrow spectral linewidth, which may allow a Raman spectroscopy system 100 to measure a Raman spectrum with a high degree of spectral resolution. For example, the spectral linewidth $\Delta\nu_{pr}$ of the probe beam 120_{pr} in FIG. 7 may be less than 1 MHz.

[0079] An optical receiver 200 of a coherent Raman spectroscopy system 100 may include one or more detectors 220, where each detector is configured to coherently mix a portion of a Raman signal 160 with at least a portion of a probe beam of light 120_{pr} to produce an electronic signal. For an optical receiver 200 with a single detector 220, all or most of the probe beam of light 120_{pr} may be mixed with the Raman signal 160. For an optical receiver 200 with multiple detectors 220, the probe beam of light 120_{pr} may be split so that a portion of the probe beam of light is sent to each of the detectors. For example, in an optical receiver 200 with four detectors 220, the probe beam 120_{pr} may be split into four portions, and each detector may receive one of the four portions of the probe beam. Similarly, for an optical receiver 200 with a single detector 220, all or most of the Raman signal 160 may be sent to the single detector, and for an optical receiver with multiple detectors 220, the Raman signal 160 may be split so that a portion of the Raman signal is sent to each of the detectors.

[0080] Coherent mixing of a probe beam 120_{pr} and a Raman signal 160, which may be referred to as heterodyne detection, may occur when the two optical signals are optically combined and then detected by a detector 220. Optically combining the probe beam 120_{pr} and the Raman signal 160 may refer to combining the two optical signals so that their electric fields are summed together. For example, the probe beam and Raman signal may be combined (e.g., with an optical combiner 130) so that the two signals are substantially coaxial and travel together in the same direction and along approximately the same optical path. Additionally, the probe beam and Raman signal may be combined so that at least a portion of their polarizations have the same orientation to allow at least a portion of their electric fields to be summed together. Once the probe beam and Raman signal are optically combined to produce a combined probe-Raman signal 210, the probe beam and Raman signal may be coherently mixed at a detector 220. The detector 220 may produce a photocurrent signal i corresponding to the coherent mixing of the probe beam 120_{pr} and a portion of the Raman signal 160.

[0081] The portion of a Raman signal 160 that is coherently mixed with a probe beam of light 120_{pr} at a detector 220 to produce an electronic signal may refer to a spectral portion of the Raman signal. The spectral portion of a Raman signal 160 that is coherently mixed with a probe beam 120_{pr} may include optical frequency components of the Raman signal that are within a particular frequency range of the frequency ν_3 of the probe beam of light, where the particular frequency range is based on or depends on the

electronic bandwidth Δf of the optical detector 220. In FIG. 7, the hatched region around the probe frequency ν_3 represents the spectral portion of the Raman signal 160 that is coherently mixed with the probe beam 120_{pr}. The particular frequency range illustrated by the hatched region extends from $\nu_3 - \Delta f$ to $\nu_3 + \Delta f$, where Δf is the electronic bandwidth of the detector. The particular frequency range includes the optical frequency components of the Raman signal 160 that are coherently mixed with the probe beam 120_{pr} to produce an electronic signal.

[0082] The electronic signal produced by the detector 220 in response to the coherent mixing of the portion of the Raman signal 160 and probe beam 120_{pr} may include one or more electronic frequency components, where each electronic frequency component has a frequency less than or equal to approximately Δf . For example, the electronic bandwidth Δf of the optical detector may be 10 GHz, and the electronic signal produced by the detector may include one or more electronic frequency components having frequencies less than or equal to approximately 10 GHz. Other optical frequency components of the Raman signal 160 that are outside the hatched region (e.g., optical frequencies less than $\nu_3 - \Delta f$ and greater than $\nu_3 + \Delta f$) may produce a coherent-mixing response in the detector 220. However, since these Raman-signal optical frequency components would produce an electronic response at frequencies greater than Δf (which is outside of the electronic bandwidth of the detector), these optical frequency components will not result in any significant contribution to the electronic signal. The electronic bandwidth of the detector 220 effectively limits or filters the optical frequency components of the Raman signal 160 that are measured by the optical receiver 200 to optical frequency components that are within a particular frequency range of the probe frequency. Accordingly, the electronic bandwidth of the detector 220, in combination with the relatively narrow spectral linewidth of the probe beam 120_{pr}, may allow a Raman spectroscopy system 100 to measure a Raman signal 160 with a high degree of spectral resolution. Herein, an optical frequency or an optical frequency component refers to a signal in the optical domain between approximately 60 THz and approximately 1,000 THz, and an electronic frequency or an electronic frequency component refers to a signal in the electronic domain between 0 Hz and approximately 50 GHz.

[0083] The electronic signal that results from coherent mixing of a Raman signal 160 and a probe beam 120_{pr} may include a coherent-mixing term that is proportional to a product of (i) E_R , the amplitude of the electric field of the Raman signal and (ii) E_{pr} , the amplitude of the electric field of the probe beam. The photocurrent signal i produced by a detector 220 in response to the coherent mixing of a Raman signal 160 and a probe beam 120_{pr} may be proportional to the square of the summed electric fields of the probe beam 120_{pr} and a spectral portion of the Raman signal 160. This type of detector 220 that produces a photocurrent signal i that is proportional to the square of a received electric field may be referred to as a square-law detector. The photocurrent signal i may be expressed as $i(t) = k |E_R(t) + E_{pr}(t)|^2$, where k is a constant (e.g., k may account for the responsivity of the detector 220 as well as other constant parameters or conversion factors). For clarity, the constant k or other constants (e.g., conversion constants or factors of 2 or 4) may be excluded from expressions herein related to the photocurrent i or the voltage signal 234. In the above

expression for $i(t)$, $E_R(t)$ is the electric field of the Raman signal **160**, and $E_{pr}(t)$ is the electric field of the probe beam **120pr**. The electric field of the Raman signal **160** may be expressed as $E_R \cos[2\pi\nu_R t + \phi_R]$, where E_R is the amplitude of the electric field of the Raman signal. The electric field of the probe beam **120pr** may be expressed as $E_{pr} \cos[2\pi\nu_3 t + \phi_{pr}]$, where E_{pr} is the amplitude of the electric field of probe beam. The frequency ν_R is the optical frequency of the electric field of the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120pr**. The frequency ν_R may include the optical frequency components of the Raman signal **160** from $\nu_3 - \Delta f$ to $\nu_3 + \Delta f$, where Δf is the electronic bandwidth of the detector. The frequency ν_3 is the optical frequency of the electric field of the probe beam **120pr**. The term ϕ_R is the phase of the electric field of the Raman signal **160**, and the term ϕ_{pr} is the phase of the electric field of the probe beam **120pr**.

[0084] The above expression for the photocurrent signal i may be expanded and written as $i(t) = E_R^2 + E_{pr}^2 + 2E_R E_{pr} \cos[2\pi(\nu_R - \nu_3)t + \Delta\phi]$, where, for clarity, the constant k is not included. In this expanded expression for the photocurrent signal $i(t)$, the first term E_R^2 corresponds to the optical power (P_R) of the Raman signal **160**, and the second term E_{pr}^2 corresponds to the optical power (P_{pr}) of the probe beam **120pr**. The third term in the above expression is $2E_R E_{pr} \cos[2\pi(\nu_R - \nu_3)t + \Delta\phi]$ and may be referred to as a coherent-mixing term that represents coherent mixing between the electric fields of the Raman signal **160** and probe beam **120pr**. The phase difference $\Delta\phi$ is the phase difference between the electric fields of the Raman signal and the probe beam (e.g., $\Delta\phi = \phi_R - \phi_{pr}$). The coherent-mixing term is proportional to $E_R \times E_{pr}$, which is the product of the electric-field amplitudes of the Raman signal **160** and the probe beam **120pr**. Additionally, the coherent-mixing term includes a cosine function that varies in time based on the frequency difference $(\nu_R - \nu_3)$ between the Raman signal **160** and the probe beam **120pr**. Since the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120pr** includes the optical frequency components of the Raman signal from $\nu_3 - \Delta f$ to $\nu_3 + \Delta f$, the frequency-difference term $(\nu_R - \nu_3)$ may include frequency components from zero to Δf . Accordingly, the coherent-mixing term may be referred to as including multiple electronic frequency components, where each electronic frequency component is proportional to $E_R E_{pr} \cos[2\pi f t + \Delta\phi]$. The frequency f , which may be referred to as an electronic frequency, is equal to the frequency difference $(\nu_R - \nu_3)$ and has a value between zero and Δf . The coherent-mixing term may also be expressed as $2\sqrt{P_R P_{pr}} \cos[2\pi f t + \Delta\phi]$, where P_R is the optical power of the Raman signal **160** and P_{pr} is the optical power of the probe beam **120pr**.

[0085] FIGS. 8-10 illustrate time-domain and frequency-domain plots of an example electronic signal resulting from coherent mixing of the Raman signal **160** and probe beam **120pr** of FIG. 7. The electronic signal produced by a detector **220** may be a photocurrent signal i , and an electronic amplifier **232** may produce a voltage signal **234** that corresponds to the photocurrent signal i . The voltage signal **234** may be approximately proportional to the photocurrent signal i , and both the voltage and photocurrent signals may be proportional to $P_R + P_{pr} + 2\sqrt{P_R P_{pr}} \cos[2\pi f t + \Delta\phi]$, where the electronic frequency f is equal to the frequency difference $(\nu_R - \nu_3)$. The electronic frequency f may take on multiple values (or a continuous range of values) between

zero and Δf , and so, the voltage signal **234** in the time domain may be viewed as a summation over these multiple frequency components. In the frequency domain, the voltage signal **234** may have frequency components that extend from DC to Δf (or, for an amplifier **232** with a low-frequency cutoff filter, from the low-frequency cutoff to Δf). The two terms P_R and P_{pr} represent the optical powers of the Raman signal and probe beam. Over the time duration of a measurement, the powers of the Raman signal and probe beam may be approximately constant, and so the contribution of $P_R + P_{pr}$ in the above expression may correspond to a DC offset with little or no time variation. An electronic amplifier **232** may include a high-pass or band-pass filter that removes or attenuates the DC or low-frequency components, resulting in an electronic signal with little or no DC offset (e.g., as illustrated by the time-domain voltage signal **234** in FIG. 8).

[0086] The signal characteristic **162** in each of FIGS. 7-10 represents a characteristic of an electronic signal (e.g., a characteristic of a photocurrent signal i or a characteristic of a corresponding voltage signal **234**) associated with a Raman signal **160**. A signal characteristic **162** may be determined from a digital signal **240**, where the digital signal is a digital representation of a photocurrent signal i or a voltage signal **234**. A characteristic **162** of an electronic signal may be determined by a processor of a Raman spectroscopy system **100** and may include one or more data points **163**, where each data point represents: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency (e.g., an electronic frequency or an optical frequency), a phase, or a polarization. A Raman spectroscopy system **100** may measure one or more signal characteristics **162** associated with one or more Raman signals **160**, and based on the measured characteristics, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0087] FIG. 8 illustrates an example time-domain plot of a voltage signal **234** resulting from coherent mixing of the Raman signal **160** and probe beam **120pr** of FIG. 7. A digitizer **236** (e.g., an ADC) may produce a digital signal **240** that represents the voltage signal **234** in the time domain, and a processor may determine a characteristic **162** of the electronic signal based on the digital signal. The signal characteristic **162** in FIG. 8 includes a single data point **163** which represents a peak amplitude A_p of the time-domain voltage signal **234**. The signal characteristic **162** may also include the optical frequency ν_3 that the probe beam **120pr** was set to when the electronic signal was obtained. Other time-domain-type data points **163** associated with the signal characteristic **162** in FIG. 8 may include (i) an amplitude of the time-domain voltage signal **234** at a particular time (e.g., at a temporal center) or (ii) an area associated with the voltage signal. For an area associated with the voltage signal **234**, a processor may first convert the voltage signal into a non-negative signal (e.g., by taking the absolute value or by squaring the values of the voltage signal) and then integrate the non-negative signal to determine an area under the curve.

[0088] A processor of a Raman spectroscopy system **100** may determine a Fourier transform of a digital signal **240**. A voltage signal **234** may be a time-domain signal, and the digital signal **240** may be a time-domain digital representa-

tion of the voltage signal **234**. The Fourier transform of the digital signal **240** may produce a frequency-domain representation of the voltage signal **234**. From the Fourier transform, the processor may determine one or more electronic frequency components of the voltage signal **234**. Determining a frequency component of a voltage signal **234** may include determining an amplitude of the frequency-domain voltage signal at a particular frequency (e.g., at 4 GHz). FIGS. 9-10 each illustrates an example frequency-domain plot of a voltage signal **234** resulting from coherent mixing of the Raman signal **160** and probe beam **120pr** of FIG. 7. The frequency-domain plots may be determined by taking a Fourier transform (e.g., a discrete Fourier transform or a fast Fourier transform) of a time-domain digital signal **240** that represents a time-domain voltage signal **234**. The x-axis of each of the frequency-domain plots in FIGS. 9-10 is labeled as “electronic frequency” to clarify and distinguish it from the x-axis of the frequency-domain plots in FIGS. 3-7. The x-axis of each of the frequency-domain plots in FIGS. 3-7 may be referred to as an “optical frequency,” where the pump beam **120pu**, Stokes beam **120S**, Raman signal **160**, and probe beam **120pr** each have frequencies in the optical domain (e.g., in the range of 60-1,000 THz). In contrast, the x-axis of each of the frequency-domain plots in FIGS. 9-10 includes frequencies in the electronic domain (e.g., in the range of DC to 50 GHz). Coherent mixing of a Raman signal **160** and probe beam **120pr** (which are signals in the optical domain) produces an electronic signal that can be detected and analyzed using electronic techniques. The y-axis of each of the plots in FIGS. 8-10 is labeled “electronic signal amplitude” and may have units of voltage, current, or electrical power (e.g., watts).

[0089] In FIG. 9, the frequency-domain voltage signal **234** has non-zero frequency components that extend from DC (i.e., zero hertz) to the detector-bandwidth frequency Δf . In FIG. 10, the frequency-domain voltage signal drops off before reaching zero hertz, indicating that the corresponding voltage signal **234** may have been AC-coupled or high-pass filtered to remove the DC or low-frequency components. The signal characteristic **162** in FIG. 9 includes a single data point **163f** which represents a peak amplitude A_p of the frequency-domain voltage signal **234**. The data point **163f** may include the peak amplitude A_p or the frequency f_p at which the peak amplitude is located. The signal characteristic **162** in FIG. 10 includes five data points: data point **163a** at frequency f_a , data point **163b** at frequency f_b , data point **163c** at frequency f_c , data point **163d** at frequency f_d , and data point **163e** at frequency f_e . Each data point **163** in FIG. 10 may include an amplitude value (e.g., amplitude A_a may be associated with data point **163a**) or a frequency (e.g., frequency f_a may be associated with data point **163a**). The frequencies associated with the data points **163** in FIG. 10 may have particular values. For example, the detector bandwidth Δf may be 5 GHz, and the frequencies f_a , f_b , f_c , f_d , and f_e may have respective values of approximately 1 GHz, 2 GHz, 3 GHz, 4 GHz, and 5 GHz. Other frequency-domain-type data points **163** associated with the signal characteristic **162** in FIG. 9 or 10 may include (i) an amplitude of the frequency-domain voltage signal **234** at a particular frequency or at a center frequency or (ii) an area or an average amplitude associated with the frequency-domain voltage signal **234**. Additionally, the signal characteristic **162** in FIG.

9 or 10 may include the optical frequency ν_3 that the probe beam **120pr** was set to when the voltage signal **234** was obtained.

[0090] A processor of a coherent Raman spectroscopy system **100** may associate a determined signal characteristic **162** with a Raman frequency shift. For example, the signal characteristic **162** in FIG. 7 may be associated with a Raman frequency shift having a frequency $\nu_1 - \nu_3$, which is the difference between the pump-beam frequency ν_1 and the probe-beam frequency ν_3 . A sample **150** under investigation may include a material with a Raman spectrum having a peak at a frequency Ω , which is approximately equal to $\nu_1 - \nu_{RP}$, the difference between the pump-beam frequency ν_1 and the Raman-signal peak frequency ν_{RP} . The peak frequency Q may be approximately equal to a vibrational frequency of the material and may correspond to a Raman frequency shift of Q (e.g., when excited with light at a frequency ν , the material may produce Raman-shifted light at the frequencies $\nu + \Omega$ and $\nu - \Omega$). In FIG. 7, the probe beam **120pr** is being used to measure the Raman signal **160** at the probe-frequency ν_3 , and the resulting signal characteristic **162** that is determined may be associated with a Raman spectrum of the material at the frequency $\nu_1 - \nu_3$, which may be referred to as a Raman frequency shift of $\nu_1 - \nu_3$.

[0091] FIG. 11 illustrates an example Raman signal **160** that is measured at multiple probe frequencies ν_3 . The probe beam **120pr** is tuned to multiple different probe frequencies (ν_{3-1} , ν_{3-2} , ν_{3-3} , $\dots$, ν_{3-n}) to measure multiple respective signal characteristics (**162-1**, **162-2**, **162-3**, $\dots$, **162-n**) of the Raman signal **160**. The Raman signal **160** is produced by coherent Raman scattering of pump and Stokes beams within a sample **150**. Throughout the measurements of the multiple signal characteristics **162** in FIG. 11, the pump-beam frequency ν_1 and the Stokes-beam frequency ν_2 may remain substantially constant so that the resulting Raman signal **160** also remains substantially constant (e.g., the Raman signal may exhibit substantially the same shape and amplitude throughout the measurements). The number n of signal characteristics **162** that are measured may be 1, 5, 10, 50, 100, 500, 1,000, or any other suitable number of signal characteristics. At each of the n probe frequencies, the probe beam **120pr** may be coherently mixed with a spectral portion of the Raman signal **160** that is within a particular frequency range of the probe frequency ν_3 (e.g., within Δf of the probe frequency, as illustrated in FIG. 7) to measure one signal characteristic **162**. Each signal characteristic **162** may provide information about the Raman signal **160** in the frequency region from $\nu_3 - \Delta f$ to $\nu_3 + \Delta f$. Based on one or more measured signal characteristics **162**, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0092] At each of the n probe frequencies, a single measurement may be performed, or multiple measurements may be performed. For example, with the probe beam **120pr** in FIG. 11 set to the probe-beam frequency ν_{3-1} , an optical receiver **200** may measure a single photocurrent signal i and produce a corresponding single digital output signal **240**. From that one output signal **240**, a processor may determine a signal characteristic **162-1**. Alternatively, with the probe beam **120pr** in FIG. 11 set to the probe-beam frequency ν_{3-1} , an optical receiver **200** may measure a series of multiple photocurrent signals i (e.g., 2, 4, 10, 20, 50, 100, or any other suitable number of signals) and produce multiple corre-

sponding digital output signals **240**. A processor may determine one signal characteristic **162-1** from the multiple digital output signals. For example, the processor may average or otherwise combine the multiple digital output signals **240** to produce one signal characteristic **162-1**. Measurement of a series of multiple photocurrent signals **i** may improve the measurement accuracy by averaging out noise or removing outliers from the measurements.

[0093] By tuning the probe-beam frequency ν_3 to multiple frequencies across at least a portion of a Raman signal **160**, a coherent Raman spectroscopy system **100** may measure the Raman signal at multiple points **162**. For example, the probe-beam frequency ν_3 in FIG. **6** may be tuned across at least a portion of the frequency range between ν_2-F_2 and ν_2-F_1 or between ν_2+F_1 and ν_2+F_2 . Each of the different probe-beam frequencies to which the probe beam **120pr** is changed may be offset from an adjacent probe-beam frequency by a frequency increment ΔF between approximately 10 MHz and approximately 10 GHz. For example, the probe-beam frequency in FIG. **11** may be changed in frequency increments ΔF of approximately 5 GHz as it is tuned across at least a portion of the Raman signal **160**. Each signal characteristic **162** in FIG. **11** is plotted along the x-axis at the probe frequency ν_3 at which the signal characteristic was obtained. The y-axis in FIG. **11** may represent an amplitude or area associated with the signal characteristics **162**. For example, each signal characteristic **162** in FIG. **11** may be plotted along the y-axis at a value corresponding to an amplitude or an area of an electronic signal from which the signal characteristic was obtained.

[0094] A probe light source **110pr** of a coherent Raman spectroscopy system **100** may include a wavelength-tunable laser, where the frequency ν_3 of the probe beam **120pr** is adjustable by changing the wavelength of light produced by the wavelength-tunable laser. For example, a wavelength-tunable laser may be adjustable over a wavelength range having a width between approximately 10 nm and approximately 100 nm. As another example, the wavelength-tuning range of a wavelength-tunable laser may be between approximately 1000 nm and approximately 1100 nm, between approximately 1490 nm and approximately 1570 nm, or between approximately 1600 nm and approximately 1690 nm. A wavelength-tunable laser may be continuously tunable over a wavelength-tuning range or may be tunable to multiple discrete wavelengths within a wavelength-tuning range. For example, a wavelength-tunable laser may be continuously tunable to any wavelength between 1530 nm and 1570 nm. Alternatively, a wavelength-tunable laser may be tunable to a set of approximately 10, 100, or 1,000 discrete wavelengths between 1530 nm and 1570 nm (e.g., the wavelengths may be separated from one another by approximately 4 nm, 0.4 nm, or 0.04 nm, respectively). A probe light source **110pr** may include a wavelength-tunable laser that sequentially changes the probe-beam frequency ν_3 to multiple different frequencies. For example, the probe-beam frequency in FIG. **6** may be tuned to approximately 100 different frequencies between the frequencies ν_2-F_2 and ν_2-F_1 . At each of the different probe-beam frequencies, the probe beam **120pr** and a spectral portion of the Raman signal **160** may be coherently mixed at a detector **220** to produce a corresponding electronic signal, and a processor may determine a signal characteristic **162** based on the electronic

[0095] In FIG. **11**, the probe beam **120pr** may initially be set to the probe-beam frequency ν_{3-1} and coherently mixed at a detector **220** with a spectral portion of the Raman signal **160** around the probe frequency ν_{3-1} . For example, the spectral portion of the Raman signal **160** may include optical frequency components of the Raman signal from $\nu_{3-1}-\Delta f$ to $\nu_{3-1}+\Delta f$, where Δf is the electronic bandwidth of the optical detector **220**. A processor may determine a signal characteristic **162-1** based on the electronic signal resulting from the coherent mixing of the probe beam **120pr** at frequency ν_{3-1} and the associated spectral portion of the Raman signal **160**. After measuring the Raman signal **160** around the probe frequency ν_{3-1} , the probe light source **110pr** may change the probe-beam frequency by a frequency change ΔF to the frequency ν_{3-2} . The probe frequency ν_{3-2} is equal to $\nu_{3-1}+\Delta F$, and the frequency change ΔF between adjacent frequencies may be between approximately 10 MHz and approximately 10 GHz. The frequency change ΔF may be a fixed value or may be dynamically adjusted during a measurement of a Raman signal **160**. After the probe-beam frequency is changed to ν_{3-2} , the probe beam **120pr** may be coherently mixed with the spectral portion of the Raman signal **160** around the probe frequency ν_{3-2} (e.g., the spectral portion may include optical frequency components of the Raman signal from $\nu_{3-2}-\Delta f$ to $\nu_{3-2}+\Delta f$). A processor may determine a signal characteristic **162-2** based on the electronic signal resulting from the coherent mixing of the probe beam **120pr** at frequency ν_{3-2} and the associated spectral portion of the Raman signal **160**. After measuring the Raman signal **160** around the probe frequency ν_{3-2} , the probe light source **110pr** may change the probe-beam frequency by the frequency change ΔF to the frequency ν_{3-3} , and a measurement of the Raman signal **160** around the frequency ν_{3-3} may be performed. The Raman spectroscopy system **100** may sequentially change the frequency ν_3 of the probe light source and measure a spectral portion of the Raman signal **160** at each frequency until reaching the final probe frequency ν_{3-n} . During the measurements, the probe-beam frequency ν_3 may be tuned so that it avoids overlapping with the frequency ν_2 of the Stokes beam to prevent mixing between the probe and Stokes beams.

[0096] FIG. **12** illustrates an example Raman spectrum corresponding to the Raman signal of FIG. **11**. A Raman spectroscopy system **100** may measure a sample **150** and determine the signal characteristics **162** of the Raman signal **160** in FIG. **11**, and the corresponding Raman spectrum in FIG. **12** may represent the Raman spectrum of one or more materials that are part of the sample **150**. For example, the sample **150** may include glucose and the peak frequency $\nu_1-\nu_{RP}$ of the Raman spectrum may be approximately equal to a frequency Q of a molecular vibration of glucose. A processor of the Raman spectroscopy system **100** may determine the Raman spectrum in FIG. **12** based on the signal characteristics **162** (and the associated probe-beam frequencies ν_3) of the Raman signal **160** in FIG. **11**. Each signal characteristic **162** of a Raman signal **160** measured at a probe frequency ν_3 may be associated with a Raman frequency shift having a frequency $\nu_1-\nu_3$, where ν_1 is the pump-beam frequency. To determine the Raman spectrum, each signal characteristic **162** may be transformed from its Raman-signal frequency ν_3 to a corresponding Raman-shift frequency $\nu_1-\nu_3$. For example, the processor may associate the signal characteristic **162-1** at frequency ν_{3-1} in FIG. **11** with a Raman shift having the frequency $\nu_1-\nu_{3-1}$ in FIG. **12**.

Similarly, the signal characteristic **162-2** at frequency ν_{3-2} in FIG. **11** may be associated with a Raman shift having the frequency $\nu_1 - \nu_{3-2}$ in FIG. **12**, and the signal characteristic **162-3** at frequency ν_{3-3} in FIG. **11** may be associated with a Raman shift having the frequency $\nu_1 - \nu_{3-3}$ in FIG. **12**. Additionally, the peak frequency ν_{RP} of the Raman signal **160** in FIG. **11** may correspond to the Raman-signal peak in FIG. **12** with a frequency of $\nu_1 - \nu_{RP}$, which in turn may correspond to the vibrational frequency Q of a particular material.

[0097] Based on the Raman spectrum in FIG. **12**, a processor may determine (i) whether a particular material is present in a sample **150** or (ii) an amount or a concentration of the particular material in the sample **150**. For example, the processor may compare one or more peaks, signal characteristics **162**, or other features of the Raman spectrum in FIG. **12** to a previously determined Raman spectrum for glucose. If one or more peaks or characteristics **162** of the Raman spectrum in FIG. **12** match or line up with peaks or characteristics of the Raman spectrum for glucose, then the processor may determine that glucose is present in the sample **150**. Alternatively, if the Raman spectrum in FIG. **12** is missing one or more peaks or characteristics of the Raman spectrum for glucose, then the processor may determine that little or no glucose is present in the sample **150**. As another example, the processor may determine the amount or concentration of glucose in the sample **150** based on the Raman spectrum in FIG. **12**. The concentration of glucose in the sample **150** may be related to the amplitude or height of one or more peaks of the Raman spectrum in FIG. **12** (e.g., the glucose concentration may be approximately proportional to the height of one or more Raman peaks). The concentration of glucose may then be determined based at least in part on the height of the Raman peak located at the Raman-shift frequency $\nu_1 - \nu_{RP}$.

[0098] FIG. **13** illustrates another example Raman signal **160** that is measured at multiple probe frequencies. The probe beam **120pr** is tuned to multiple different probe frequencies (ν_{3-1} , ν_{3-2} , ν_{3-3} , . . . ν_{3-n}) to measure multiple respective signal characteristics (**162-1**, **162-2**, **162-3**, . . . **162-n**) of the Raman signal **160**. The Raman signal **160** in FIG. **13** has a single peak centered at frequency ν_{RP} with a spectral linewidth of $\Delta\nu_R$. The Raman signal **160** in FIG. **11** has one main peak located at frequency ν_{RP} along with multiple smaller peaks located on either side of the main peak. A Raman signal **160** may include a single peak (e.g., as illustrated in FIG. **13**) or may include multiple peaks (e.g., as illustrated in FIG. **11**).

[0099] FIGS. **14-15** each illustrate a second example Raman signal obtained by changing the frequency offset Ω between a pump beam **120pu** and a Stokes beam **120S**. A coherent Raman spectroscopy system **100** may include a pump light source **110pu** or a Stokes light source **110S** with a wavelength-tunable laser, and the frequency offset Ω may be adjustable by changing the wavelength of the wavelength-tunable laser. A wavelength-tunable pump or Stokes laser may be continuously tunable over a wavelength-tuning range or may be tunable to multiple discrete wavelengths within a wavelength-tuning range. For example, Stokes beam **120** in FIG. **14** may be produced by a continuously tunable laser diode that can produce light at the frequencies ν_2 and ν_2' . As another example, a Stokes laser **110S** may include two fixed-wavelength laser diodes that operate at the respective frequencies ν_2 and ν_2' . Similarly, pump beam

120pu in FIG. **15** may be produced by a continuously tunable laser diode that can produce light at the frequencies ν_1 and ν_1' , or a pump laser **110pu** may include two fixed-wavelength laser diodes that operate at the respective frequencies ν_1 and ν_1' .

[0100] In FIG. **14**, the pump frequency ν_1 is fixed, and the Stokes frequency is changed from ν_2 to ν_2' to change the frequency offset from Ω_1 to Ω_2 . Frequency offset Ω_1 equals $\nu_1 - \nu_2$, and frequency offset Ω_2 equals $\nu_1 - \nu_2'$. In FIG. **15**, the Stokes frequency ν_2 is fixed, and the pump frequency is changed from ν_1 to ν_1' to change the frequency offset from Ω_1 to Ω_2 . Frequency offset Ω_1 equals $\nu_1 - \nu_2$, and frequency offset Ω_2 equals $\nu_1' - \nu_2$. In other embodiments, both the pump and Stokes frequencies may be changed to change the frequency offset Ω from one value to another.

[0101] FIG. **14** includes two Raman signals **160a** and **160b**. Initially, a Stokes light source **110S** may produce a Stokes beam **120S** at the frequency ν_2 to produce the frequency offset Ω_1 between the Stokes beam and the pump beam **120pu**, where $\Omega_1 = \nu_1 - \nu_2$. For example, the Stokes beam **120S** may have a frequency ν_2 of 193 THz, and the pump beam **120pu** may have a frequency ν_1 of 207 THz, which corresponds to a frequency offset Ω_1 of 14 THz (or, 467 cm^{-1}). The Raman signal **160a** is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu** within a sample **150**. The Raman signal **160a** is centered at or near the frequency ν_2 of the Stokes beam **120S**, and the frequency ν_3 of the probe beam **120pr** overlaps the Raman signal **160a** and is relatively close to the frequency ν_2 of the Stokes beam **120S** (e.g., the probe-beam frequency ν_3 may be within 200 GHz of the Stokes-beam frequency ν_2). The probe beam **120pr** and a spectral portion of the Raman signal **160a** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency ν_3 of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160a** to measure multiple signal characteristics **162** associated with the Raman signal.

[0102] After measuring the first Raman signal **160a**, the Stokes light source **110S** may change the frequency of the Stokes beam to produce a Stokes beam **120S'** at the frequency ν_2' , resulting in a frequency offset of Ω_2 between the Stokes beam **120S'** and the pump beam **120pu**, where $\Omega_2 = \nu_1 - \nu_2'$. For example, the Stokes beam **120S** may be changed to a frequency ν_2' of 182 THz, and the pump-beam frequency ν_1 may remain at 207 THz, which corresponds to a frequency offset Ω_1 of 25 THz (or, 834 cm^{-1}). The second Raman signal **160b** is produced by coherent Raman scattering of the Stokes beam **120S'** and pump beam **120pu** within the sample **150**. The Raman signal **160b** is centered at or near the frequency ν_2' of the Stokes beam **120S'**. Additionally, a probe light source **110pr** may change the frequency of the probe beam to produce a probe beam **120pr'** at a frequency ν_3' that overlaps the Raman signal **160b** and is relatively close to the frequency ν_2' of the Stokes beam **120S'** (e.g., the probe-beam frequency ν_3' may be within 200 GHz of the Stokes-beam frequency ν_2').

[0103] The probe beam **120pr'** and a spectral portion of the Raman signal **160b** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency ν_3' of the probe beam **120pr'** may be tuned across

at least a portion of the second Raman signal **160b** to measure multiple signal characteristics **162** associated with the Raman signal.

[0104] The two Raman signals **160a** and **160b** in FIG. **14** may correspond to two respective peaks of a Raman spectrum of a material that is in the sample **150**. One Raman peak, associated with Raman signal **160a**, is located at a frequency of approximately Ω_1 , and the frequency Ω_1 may correspond to a vibrational frequency of the material. The other Raman peak, associated with Raman signal **160b**, is located at a frequency of approximately Ω_2 , and the frequency Ω_2 may correspond to another vibrational frequency of the material. A Raman spectroscopy system **100** may perform measurements of signal characteristics **162** at 1, 2, 4, 10, 20, or 50 different values of the frequency offset Ω . The frequency offset Ω between the Stokes beam **120S** and the pump beam **120pu** may be set to each of the different frequency-offset values to produce an associated Raman signal **160**. At each value of Ω , the frequency ν_3 of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal. Then, the frequency of the Stokes beam or pump beam may be adjusted to the next value of Ω , where another measurement of an associated Raman signal **160** is performed. Based on the signal characteristics **162** associated with each of the values of the frequency offset Ω , a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0105] FIG. **15** includes the Raman signal **160c**. A first Raman signal (similar to Raman signal **160a** in FIG. **14**) may be approximately overlapped with the second Raman signal **160c** in FIG. **15**, and for clarity, the first Raman signal is not included in FIG. **15**. Initially, a Stokes light source **110S** may produce a Stokes beam **120S** at the frequency ν_2 , and a pump light source **110pu** may produce a pump beam **120pu** at the frequency ν_1 , which results in the frequency offset Ω_1 between the Stokes and pump beams in FIG. **15**. For example, the Stokes beam **120S** may have a frequency ν_2 of 193 THz, and the pump beam **120pu** may have a frequency ν_1 of 207 THz, which corresponds to a frequency offset Ω_1 of 14 THz (or, 467 cm^{-1}). A first Raman signal (similar to Raman signal **160a** in FIG. **14**) is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu** within a sample **150**. The first Raman signal may be centered at or near the frequency ν_2 of the Stokes beam **120S**, and the frequency ν_3 of the probe beam **120pr** may overlap the first Raman signal and may be relatively close to the frequency ν_2 of the Stokes beam of light **120S**. The probe beam **120pr** and a spectral portion of the first Raman signal may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency ν_3 of the probe beam **120pr** may be tuned across at least a portion of the first Raman signal to measure multiple signal characteristics **162** associated with the first Raman signal.

[0106] In FIG. **15**, after measuring the first Raman signal, the pump light source **110pu** may change the frequency of the pump beam **120pu** from ν_1 to ν_1' . This results in a frequency offset of Ω_2 between the Stokes beam **120S** and the pump beam **120pu'**, where $\Omega_2 = \nu_1' - \nu_2$. For example, the pump beam may be changed to a frequency ν_1' of 218 THz, and the Stokes-beam frequency ν_2 may remain at 193 THz,

which corresponds to a frequency offset Ω_1 of 25 THz (or, 834 cm^{-1}). The second Raman signal **160c** is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu'** within the sample **150**. As with the first Raman signal, the Raman signal **160c** is centered at or near the frequency ν_2 of the Stokes beam **120S**, and the frequency ν_3 of the probe beam **120pr** overlaps the Raman signal **160c** and is relatively close to the frequency ν_2 of the Stokes beam of light **120S** (e.g., the probe-beam frequency ν_3 may be within 200 GHz of the Stokes-beam frequency ν_2). The probe beam **120pr** and a spectral portion of the Raman signal **160c** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency ν_3 of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160c** to measure multiple signal characteristics **162** associated with the Raman signal.

[0107] In FIG. **15**, the first Raman signal (which may be similar to Raman signal **160a** in FIG. **14**) and the second Raman signal **160c** may correspond to two different peaks of a Raman spectrum of a material that is in the sample **150**. One Raman peak, associated with the first Raman signal, is located at a frequency of approximately Ω_1 , and the frequency Ω_1 may correspond to a vibrational frequency of the material. The other Raman peak, associated with Raman signal **160c**, is located at a frequency of approximately Ω_2 , and the frequency Ω_2 may correspond to another vibrational frequency of the material. Based at least in part on the signal characteristics **162** associated with the two frequency offsets Ω_1 and Ω_2 , a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0108] FIG. **16** illustrates an example Raman signal **160** along with two probe beams of light **120pr-1** and **120pr-2**. The Raman signal **160** may be produced by coherent Raman scattering of the pump beam **120pu** and Stokes beam **120S** within a sample **150**. The frequency of the pump beam **120pu** is ν_1 , and the frequency of the Stokes beam **120S** is ν_2 , which corresponds to a frequency offset Ω equal to $\nu_1 - \nu_2$. The Raman signal **160** is centered at or near the frequency ν_2 of the Stokes beam **120S** (e.g., the center frequency of the Raman signal **160** may be within 200 GHz of ν_2). The first probe beam **120pr-1** may be used to measure the Raman signal **160**, and the second probe beam **120pr-2** may be used to measure the pump beam **120pu**.

[0109] The frequency ν_3 of first probe beam **120pr-1** overlaps the Raman signal **160** and is relatively close to the frequency ν_2 of the Stokes beam **120S** (e.g., ν_3 may be within 200 GHz of ν_2). The probe beam **120pr-1** and a spectral portion of the Raman signal **160** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency ν_3 of the probe beam **120pr-1** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal.

[0110] The frequency ν_4 of the second probe beam **120pr-2** is relatively close to the frequency ν_1 of the pump beam **120pu** (e.g., ν_4 may be within 50 GHz of ν_1). For example, the frequency ν_4 of the second probe beam **120pr-2** may be offset from the frequency ν_1 of the pump beam **120pu** by approximately 10 GHz, 5 GHz, or 1 GHz. Alternatively, the frequency ν_4 of the second probe beam **120pr-2** may be approximately equal to the frequency ν_1 of

the pump beam **120pu**. After the pump beam **120pu** has interacted with the sample, the probe beam **120pr-2** may be coherently mixed with the pump beam. For example, after the pump and Stokes beams have produced the Raman signal **160** and after the pump beam has exited the sample, the pump and probe beams may be coherently mixed together at a detector **220** to produce an electronic signal, from which a signal characteristic may be determined.

[0111] FIG. 17 illustrates an example optical receiver **200** for measuring the Raman signal **160** and pump beam of light **120pu** from FIG. 16. The combined pump-Stokes beam **140** (which includes pump beam **120pu** and Stokes beam **120S**) is directed to a sample **150**, which produces a Raman signal **160** in response to the pump and Stokes beams. The Raman signal **160** may be collected by one or more optical elements and directed to the optical receiver **200**. Additionally, residual light from the pump beam **120pu** may be collected and directed to the optical receiver **200** as a residual pump beam **120pu-2**. The residual pump beam **120pu-2** may include light from the pump beam **120pu** after the pump beam has interacted with and exited the sample **150**. The optical receiver **200** in FIG. 17 may be referred to as a two-channel optical receiver that includes two parallel measurement channels for separately detecting and measuring the Raman signal **160** and the residual pump beam **120pu-2**. The first measurement channel detects the Raman signal **160** and includes probe laser **110pr-1**, detector **220-1**, and detection electronics **230-1**. The second measurement channel detects the residual pump beam **120pu-2** and includes probe laser **110pr-2**, detector **220-2**, and detection electronics **230-2**.

[0112] The Raman signal **160** and the residual pump beam **120pu-2** are directed to the combiner **130c**, which may be a dichroic beamsplitter, and the combiner **130c** reflects the Raman signal **160** and transmits the residual pump beam **120pu-2**. The combiner **130c** also transmits at least a portion of the probe beam **120pr-1** produced by the probe laser **110pr-1** and combines the probe beam **120pr-1** with the Raman signal **160** to produce a combined probe-Raman signal **210-1**. The probe-Raman signal **210-1** is sent to the detector **220-1**, where the probe beam **120pr-1** and a spectral portion of the Raman signal **160** are coherently mixed to produce a photocurrent signal i_1 . The detection electronics **230-1** receives the photocurrent signal i_1 and produces a digital output signal **240-1** that corresponds to the photocurrent signal i_1 .

[0113] The combiner **130d** (which may be a dichroic or a non-dichroic beamsplitter) reflects at least a portion of the residual pump beam **120pu-2** and transmits at least a portion of the probe beam **120pr-2** produced by the probe laser **110pr-2**. The combiner **130d** combines the probe beam **120pr-2** with the residual pump beam **120pu-2** to produce a combined probe-pump signal **210-2**, which is sent to the detector **220-2**. The probe beam **120pr-2** and the residual pump beam **120pu-2** are coherently mixed at the detector **220-2** to produce a photocurrent signal i_2 , and the detection electronics **230-2** produces a digital output signal **240-2** that corresponds to the photocurrent signal i_2 .

[0114] The two digital output signals **240-1** and **240-2** may be sent to a processor which determines a signal characteristic **162** of each of the photocurrent signals i_1 and i_2 based on the digital output signals. Additionally, the frequency of the first probe beam **120pr-1** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal

characteristics **162** associated with the Raman signal. If the frequency ν_1 of the pump beam **120pu** remains fixed, the frequency ν_4 of the second probe beam **120pr-2** may also remain fixed. Alternatively, if the frequency ν_1 of the pump beam **120pu** is changed (e.g., to switch to a different frequency offset Ω), the frequency of the probe beam **120pr-2** may also be switched to maintain a particular frequency offset between the pump and probe frequencies.

[0115] Measurement of the residual pump beam **120pu-2** may be performed two or more times to determine how the power of the pump beam changes when the Raman signal **160** is produced. For example, the residual pump beam **120pu-2** may be measured once when the Stokes beam **120S** is turned off (and no Raman signal **160** is produced) and another time when the Stokes beam is turned on (and the Raman signal **160** is produced). A processor may determine the change in the power of the residual pump beam **120pu-2** associated with the Stokes beam **120S** being turned off and on. Since at least part of the Raman signal **160** may be produced by Stokes-shifted photons from the pump beam **120pu**, a decrease in the power of the residual pump beam **120pu-2** may correspond to the power of the Raman signal.

[0116] In another embodiment of a two-channel optical receiver, the optical receiver may not include a second probe laser **110pr-2**. Instead, the residual pump beam **120pu-2** may be sent to a detector **220** for direct detection without mixing the residual pump beam with another signal.

[0117] FIG. 18 illustrates an example Raman spectroscopy system **100** for measuring a Raman signal **160** produced by spontaneous Raman scattering. Instead of producing a Raman signal by coherent Raman scattering of pump and Stokes beams within a sample (e.g., as illustrated in FIGS. 1-2), the Raman signal **160** in FIG. 18 is produced by spontaneous Raman scattering of the pump beam of light **120pu** within the sample **150**. While the production of the Raman signal **160** in FIG. 18 is different from that of the Raman spectroscopy systems of FIGS. 1-2, the optical receiver **200** and the Raman-signal detection technique in FIG. 18 is similar to that of FIGS. 1-2. In FIG. 18, the Raman signal **160** is detected by coherently mixing the Raman signal with a probe beam of light **120pr**.

[0118] The Raman spectroscopy system **100** in FIG. 18 includes a pump light source **110pu** that produces a pump beam of light **120pu** at a pump frequency ν_1 . The pump beam **120pu** is directed to a sample **150** (e.g., by one or more optical elements), and the sample **150** produces a Raman signal **160** by spontaneous Raman scattering of light from the pump beam. The spontaneous Raman signal **160** is collected (e.g., by one or more optical elements) and directed to the optical receiver **200**. The optical receiver **200** includes a probe light source **110pr** that produces a probe beam of light **120pr** at a probe frequency ν_3 , where the probe frequency overlaps the Raman signal. The optical combiner **130b**, which may be a dichroic or a non-dichroic beamsplitter, combines the Raman signal **160** and the probe beam **120pr** to produce a combined probe-Raman signal **210** that is directed to a detector **220**. The detector **220** coherently mixes a spectral portion of the Raman signal **160** with the probe beam **120pr** to produce a photocurrent signal i . The detection electronics **230** may produce (i) an analog voltage signal that corresponds to the photocurrent signal i and (ii) a digital output signal **240** that corresponds to the photocurrent signal or the voltage signal. The digital output signal **240** may be sent to a processor, and the processor may

determine a signal characteristic **162** of the photocurrent signal or voltage signal based on the digital output signal **240**.

[0119] FIG. 19 illustrates an example Raman signal produced by the Raman spectroscopy system of FIG. 18. The Raman signal **160** produced by spontaneous Raman scattering of the pump beam **120_{pu}** has a peak frequency of ν_{RP} . The frequency offset Ω between the pump beam and the peak frequency of the Raman signal equals $\nu_1 - \nu_{RP}$, and the frequency offset Ω may correspond to a vibrational frequency of a material that is part of the sample **150**. In FIG. 19, the probe beam **120_{pr}** may be coherently mixed with a spectral portion of the Raman signal **160** that is within a particular frequency range of the probe frequency ν_3 to measure one signal characteristic **162**. Additionally, the probe light source **110_{pr}** in FIG. 18 may include a wavelength-tunable laser that tunes the probe beam **120_{pr}** to multiple frequencies across at least a portion of the Raman signal **160**, and the optical receiver **200** may measure multiple respective signal characteristics **162** associated with the Raman signal.

[0120] FIG. 20 illustrates an example laser diode **110** that produces a free-space beam of light **120**. The laser diode **110** in FIG. 20 may be part of a pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}**, and the free-space beam **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**. For example, the probe laser **110_{pr}** in FIG. 2 may be similar to the laser diode **110** in FIG. 20, and the probe laser **110_{pr}** may produce a free-space probe beam **120_{pr}** that is combined with the Raman signal **160** by a free-space beam combiner **130_b**. In FIG. 20, the electronic driver **112** supplies laser-diode drive current to the laser diode **110**, and the laser diode produces output light that is collimated by a lens **114** to produce a collimated free-space beam **120**. In other embodiments, a lens **114** may produce a focused free-space beam **120** (e.g., the lens may focus the free-space beam onto a sample **150**). The laser current/supplied to the laser diode **110** may be a substantially constant DC current resulting in an output beam of light **120** having a substantially constant optical power. Additionally or alternatively, the laser current/may include pulses of current resulting in an output beam **120** that includes corresponding pulses of light.

[0121] The output beam of light **120** produced by the laser diode **110** may have a spectral linewidth of less than approximately 200 MHz, 100 MHz, 50 MHz, 10 MHz, 1 MHz, or 100 KHz. The laser diode **110** in FIG. 20 may be a wavelength-tunable laser diode where the wavelength of the output beam **120** is adjustable over a wavelength range having a width between approximately 10 nm and approximately 100 nm. For example, the operating wavelength of the laser diode **110** may be tunable over at least a portion of one of the following wavelength ranges: 1000 nm to 1100 nm; 1220 nm to 1450 nm; 1490 nm to 1570 nm; 1600 nm to 1690 nm. Alternatively, the laser diode **110** in FIG. 20 may be a fixed-wavelength laser diode. For example, the laser diode **110** in FIG. 20 may be a distributed feedback (DFB) laser diode with a spectral linewidth of less than 1 MHz, and the output beam **120** may have any suitable substantially fixed wavelength between approximately 600 nm and approximately 2000 nm.

[0122] FIG. 21 illustrates an example laser diode **110** that produces seed light **122** that is amplified by a semiconductor optical amplifier (SOA) **124**. Instead of directly emitting an

output beam **120** (e.g., as illustrated in FIG. 20), the light from a laser diode **110** may first be amplified by an optical amplifier. The laser diode **110** in FIG. 21 acts as a seed laser that produces seed light **122** that is coupled into the input end of the waveguide **125** of the SOA **124**. The SOA waveguide **125** in FIG. 21 is indicated by the cross-hatched region within the SOA **124**. The SOA **124** amplifies the seed light as it propagates within the waveguide from the input end to the output end, and the output beam of light **120** is emitted from the output end of the SOA. The optical gain provided by the SOA may come from electrical current that is supplied to the SOA by an electronic driver (not illustrated in FIG. 21). For example, an electronic driver **112** may supply substantially constant DC electrical current to the laser diode **110** and to the SOA **124**, and the resulting output beam **120** may have substantially constant optical power. The output beam **120** may be a free-space beam, or the output beam **120** may be coupled into an optical fiber or into a waveguide of a photonic integrated circuit (PIC).

[0123] The laser diode **110** and the SOA **124** in FIG. 21 may be fabricated or integrated together on the same chip so that seed light **122** from the laser diode is directly coupled into the waveguide **125** of the SOA. The waveguide **125** of the SOA **124** may be a tapered optical waveguide (as illustrated in FIG. 21) with a width that increases along a lateral direction from the input end that receives the seed light **122** to the output end that emits the output beam **120**. A light source that includes a seed laser diode **110** that supplies seed light **122** that is amplified by a SOA **124** (as illustrated in FIG. 21) may be referred to as a master-oscillator power-amplifier laser (MOPA laser). The seed laser diode **110** may be referred to as a master oscillator, and the SOA **124** may be referred to as a power amplifier. The MOPA laser in FIG. 21 may be part of a pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}**, and the output beam of light **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**.

[0124] FIG. 22 illustrates an example laser diode **110** that produces seed light **122** that is amplified by a fiber-optic amplifier **126**. The laser diode **110** in FIG. 22 acts as a seed laser that produces seed light **122** that is coupled into an optical fiber **116**. The optical fiber **116** directs the seed light **122** to the fiber-optic amplifier **126**, and the fiber-optic amplifier amplifies the seed light as it propagates through an optical gain fiber of the fiber-optic amplifier. The optical gain fiber may be doped with rare-earth ions (e.g., neodymium, erbium, or ytterbium) or bismuth that provide the optical gain to the seed light. One or more pump lasers may optically pump the active material (e.g., rare-earth ions or bismuth ions) in the optical gain fiber, which in turn provide optical amplification to the seed light **122** propagating through the gain fiber. The amplified seed light produced by the fiber-optic amplifier **126** propagates in an optical fiber as a fiber-coupled output beam **120**. The laser diode **110** and fiber-optic amplifier **126** in FIG. 22 may be part of a pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}**, and the fiber-coupled output beam **120** may be directed to an optical combiner **130**, a sample **150**, or a detector **220**.

[0125] A pump light source **110_{pu}**, a Stokes light source **110_S**, or a probe light source **110_{pr}** may include a seed laser diode **110** followed by an optical amplifier. The seed laser diode **110** may produce seed light **122** that is amplified by the optical amplifier to produce an output beam of light **120**.

An optical amplifier may include a SOA 124 (e.g., as illustrated in FIG. 21) or a fiber-optic amplifier 126 (e.g., as illustrated in FIG. 22). In some embodiments, an optical amplifier may include a SOA 124 followed by a fiber-optic amplifier 126. For example, a seed laser diode 110 may produce seed light 122 that is first amplified by a SOA 124 and then further amplified by a fiber-optic amplifier 126.

[0126] In some embodiments, a pump light source 110_{pu}, a Stokes light source 110_S, or a probe light source 110_{pr} may include a laser diode 110 and an optical fiber 116 and may not include a fiber-optic amplifier. For example, light produced by a laser diode 110 may be coupled into an optical fiber 116 to produce a fiber-coupled beam 120, and the optical fiber may direct the laser-diode light to an optical combiner 130, a sample 150, or a detector 220. A laser diode 110 that produces a fiber-coupled beam 120 may be referred to as a fiber-coupled laser diode.

[0127] FIG. 23 illustrates an example sampled-grating distributed Bragg reflector (SG-DBR) laser 110. An SG-DBR laser 110 is a wavelength-tunable laser diode that produces an output beam 120 that can be tuned over a wavelength range having a width of between 20 nm and 50 nm. For example, an SG-DBR laser 110 may have a 40-nm wavelength-tuning range from approximately 1530 nm to approximately 1570 nm or from approximately 1630 nm to approximately 1670 nm. The SG-DBR laser 110 in FIG. 23 may be part of a pump light source 110_{pu}, a Stokes light source 110_S, or a probe light source 110_{pr}, and the output beam 120 may be a pump beam 120_{pu}, a Stokes beam 120_S, or a probe beam 120_{pr}. For example, an SG-DBR laser 110 may be part of a probe light source 110_{pr} that produces the probe beam 120_{pr} in FIG. 6, and the SG-DBR laser may tune the frequency ν_3 of the probe beam across at least a portion of the Raman signal 160. An SG-DBR laser 110 may produce a free-space beam 120, a fiber-coupled beam 120, or a beam that is coupled into a waveguide of a PIC. Alternatively, an SG-DBR laser 110 may be integrated with a SOA 124 that amplifies the light produced by the SG-DBR laser (e.g., as illustrated in FIG. 21).

[0128] The SG-DBR laser 110 in FIG. 23 includes a back mirror 180, a phase section 182, a gain section 184, and a front mirror 186, where the phase and gain sections are located between the front and back mirrors. The laser diode current/supplied to the SG-DBR laser 110 includes the following: current/supplied to the back mirror 180, current/supplied to the phase section 182, current/g supplied to the gain section 184, and current #supplied to the front mirror 186. The gain current/g provides optical gain to the optical waveguide 188 of the SG-DBR laser 110, and the other currents may be used to set the wavelength of the output beam 120 produced by the SG-DBR laser. The electronic driver 112 may supply particular combinations of electrical currents to the back mirror 180, phase section 182, gain section 184, and front mirror 186, where each particular combination of electrical currents causes the SG-DBR laser 110 to produce an output beam 120 at a particular wavelength. For example, an SG-DBR laser 110 may be part of a wavelength-tunable probe light source 110_{pr} that produces the probe beam 120_{pr} in FIG. 11, and the electronic driver 112 may supply particular combinations of the electrical currents I_b , I_p , I_g , and I_f to produce the different probe frequencies ν_{3-1} , ν_{3-2} , ν_{3-3} , . . . and ν_{3-n} to tune the probe beam across at least a portion of the Raman signal 160 in FIG. 11.

[0129] FIG. 24 illustrates an example light source 110 with multiple laser diodes 110 and an optical multiplexer 118 that combines light produced by the laser diodes into a single output beam of light 120. Each of the laser diodes 110-1, 110-2, . . . 110-N produces a respective output beam 120-1, 120-2, . . . 120-N, and the optical multiplexer 118 combines the output beams into the output beam of light 120. The light source 110 may be configured to switch between operating the N laser diodes one at a time so that, at any given time, only one laser diode produces light, and the output beam 120 includes just the light produced by that one laser diode.

[0130] The optical multiplexer 118 may be a free-space device, a fiber-optic device, a waveguide-based device, or a metamaterial-based device, and the multiplexer may combine N different wavelengths of light from the N laser diodes into a single output beam 120. The optical multiplexer 118 may include one or more of the following: a free-space diffraction grating; an arrayed waveguide grating (AWG); a metamaterial that acts as a diffractive optical element; one or more optical filters; one or more optical combiners; one or more optical switches (e.g., thermo-optic switches, liquid crystal switches, or microelectromechanical systems (MEMS) switches); a series of two or more fiber Bragg gratings with optical circulators.

[0131] The light source 110 in FIG. 24 may be a pump light source 110_{pu}, a Stokes light source 110_S, or a probe light source 110_{pr}, and the output beam 120 may be a pump beam 120_{pu}, a Stokes beam 120_S, or a probe beam 120_{pr}. The output beam 120 may be a free-space beam, or the output beam 120 may be coupled into an optical fiber or into a waveguide of a photonic integrated circuit (PIC). For example, each of the laser diodes 110-1, 110-2, . . . 110-N may produce a fiber-coupled beam 120-1, 120-2, . . . 120-N, and the optical multiplexer 118 may be a fiber-optic device that produces a fiber-coupled output beam 120. The output beam 120 may be sent to an optical combiner 130, a sample 150, or a detector 220. Alternatively, the light source 110 may include an optical amplifier (e.g., an SOA or a fiber-optic amplifier) located after the optical multiplexer 118, and the output beam 120 may be coupled from the multiplexer to an optical amplifier that provides optical amplification to the output beam.

[0132] The light source 110 in FIG. 24 includes N laser diodes 110-1, 110-2, . . . 110-N, where N is an integer greater than or equal to 2. Each of the N laser diodes 110 in FIG. 24 may be a wavelength-tunable laser diode or a fixed-wavelength laser diode. For example, the light source in FIG. 24 may include (i) N wavelength-tunable laser diodes, (ii) N fixed-wavelength laser diodes, or (iii) one or more wavelength-tunable laser diodes and one or more fixed-wavelength laser diodes. The light source 110 in FIG. 24 may be referred to as a wavelength-tunable light source, a frequency-tunable light source, or a tunable light source. A wavelength-tunable light source may include one or more continuously tunable laser diodes (e.g., SG-DBR laser diodes); multiple fixed-wavelength laser diodes (e.g., multiple DFB laser diodes), each laser diode operating at a different wavelength; or any combination thereof.

[0133] The light source 110 in FIG. 24 may be a pump light source 110_{pu} or a Stokes light source 110_S that includes N fixed-wavelength laser diodes, each laser diode having a different operating wavelength. The wavelength of the output beam 120 produced by the wavelength-tunable

light source **110** in FIG. **24** may be adjustable to any wavelength of N different wavelengths by selecting one of the N fixed-wavelength laser diodes for operation. The frequency offset Ω between the pump beam **120_{pu}** and Stokes beam **120_S** may be adjustable by selecting one of the fixed-wavelength laser diodes for operation. For example, the Stokes beams **120_S** and **120_{S'}** in FIG. **14** may be produced by the light source **110** in FIG. **24**. Laser diode **110-1** may be a fixed-wavelength laser diode operating at the frequency ν_2 , and laser diode **110-2** may be a fixed-wavelength laser diode operating at the frequency ν_2' . Selecting laser diode **110-1** for operation produces the frequency offset Ω_1 in FIG. **14**, and selecting laser diode **110-2** for operation produces the frequency offset Ω_2 .

[0134] The light source **110** in FIG. **24** may operate only one of the N laser diodes **110** at any given time. Each of the laser diodes **110** may operate at a particular wavelength or over a particular range of wavelengths, and one of the laser diodes may be selected for operation based on the wavelength that is needed to perform a particular measurement. For example, the pump beams **120_{pu}** and **120_{pu'}** in FIG. **15** may be produced by the light source **110** in FIG. **24**. Laser diode **110-1** may be a fixed-wavelength laser diode operating at the frequency ν_1 , and laser diode **110-2** may be a fixed-wavelength laser diode operating at the frequency ν_1' . During a first measurement period, laser diode **110-1** may be operated to produce output beam **120-1** at the frequency ν_1 , and the other laser diodes **110-2** to **110-N** may be turned off or otherwise configured to not produce light. The multiplexer **118** receives the output beam **120-1** from the laser diode **110-1** and directs it to the output of the multiplexer to produce the output beam **120** having a frequency ν_1 . During a second measurement period, the Raman signal **160_c** in FIG. **15** may be measured, and laser diode **110-2** may be operated to produce output beam **120-2** at the frequency ν_1' . The other laser diodes (i.e., laser diodes **110-1** to **110-N**, excluding laser diode **110-2**) may be turned off or otherwise configured to not produce light. The multiplexer **118** receives the output beam **120-2** from the laser diode **110-2** and directs it to the output of the multiplexer to produce the output beam **120** having a frequency ν_1 .

[0135] The light source **110** in FIG. **24** may be a probe light source that includes N wavelength-tunable laser diodes. The probe light source may be configured to tune over one or more wavelength ranges having a total width between $p \cdot N \cdot \Delta\lambda_{av}$ and $N \cdot \Delta\lambda_{av}$, where $\Delta\lambda_{av}$ is an average wavelength-tuning range of the N laser diodes **110**, and p is a wavelength-overlap parameter between 0.5 and 1. For example, if the overlap parameter p has a value of 0.7, then the combined wavelength-tuning range of the N wavelength-tunable laser diodes may be between $(0.7)N \cdot \Delta\lambda_{av}$ and $N \cdot \Delta\lambda_{av}$. The wavelength-overlap parameter p represents the amount of wavelength overlap between adjacent wavelength-tuning ranges (e.g., an overlap value p of 1 indicates that there is no wavelength overlap between the wavelength-tuning ranges). For example, the light source **110** in FIG. **24** may be a probe light source that includes three SG-DBR laser diodes having respective wavelength-tuning ranges of 1490-1530 nm, 1520-1560 nm, and 1550-1590 nm. Each of the SG-DBR laser diodes has a 40-nm tuning range with a 10-nm overlap between adjacent tuning ranges, which results in the light source **110** in FIG. **24** having a 100-nm wavelength-tuning range from 1490 nm to 1590 nm. In this case, the average wavelength-tuning range $\Delta\lambda_{av}$ of the three

laser diodes is 40 nm, and the total wavelength-tuning range has a width of 100 nm, which corresponds to the wavelength-overlap parameter p having a value of approximately 0.83. The probe light source **110** may produce an output beam **120** having any wavelength from 1490 nm to 1590 nm by selecting one of the three SG-DBR laser diodes for operation and tuning the laser diode to the desired wavelength. As another example, the light source **110** in FIG. **24** may be a probe light source **110** that includes three SG-DBR laser diodes having respective wavelength-tuning ranges of 1490-1530 nm, 1530-1560, and 1640-1680 nm. Each of the SG-DBR laser diodes has a 40-nm tuning range with no wavelength overlap between adjacent tuning ranges. The average wavelength-tuning range $\Delta\lambda_{av}$ of the three laser diodes is 40 nm, and the total wavelength-tuning range has a width of 120 nm. This corresponds to the wavelength-overlap parameter p having a value of 1, indicating that there is no wavelength overlap between the tuning ranges of the three laser diodes.

[0136] FIG. **25** illustrates an example pump laser **110_{pu}** and Stokes laser **110_S** with a fiber-optic combiner **130** that produces a combined pump-Stokes beam **140** coupled into an optical fiber **116**. A Raman spectroscopy system **100** may include one or more optical elements that direct the pump and Stokes beams to a sample **150**. The optical elements may include a combiner **130** that combines the pump beam **120_{pu}** and the Stokes beam **120_S** to produce a combined pump-Stokes beam **140** that is directed to a sample **150**. In FIG. **1**, the combiner **130_a** may be a free-space optical combiner, and the pump beam **120_{pu}**, Stokes beam **120_S**, and combined beam **140** may each be free-space beams. The combiner **130** in FIG. **25** is a fiber-optic combiner that receives the pump and Stokes beams via two input optical fibers **116** and combines the two beams into a combined pump-Stokes beam **140** that propagates in an output optical fiber **116**. The pump laser **110_{pu}** may be a fiber-coupled laser diode that produces a pump beam **120_{pu}** that is directed to the fiber-optic combiner **130** via an input optical fiber **116**. Similarly, the Stokes laser **110_S** may be a fiber-coupled laser diode that produces a Stokes beam **120_S** that is directed to the fiber-optic combiner **130** via another input optical fiber **116**. The pump laser **110_{pu}** or the Stokes laser **110_S** may be followed by an optical amplifier (not illustrated in FIG. **25**) that amplifies the pump beam **120_{pu}** or Stokes beam **120_S** prior to directing the light to the combiner **130**. After the fiber-optic combiner **130** combines the pump and Stokes beams, the output optical fiber **116** may direct the combined pump-Stokes beam **140** to a sample **150**.

[0137] The fiber-optic combiner **130** in FIG. **25** may include a fiber-optic wavelength division multiplexer (WDM) with two input optical fibers (for the pump and Stokes beam) and one output optical fiber for the output beam **120**. The WDM may include a dichroic beamsplitter or a fused fiber coupler. Each of the input or output optical fiber **116** may be a single-mode optical fiber or a multi-mode optical fiber.

[0138] In some embodiments, instead of using a single pump laser or a single Stokes laser (as illustrated in FIG. **25**), a Raman spectroscopy system may use a pump or Stokes light source **110** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. **24**). For example, the light source **110** in FIG. **24** may be a pump light source that produces a fiber-coupled output beam **120** that is coupled to an input fiber of the fiber-optic combiner **130** in

FIG. 25. One of the laser diodes of the pump light source may be selected for operation, and the light from the selected laser diode may be directed by the multiplexer 118 to an optical fiber that is coupled to the combiner 130. Additionally or alternatively, the light source 110 in FIG. 24 may be a Stokes light source that produces a fiber-coupled output beam 120 that is coupled to an input fiber of the fiber-optic combiner 130 in FIG. 25.

[0139] FIG. 26 illustrates an example laser diode 110 coupled to a waveguide 172 of a photonic integrated circuit (PIC) 170. A PIC 170 (which may be referred to as a planar lightwave circuit (PLC), a waveguide-based device, an integrated-optic device, an integrated optoelectronic device, or a silicon optical bench) may be fabricated from a substrate that includes silicon, indium phosphide, glass (e.g., silica), a polymer, or an electro-optic material (e.g., lithium niobate (LiNbO_3) or lithium tantalate (LiTaO_3)). A PIC 170 may include one or more optical waveguides 172 that confine and guide a beam of light. An optical waveguide 172 that is part of a PIC 170 may be a passive optical waveguide formed in the PIC, and the waveguide may convey light from one optical element to another with relatively low optical loss.

[0140] In FIG. 26, light from the laser diode 110 is coupled into the PIC waveguide 172 to produce a waveguide-coupled beam 120. The waveguide 172 may convey the beam of light 120 from the laser diode 110 to another optical element (e.g., an optical combiner 130, a sample 150, or a detector 220). Light from the laser diode 110 in FIG. 26 may be coupled into the waveguide 172 using one or more lenses, or the laser diode 110 may be butt-coupled to an input of the waveguide so that the light from the laser diode is directly coupled into the waveguide. The laser diode 110 may be mechanically attached or connected to the PIC 170 or to a substrate to which the PIC is also attached. For example, the laser diode 110 may be attached using epoxy, adhesive, or solder. Alternatively, the laser diode 110 in FIG. 26 may be located apart from the PIC 170, and the laser diode may send a beam of light 120 to the PIC 170 via optical fiber. An output end of the optical fiber may be attached or connected to the PIC 170 so that the light is coupled into the PIC waveguide 172. Light from the laser diode 110 may be amplified by an optical amplifier (not illustrated in FIG. 26) prior to being coupled into the PIC waveguide 172. For example, the laser diode 110 in FIG. 26 may be a MOPA laser similar to that illustrated in FIG. 21, and the output beam 120 produced by the MOPA laser may be directly coupled into the waveguide 172.

[0141] FIG. 27 illustrates an example pump laser 110_{pu} and Stokes laser 110_S with a photonic integrated circuit (PIC) 170 that produces a combined pump-Stokes beam 140 coupled into an optical waveguide 172 of the PIC. The PIC 170 includes a waveguide combiner 130 and three optical waveguides 172 (two input waveguides for the pump and Stokes beams and one output waveguide for the combined pump-Stokes beam 140). The waveguide combiner 130 is a waveguide-based optical combiner that combines the pump beam 120_{pu} and the Stokes beam 120_S to produce a combined pump-Stokes beam 140. The combined pump-Stokes beam 140 is coupled to an output optical waveguide 172 of the PIC 170, and the output waveguide may direct the beam to a sample 150. The pump laser 110_{pu} or the Stokes laser 110_S may be a laser diode that is mechanically attached or connected to the PIC 170 or to a substrate to which the PIC is also attached. Alternatively, the pump laser 110_{pu} or

the Stokes laser 110_S may be a fiber-coupled laser diode that sends a beam of light to the PIC via optical fiber (e.g., an output end of the optical fiber may be attached to the PIC so that the light is coupled into an input optical waveguide 172).

[0142] In some embodiments, instead of using a single pump laser or a single Stokes laser (as illustrated in FIG. 27), a Raman spectroscopy system may use a pump or Stokes light source 110 with multiple laser diodes and an optical multiplexer 118 (e.g., as illustrated in FIG. 24). For example, the light source 110 in FIG. 24 may be a pump or Stokes light source that produces a fiber-coupled output beam 120 that is coupled to an input optical waveguide 172 of the PIC 170 in FIG. 27. Alternatively, the multiplexer 118 in FIG. 24 may be a waveguide-based device and the output beam 120 may propagate in a PIC waveguide that directs the light to an input optical waveguide 172 of the PIC 170 in FIG. 27.

[0143] FIG. 28 illustrates an example fiber-optic combiner 130 that combines a Raman signal 160 with a probe beam 120_{pr}. An optical receiver 200 may include an optical combiner 130 that combines a Raman signal 160 and a probe beam of light 120_{pr} to produce one or more combined probe-Raman signals 210 that are each directed to a detector 220. In FIG. 1, the combiner 130_b may be a free-space optical combiner, and the Raman signal 160, probe beam 120_{pr}, and the combined probe-Raman signal 210 may each be free-space beams. The combiner 130 in FIG. 28 is a fiber-optic combiner that receives the Raman signal 160 and the probe beam 120_{pr} via two input optical fibers 116 and combines the two beams into a combined probe-Raman signal 210 that is directed to a detector 220 via an output optical fiber 116. Each of the input or output optical fiber 116 in FIG. 28 may be a single-mode optical fiber or a multi-mode optical fiber.

[0144] The Raman signal 160 in FIG. 28 may be a free-space beam that is coupled into an input optical fiber 116 using one or more lenses. The probe laser 110_{pr} may be a fiber-coupled laser diode that directs the probe beam 120_{pr} to the fiber-optic combiner 130 via optical fiber 116. The probe beam 120_{pr} may be amplified by an optical amplifier (not illustrated in FIG. 28) prior to being directed to the combiner 130. In some embodiments, instead of using a single probe laser 110_{pr} (as illustrated in FIG. 28), an optical receiver 200 may use a light source 110 with multiple laser diodes and an optical multiplexer 118 (e.g., as illustrated in FIG. 24). For example, the light source 110 in FIG. 24 may be a probe light source that produces a fiber-coupled output beam 120 that is coupled to an input optical fiber 116 of the fiber-optic combiner 130 in FIG. 28.

[0145] FIG. 29 illustrates an example photonic integrated circuit (PIC) 170 with a waveguide combiner 130 that combines a Raman signal 160 with a probe beam 120_{pr}. An optical combiner 130 may be a waveguide combiner 130 that is part of a PIC 170 and may combine a Raman signal 160 and a probe beam of light 120_{pr} to produce one or more combined probe-Raman signals 210 that are each directed to a detector 220. The waveguide combiner 130 in FIG. 29 receives the Raman signal 160 and the probe beam 120_{pr} via two input waveguides 172. A waveguide combiner 130 may produce 1, 2, or 4 combined output beams 210. The waveguide combiner 130 in FIG. 29 combines the Raman signal 160 and the probe beam 120_{pr} to produce two combined probe-Raman signals 210_a and 210_b which are each directed

to a respective detector **220a** and **220b** via two output waveguides **172**. The PIC **170** in FIG. **29** may be part of an optical receiver **200**.

[0146] The Raman signal **160** in FIG. **29** may be a free-space beam that is coupled into an input waveguide **172** using one or more lenses. The probe laser **110pr** may be a laser diode that is mechanically attached or connected to the PIC **170** or to a substrate to which the PIC is also attached. Alternatively, the probe laser **110pr** may be a fiber-coupled laser diode that sends the probe beam **120pr** to the PIC **170** via optical fiber (e.g., an output end of the optical fiber may be attached to the PIC so that the light is coupled into an input optical waveguide **172**). The probe beam **120pr** may be amplified by an optical amplifier (not illustrated in FIG. **29**) prior to being coupled into an input optical waveguide **172**. In some embodiments, instead of using a single probe laser **110pr** (as illustrated in FIG. **29**), an optical receiver **200** may use a light source **110** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. **24**), and the probe beam **120pr** may be delivered from the multiplexer to the PIC **170** via optical fiber or via a waveguide **172** of the PIC.

[0147] A Raman spectroscopy system **100** may include one or more optical elements that (i) direct a pump beam **120pu** and a Stokes beam **120S** to a sample **150** and (ii) direct a Raman signal **160** and a probe beam **120pr** to one or more detectors **220**. The optical elements may include one or more PICs **170** that each include one or more optical waveguides **172**. One or more of the optical waveguides **172** may direct the pump beam **120** and the Stokes beam **120s** to the sample **150**. For example, a PIC **170** may include an optical combiner **130** that produces a combined pump-Stokes beam **140** that is directed to the sample by an optical waveguide **172** of the PIC **170**. One or more other optical waveguides **172** may direct the Raman signal **160** and the probe beam **120pr** to one or more detectors **220**. For example, a PIC **170** may include an optical combiner **130** that combines the Raman signal **160** and the probe beam **120pr** to produce one or more combined probe-Raman signals **210** that are each directed to a detector **220** by an optical waveguide **172** of the PIC **170**.

[0148] FIGS. **30-35** each illustrate example frequency ranges of a pump beam **120pu** and a Stokes beam **120S**. The pump beam **120pu** may be produced by a pump light source **110pu**, and the Stokes beam **120S** may be produced by a Stokes light source **110S**. The pump and Stokes light sources may each include one or more fixed wavelength laser diodes or one or more wavelength-tunable laser diodes. In each of FIGS. **30-35**, the pump beam **120pu** and the Stokes beam **120S** each have one or more fixed frequencies or one or more frequencies that are adjustable over a particular frequency range. The corresponding frequency offset Ω between the pump and Stokes beams is indicated as a range of frequencies or a set of discrete frequencies that the frequency offset can be set to, based on the frequencies that are available to the pump and Stokes beams. The frequency offset Ω is determined from $\nu_1 - \nu_2$, where ν_1 is the range or set of fixed frequencies for the pump beam **120pu**, and ν_2 is the range or set of fixed frequencies for the Stokes beam **120S**. The frequency offsets Ω_1 , Ω_2 , Ω_3 , and Ω_4 in FIGS. **30-35** may have any suitable value between approximately 5 THz and approximately 100 THz. The frequency range $\Delta\Omega$

over which a frequency offset may be varied may have any suitable value between approximately 5 THz and approximately 80 THz.

[0149] In FIG. **30**, the pump beam **120pu** has a single fixed frequency ν_1 , and the Stokes beam **120S** has a frequency ν_2 that is adjustable over a frequency range of width $\Delta\nu_2$. The adjustable frequency range $\Delta\nu_2$ of the Stokes beam **120S** extends from a low frequency ν_{2L} to a high frequency ν_{2H} , where $\Delta\nu_2 = \nu_{2H} - \nu_{2L}$. The pump laser **110pu** that produces the pump beam **120pu** in FIG. **30** may be a fixed-wavelength laser diode, and the Stokes laser **110S** that produces the Stokes beam **120S** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between Ω_1 and Ω_2 , and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_2 - \Omega_1$. In FIG. **30**, the frequency range $\Delta\Omega$ is also equal to the frequency range $\Delta\nu_2$ of the Stokes beam **120S**. When the Stokes beam **120S** is set to the lower frequency ν_{2L} , the frequency offset between the pump and Stokes beams is $\nu_1 - \nu_{2L}$, which is equal to Ω_2 . Similarly, when the Stokes beam **120S** is set to the upper frequency ν_{2H} , the frequency offset between the pump and Stokes beams is $\nu_1 - \nu_{2H}$, which is equal to Ω_1 .

[0150] For example, the pump beam **120pu** in FIG. **30** may have a frequency ν_1 of 250 THz (corresponding to a wavelength of approximately 1200 nm), and the Stokes beam **120S** may be adjustable from a low frequency ν_{2L} of 195 THz to a high frequency ν_{2H} of 200 THz. This corresponds to a frequency-tuning range $\Delta\nu_2$ of the Stokes beam **120S** of 5 THz (167 cm^{-1} in wavenumbers) and a 38-nm wavelength-tuning range from approximately 1499 nm to approximately 1537 nm. The resulting frequency offset Ω between the pump and Stokes beams may be set to a value between the lower value Ω_1 of 50 THz (1668 cm^{-1} in wavenumbers) and the upper value Ω_2 of 55 THz (1835 cm^{-1} in wavenumbers), corresponding to a frequency range $\Delta\Omega$ of 5 THz (167 cm^{-1} in wavenumbers).

[0151] In FIG. **31**, the pump beam **120pu** has a single fixed frequency ν_1 , and the Stokes beam **120S** has a frequency ν_2 that is adjustable over a frequency range of width $\Delta\nu_2$. The adjustable frequency range $\Delta\nu_2$ of the Stokes beam **120S** extends from a low frequency ν_{2L} to a high frequency ν_{2H} , where $\Delta\nu_2 = \nu_{2H} - \nu_{2L}$. The pump laser **110pu** that produces the pump beam **120pu** in FIG. **31** may be a fixed-wavelength laser diode. The Stokes light source **110S** that produces the Stokes beam **120S** may include two wavelength-tunable laser diodes. For example, the Stokes light source **110S** may be similar to the light source in FIG. **24** where two wavelength-tunable laser diodes are combined by a multiplexer **118**. A first wavelength-tunable laser diode may operate from frequency ν_{2L} to frequency ν_{2M} , and a second wavelength-tunable laser diode may operate from frequency ν_{2M} to frequency ν_{2H} . The total tuning range $\Delta\nu_2$ of the Stokes laser **110S** equals the sum of the tuning ranges $\Delta\nu_{2a}$ and $\Delta\nu_{2b}$ of the two wavelength-tunable laser diodes. In other embodiments, if the tuning ranges of the two wavelength-tunable laser diodes overlap, the total tuning range $\Delta\nu_2$ of the Stokes light source **110S** will be reduced by the amount of frequency overlap between the two lasers.

[0152] In FIG. **31**, the frequency offset Ω between the pump and Stokes beams may be set to any value between Ω_1 and Ω_2 , and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_2 - \Omega_1$. When the first wavelength-tunable laser diode (with a frequency range from ν_{2L} to ν_{2M}) is selected to

operate, the frequency offset Ω may be set to any value between Ω_M and Ω_2 . For example, when the Stokes beam **120S** is set to the lower frequency ν_{2L} , the frequency offset between the pump and Stokes beams is $\nu_1 - \nu_{2L}$, which is equal to Ω_2 . When the second wavelength-tunable laser diode (with a frequency range from ν_{2M} to ν_{2H}) is selected to operate, the frequency offset Ω may be set to any value between Ω_1 and Ω_M . For example, when the Stokes beam **120S** is set to the upper frequency ν_{2H} , the frequency offset between the pump and Stokes beams is $\nu_1 - \nu_{2H}$, which is equal to Ω_1 . The total frequency range $\Delta\Omega$ of the frequency offset Ω is equal to the sum of the two frequency ranges $\Delta\Omega_{2a}$ and $\Delta\Omega_{2b}$. The frequency range $\Delta\Omega$ is also equal to the overall frequency range $\Delta\nu_2$ of the Stokes beam **120S**.

[0153] In FIG. 32, the pump beam **120pu** can be set to two fixed frequencies via and ν_{1b} , and the Stokes beam **120S** has a frequency ν_2 that is adjustable over a frequency range of width $\Delta\nu_2$. The adjustable frequency range $\Delta\nu_2$ of the Stokes beam **120S** extends from a low frequency ν_{2L} to a high frequency ν_{2H} , where $\Delta\nu_2 = \nu_{2H} - \nu_{2L}$. The pump light source **110pu** that produces the two pump beams **120pu-a** and **120pu-b** may include two fixed-wavelength laser diodes. For example, the pump light source **110pu** may be similar to the light source in FIG. 24 where two fixed-wavelength laser diodes are combined by a multiplexer **118**. The Stokes laser **110S** that produces the Stokes beam **120S** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between Ω_1 and Ω_2 , and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_2 - \Omega_1$. When the pump laser **110pu** produces the pump beam **120pu-a** at frequency via, the Stokes beam **120S** may be tuned to a frequency between ν_{2L} and ν_{2H} to produce a frequency offset Ω between Ω_1 and Ω_M . For example, with the Stokes beam **120S** set to the upper frequency ν_{2H} , the frequency offset between the pump and Stokes beams is $\nu_{1a} - \nu_{2H}$, which is equal to Ω_1 . When the pump laser **110pu** produces the pump beam **120pu-b** at frequency ν_{1b} , the Stokes beam **120S** may be tuned to a frequency between ν_{2L} and ν_{2H} to produce a frequency offset Ω between Ω_M and Ω_2 . For example, with the Stokes beam **120S** set to the lower frequency ν_{2L} , the frequency offset between the pump and Stokes beams is $\nu_{1b} - \nu_{2L}$, which is equal to Ω_2 .

[0154] FIG. 33 is similar to FIG. 30, except in FIG. 33, the Stokes beam **120S** has a fixed frequency ν_2 and the frequency of the pump beam **120pu** is adjustable. The adjustable frequency range $\Delta\nu_1$ of the pump beam **120pu** extends from a low frequency ν_{1L} to a high frequency ν_{1H} , where $\Delta\nu_1 = \nu_{1H} - \nu_{1L}$. The Stokes laser **110S** that produces the Stokes beam **120pu** in FIG. 33 may be a fixed-wavelength laser diode, and the pump laser **110pu** that produces the pump beam **120pu** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between Ω_1 and Ω_2 , and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_2 - \Omega_1$. In FIG. 33, the frequency range $\Delta\Omega$ is also equal to the frequency range $\Delta\nu_1$ of the pump beam **120pu**. When the pump beam **120pu** is set to the lower frequency ν_{1L} , the frequency offset between the pump and Stokes beams is $\nu_{1L} - \nu_2$, which is equal to Ω_1 . Similarly, when the pump beam **120pu** is set to the upper frequency ν_{1H} , the frequency offset between the pump and Stokes beams is $\nu_{1H} - \nu_2$, which is equal to Ω_2 .

[0155] In FIG. 34, both the pump beam **120pu** and the Stokes beam **120S** have adjustable frequencies. The pump

laser **110pu** that produces the pump beam **120pu** and the Stokes laser **110S** that produces the Stokes beam **120S** may each include a wavelength-tunable laser diode. The adjustable frequency range $\Delta\nu_1$ of the pump beam **120pu** extends from a low frequency ν_{1L} to a high frequency ν_{1H} , where $\Delta\nu_1 = \nu_{1H} - \nu_{1L}$. The adjustable frequency range $\Delta\nu_2$ of the Stokes beam **120S** extends from a low frequency ν_{2L} to a high frequency ν_{2H} , where $\Delta\nu_2 = \nu_{2H} - \nu_{2L}$. The frequency offset Ω between the pump and Stokes beams may be set to any value between Ω_1 and Ω_2 , and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_2 - \Omega_1$. When the Stokes beam **120S** is set to the lower Stokes frequency ν_{2L} and the pump beam **120pu** is set to the upper pump frequency ν_{1H} , the frequency offset between the pump and Stokes beams is $\nu_{1H} - \nu_{2L}$, which is equal to Ω_2 . When the Stokes beam **120S** is set to the upper Stokes frequency ν_{2H} and the pump beam **120pu** is set to the lower pump frequency ν_{1L} , the frequency offset between the pump and Stokes beams is $\nu_{1L} - \nu_{2H}$, which is equal to Ω_1 .

[0156] In FIG. 35, both the pump beam **120pu** and the Stokes beam **120S** can be set to two different fixed frequencies. A pump light source **110pu** and a Stokes light source **110S** may each include two or more fixed wavelength laser diodes, and each light source may be similar to the light source in FIG. 24 where multiple fixed-wavelength laser diodes are combined by a multiplexer **118**. In FIG. 35, the pump light source **110pu** that produces the two pump beams **120pu-a** and **120pu-b** may include two fixed-wavelength laser diodes, and the Stokes light source **110S** that produces the two Stokes beams **120S-a** and **120S-b** may include two fixed-wavelength laser diodes. The frequencies of the pump and Stokes beams may be selected to produce four different frequency offsets Ω_1 , Ω_2 , Ω_3 , and Ω_4 . For example, selecting Stokes beam **120S-b** at frequency ν_{2b} and pump beam **120pu-a** at frequency via produces a frequency offset Ω_1 , which is equal to $\nu_{1a} - \nu_{2b}$. As another example, selecting Stokes beam **120S-a** at frequency ν_{2a} and pump beam **120pu-b** at frequency ν_{1b} produces a frequency offset Ω_4 , which is equal to $\nu_{1b} - \nu_{2a}$. The frequency offset Ω_2 may be produced by selecting Stokes beam **120S-a** at frequency ν_{2a} and pump beam **120pu-a** at frequency via, and the frequency offset Ω_3 may be produced by selecting Stokes beam **120S-b** at frequency ν_{2b} and pump beam **120pu-b** at frequency ν_{1b} .

[0157] FIG. 36 illustrates an example optical receiver **200** with two detectors **220a** and **220b**. The optical receivers **200** in FIGS. 29 and 36 are similar, except one difference is that the optical receiver in FIG. 29 is a waveguide-based optical receiver, while the optical receiver in FIG. 36 is a free-space optical receiver **200**. The optical combiner **130** in FIG. 36 may be a 50/50 free-space beamsplitter that reflects approximately 50% of an incident beam of light and transmits approximately 50% of the beam. The optical combiner **130** splits the Raman signal **160** and the probe beam **120pr** into two beams to produce two combined probe-Raman signals **210a** and **210b**. The combined probe-Raman signal **210a** is directed to detector **220a** and includes a transmitted portion of the probe beam **120pr** and a reflected portion of the Raman signal **160** (e.g., approximately 50% of the probe beam and approximately 50% of the Raman signal). Similarly, the combined probe-Raman signal **210b** is directed to detector **220b** and includes a reflected portion of the probe beam **120pr** and a transmitted portion of the Raman signal **160**. The portions of the probe beam **120pr** and the Raman signal **160** that make up the combined probe-Raman signal

210a may be coherently mixed at detector **220a** to produce the photocurrent signal i_a . Similarly, the portions of the probe beam **120pr** and the Raman signal **160** that make up the combined probe-Raman signal **210b** may be coherently mixed at detector **220b** to produce the photocurrent signal i_b .

[0158] The two detectors **220a** and **220b** are arranged so that their respective photocurrents i_a and i_b are subtracted. The anode of detector **220a** is electrically connected to the cathode of detector **220b**, and the subtracted photocurrent signal $i_a - i_b$ from the anode-cathode connection is sent to the detection electronics **230**, which produces a digital output signal **240** that corresponds to the subtracted photocurrent signal. The subtracted photocurrent signal may be expressed as $i_a - i_b = 2E_R E_{pr} \cos[2\pi(v_R - v_3)t + \Delta\phi]$, which corresponds to the coherent-mixing term discussed herein. The subtracted photocurrent signal does not include the terms E_R^2 and E_{pr}^2 corresponding to the respective optical powers of the Raman signal **160** and the probe beam **120pr**. By subtracting the two photocurrents i_a and i_b the common-mode terms E_R^2 and E_{pr}^2 (as well as common-mode noise) that appear in each of the photocurrent signals i_a and i_b are substantially removed, leaving the coherent-mixing term, which is the quantity of interest. Since subtraction may remove common-mode noise, the subtracted photocurrent signal $i_a - i_b$ may have a reduced noise compared to each of the photocurrent signals i_a and i_b alone. The dual-detector arrangement in FIG. 36 in which the photocurrents are subtracted may be referred to as a balanced optical detector. A balanced detector may be implemented as a free-space device, a fiber-optic-based device, or a waveguide-based device.

[0159] FIG. 37 illustrates an example optical receiver **200** configured for polarization-sensitive detection of a Raman signal **160**. A polarization-sensitive optical receiver **200** may be used to determine the polarization of a Raman signal **160**. The polarization of a Raman signal **160** may be determined by a processor based on one or more digital output signals **240** produced by the polarization-sensitive optical receiver **200**. Determining the polarization of a Raman signal **160** may include determining a relative size or ratio of two orthogonal polarization components of the Raman signal (e.g., horizontal and vertical polarization components of the Raman signal). For example, if the ratio of the horizontal and vertical polarization components of a Raman signal **160** is 100:1, then the Raman signal may be determined to be substantially horizontally polarized. As another example, if the ratio of the horizontal and vertical polarization components of a Raman signal **160** is 1:1, then the horizontal and vertical polarization components of the Raman signal may be determined to be approximately equal (e.g., the Raman signal may be circularly polarized or linearly polarized at a 45-degree angle to the horizontal and vertical directions).

[0160] A polarization-sensitive optical receiver **200** may include a polarization beamsplitter (PBS) **135** that splits an input beam into two output beams, where one output beam is horizontally polarized, and the other output beam is vertically polarized. The horizontally polarized output beam includes the horizontal polarization component of the input beam, and the vertically polarized output beam includes the vertical polarization component of the input beam. The Raman signal **160** in FIG. 37 is directed to a Raman-signal PBS **135R** that splits the Raman signal into a horizontal-polarization Raman signal **160-h** and a vertical-polarization Raman signal **160-v**. Similarly, the probe beam **120pr** is directed to a probe-beam PBS **135pr** that splits the probe

beam into a horizontal-polarization probe beam **120pr-h** and a vertical-polarization probe beam **120pr-v**.

[0161] A polarization-sensitive optical receiver **200** may include a waveplate **132** that changes the polarization of the probe beam **120pr** so that the probe beam is split into two polarization components. The two polarization components may each have approximately one-half the power of the probe beam **120pr**. The waveplate **132c** in FIG. 37 may be (i) a half-wave plate that rotates the polarization of the probe beam **120pr** or (ii) a quarter-wave plate that converts the probe beam **120pr** to a circular or elliptical polarization. For example, the probe laser **110pr** may produce a probe beam **120pr** that is vertically polarized, and the waveplate **132c** may be a half-wave plate that rotates the probe-beam polarization by 45 degrees so that the horizontal-polarization and vertical-polarization probe beams **120pr-h** and **120pr-v** each have approximately equal optical powers.

[0162] The optical combiner **130h** in FIG. 37 combines the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** to produce a horizontal probe-Raman signal **210h** that is directed to a horizontal-polarization optical receiver **200h**. Similarly, the optical combiner **130v** combines the vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-v** to produce a vertical probe-Raman signal **210v** that is directed to a vertical-polarization optical receiver **200v**. The horizontal-polarization optical receiver **200h** may include one or more optical detectors **200**, where each detector is configured to coherently mix at least a portion of the horizontal-polarization Raman signal **160-h** and at least a portion of the horizontal-polarization probe beam **120pr-h** to produce a horizontal-polarization electronic signal. Similarly, the vertical-polarization optical receiver **200v** may include one or more optical detectors **200**, where each detector is configured to coherently mix at least a portion of the vertical-polarization Raman signal **160-v** and at least a portion of the vertical-polarization probe beam **120pr-v** to produce a vertical-polarization electronic signal. The horizontal-polarization optical receiver **200h** and the vertical-polarization optical receiver **200v** may each include: (i) a single detector **220** (e.g., similar to the optical receiver **200** in FIG. 2), (ii) two detectors **220** (e.g., similar to the balanced optical detector arrangement in FIG. 36), or (iii) four detectors **220** (e.g., similar to the arrangement in FIG. 39). The electronic signals may include a photocurrent signal i , and each optical receiver may include an electronic amplifier **232** that produces a corresponding voltage signal **234**. The h-polarization optical receiver **200h** may produce a digital output signal **240-h** corresponding to the horizontal-polarization electronic signal, and the v-polarization optical receiver **200v** may produce a digital output signal **240-v** corresponding to the vertical-polarization electronic signal.

[0163] A processor may determine one or more characteristics of the horizontal-polarization and vertical-polarization electronic signals based on the digital output signals **240-h** and **240-v**. Additionally, a processor may determine a polarization of the Raman signal **160** based on the characteristics of the horizontal-polarization and vertical-polarization electronic signals. For example, the characteristics of the electronic signals may include an amplitude or an area associated with the electronic signals, and the polarization of the Raman signal **160** may be expressed as a relative size or ratio of the amplitudes or areas associated with the horizontal and vertical polarization components of the Raman

signal. If the horizontal digital output signal **240-h** includes an amplitude characteristic with value 100 and the vertical digital output signal **240-v** includes a corresponding amplitude characteristic with value 1, then the Raman signal **160** may be determined to be substantially horizontally polarized. If the horizontal and vertical digital output signals each include amplitude characteristics having approximately equal values, then the Raman signal **160** may be determined to have approximately equal horizontal and vertical polarization components.

[0164] A polarization-sensitive optical receiver **200** as illustrated in FIG. 37 may be implemented with free-space optical elements, fiber-optic components, waveguide-based optical elements, a metamaterial-based device, or any suitable combination thereof. For example, the two PBSs **135** in FIG. 37 may be free-space polarization beamsplitter cubes, and the Raman signal **160** and the probe beam **120pr** may be free-space optical beams. Alternatively, the two PBSs **135** may be fiber-optic components, and the Raman signal **160** and the probe beam **120pr** may be conveyed to the PBSs **135** via optical fiber (e.g., single-mode optical fiber or polarization-maintaining optical fiber). Additionally, the horizontally and vertically polarized probe-Raman signals **210h** and **210v** may be conveyed to the respective h-polarization and v-polarization optical receivers via polarization-maintaining optical fiber. The h-polarization and v-polarization optical receivers may each preserve the polarization of the respective horizontally and vertically polarized probe-Raman signals. For example, the h-polarization and v-polarization optical receivers may each include polarization-maintaining optical fiber that maintains the polarization of the beams. Alternatively, the h-polarization and v-polarization receivers may each include a PIC with optical waveguides configured to maintain the polarization of the beams.

[0165] FIG. 38 illustrates an example optical receiver **200** configured to detect in-phase and quadrature components of a Raman signal **160**. The optical receiver **200** includes a 90-degree optical hybrid **250** and four detectors **220I+**, **220I-**, **220Q+**, and **220Q-**. A 90-degree optical hybrid **250** is an optical-combiner component with two input ports and four output ports. Input light received at each of the two input ports is split, combined, and directed to each of the four output ports, and a 90-degree phase shift is imparted to one of the split beams before the Raman signal **160** and probe beam **120pr** are combined. The 90-degree optical hybrid **250** in FIG. 38 combines a Raman signal **160** and a probe beam **120pr** to produce four combined output beams: two in-phase combined beams **210I+** and **210I-**, and two quadrature combined beams **210Q+** and **210Q-**. Each of the four combined beams **210** may include a portion of the Raman signal **160** and a portion of the probe beam **120pr**, and each of the combined beams is directed to one of the four detectors of the optical receiver **200**. In FIG. 38, each of the four detectors produces a photocurrent signal that corresponds to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120pr**.

[0166] A 90-degree optical hybrid **250** may be configured so that the combined beams directed to each of the output ports have approximately the same optical power or energy. For example, the 90-degree optical hybrid **250** in FIG. 38 may split the Raman signal **160** into four approximately equal portions and direct each of the Raman-signal portions to one of the detectors. Similarly, the probe beam **120pr** may be split into four approximately equal portions directed to

each of the four detectors. In the example of FIG. 38, the combined beam **210I+**, which is directed to detector **220I+**, may include approximately one-quarter of the power of the Raman signal **160** and approximately one-quarter of the power of the probe beam **120pr**. Similarly, each of the three other combined beams (**210I-**, **210Q+**, **210Q-**) in FIG. 38 may also include approximately one-quarter of the Raman signal **160** and approximately one-quarter of the probe beam **120pr**.

[0167] A 90-degree optical hybrid **250** may be implemented as a waveguide-based device in a PIC. The 90-degree optical hybrid **250** in FIG. 38 is a waveguide-based optical device that includes two waveguide-based optical splitters (**252a**, **252b**) and two waveguide-based optical combiners (**130I**, **130Q**). Splitter **252a** may split the Raman signal **160** into two portions having substantially equal optical power, a first portion directed to combiner **130I** and a second portion directed to combiner **130Q**. Similarly, splitter **252b** may split the probe beam **120pr** into two portions having substantially equal power, a first portion directed to combiner **130I** and a second portion directed to combiner **130Q**. Each optical combiner **130** combines a portion of the Raman signal **160** with a portion of the probe beam **120pr**, and the combined portions are split into a first combined beam (e.g., combined beam **210I+**) and a second combined beam (e.g., combined beam **210I-**). The combined beam **210I+** is directed to detector **220I+** and includes portions of the Raman signal **160** and the probe beam **120pr** (e.g., approximately 25% of the Raman signal **160** and approximately 25% of the probe beam **120pr**). The combined beam **210I-** is directed to detector **220I-** and may include approximately 25% of the Raman signal **160** and approximately 25% of the probe beam **120pr**.

[0168] In other embodiments, all or part of a 90-degree optical hybrid **250** may be implemented as a free-space optical device. For example, a free-space 90-degree optical hybrid **250** may include one or more free-space beamsplitters or combiners that receive the Raman signal **160** and probe beam **120pr** as free-space beams and produce four free-space combined beams (**210I+**, **210I-**, **210Q+**, **210Q-**). Alternatively, all or part of a 90-degree optical hybrid **250** may be implemented as a fiber-optic device. For example, a 90-degree optical hybrid **250** may be contained in a package with two input optical fibers that direct the Raman signal **160** and probe beam **120pr** into the package and four output optical fibers that direct the four combined beams to four respective detectors.

[0169] A 90-degree optical hybrid **250** may include an optical phase shifter **254** that imparts a 90-degree phase change ($\Delta\Phi$) to a portion of the probe beam **120pr** or to a portion of the Raman signal **160**. The phase shifter **254** may apply the 90-degree phase change after a beam of light is split by an optical splitter **252** and prior to combining the Raman signal with the probe beam at an optical combiner **130**. For example, a splitter **252a** may split the Raman signal **160** into two portions, and a phase shifter **254** may impart a 90-degree phase change to one portion of the Raman signal with respect to the other portion, after which the two portions are sent to two different optical combiners. As another example, a splitter **252b** may split the probe beam **120pr** into two portions, and a phase shifter **254** may impart a 90-degree phase change to one portion of the probe beam with respect to the other portion. In FIG. 38, the phase shifter is located after the splitter **252b** and before the combiner

130Q. The splitter **252b** splits the probe beam **120pr** into two portions, and the phase shifter **254** imparts a 90-degree phase change to the probe-beam portion directed to combiner **130Q**. The other portion of the probe beam **120pr** directed to combiner **130I** does not pass through the phase shifter **254** and does not receive a phase shift from the phase shifter **254**.

[0170] An optical phase shifter **254** may be implemented as a part of a waveguide-based 90-degree optical hybrid **250**. For example, a phase shifter **254** may be implemented as part of an optical waveguide that only one portion of the probe beam **120pr** propagates through. That part of the optical waveguide may be temperature controlled to adjust the refractive index of the waveguide and produce a relative phase delay of approximately 90 degrees between two portions of the probe beam **120pr**. Additionally or alternatively, the 90-degree optical hybrid **250** as a whole may be temperature controlled to set and maintain a 90-degree phase delay. As another example, a phase shifter **254** may be implemented by applying an external electric field to part of an optical waveguide to change the refractive index of the waveguide and produce a 90-degree phase delay. In other embodiments, a phase shifter **254** may be implemented as a part of a free-space or fiber-coupled 90-degree optical hybrid **250**. For example, the input and output beams in a free-space 90-degree optical hybrid **250** may be reflected by or transmitted through the optical surfaces of a free-space optical hybrid **250** so that a relative phase shift of 90 degrees is imparted to one portion of the probe beam **120pr** with respect to another portion of the probe beam.

[0171] In FIG. 38, each of the four detectors produces a photocurrent signal that corresponds to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120pr**. The photocurrents are subtracted in a manner similar to that illustrated in FIG. 36. The photocurrents i_{I+} and i_{I-} from detectors **220I+** and **220I-** are subtracted to produce the subtracted in-phase photocurrent signal i_I which is equal to $i_{I+} - i_{I-}$. Similarly, the photocurrents i_{Q+} and i_{Q-} from detectors **220Q+** and **220Q-** are subtracted to produce the subtracted quadrature photocurrent signal to which is equal to $i_{Q+} - i_{Q-}$. Each of the subtracted photocurrent signals represents a coherent-mixing term corresponding to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120pr**. The two subtracted photocurrent signals i_I and i_Q are similar, except the in-phase photocurrent signal i_I includes a cosine function, while the quadrature photocurrent signal i_Q includes a sine function. This difference between the two subtracted photocurrent signals arises from the 90-degree phase shift provided by the phase shifter **254**. Because a 90-degree phase shift is imparted to the probe beam **120pr** directed to the combiner **130Q**, the subtracted quadrature photocurrent signal i_Q includes a sine function (which has a 90-degree phase offset with respect to a cosine function).

[0172] Each of the subtracted photocurrent signals i_I and i_Q may be sent to detection electronics **230** that produce voltage signals and digital output signals corresponding to the subtracted photocurrent signals. Based on the digital output signals (which result from the four photocurrent signals i_{I+} , i_{I-} , i_{Q+} , and i_{Q-}), a processor may determine an in-phase portion IP associated with the Raman signal **160** and a quadrature portion Q associated with the Raman signal. Additionally or alternatively, the processor may determine a phase associated with the Raman signal **160**. For

example, the processor may determine a phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr**. A phase difference may be referred to as a phase offset or a relative phase between the Raman signal **160** and the probe beam **120pr**.

[0173] The in-phase portion IP associated with the Raman signal **160** may be determined from a characteristic (e.g., an amplitude or an area) of an electronic signal associated with the in-phase photocurrent signal i_I , and the quadrature portion Q may be determined from a characteristic associated with the quadrature photocurrent signal i_Q . The in-phase portion IP may correspond to an amount of the Raman signal **160** that is in-phase with the probe beam **120pr**, and the quadrature portion Q may represent an amount of the Raman signal that is out of phase (i.e., 90-degrees phase-shifted) with the probe beam. For example, the in-phase portion IP and the quadrature portion Q, may each have values from -1 to 1. If the Raman signal **160** is in-phase with the probe beam **120pr**, then the in-phase portion IP may have a value of approximately 1, and the quadrature portion Q may have a value of approximately 0. Similarly, if the Raman signal **160** is out of phase by +90 degrees with respect to the probe beam **120pr**, then the in-phase portion IP may have a value of 0, and the quadrature portion Q may have a value of +1. The phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr** may be determined from the expression $\Delta\phi = \arctan(Q/IP)$. For example, if Q is 0 and IP is 1, then the Raman signal and the probe beam are substantially in phase, with a phase difference $\Delta\phi$ of 0 degrees. As another example, if Q is 1 and IP is 0, then the Raman signal and the probe beam are substantially out of phase, with a phase difference $\Delta\phi$ of 90 degrees.

[0174] FIG. 39 illustrates an example optical receiver **200** configured to detect polarization as well as in-phase and quadrature components of a Raman signal **160**. The optical receiver **200** in FIG. 39 is similar the optical receiver **200** in FIG. 37, where the horizontal-polarization optical receiver **200h** and the vertical-polarization optical receiver **200v** each includes a 90-degree optical hybrid **250**. Each of the 90-degree optical hybrids in FIG. 39 may be similar to the 90-degree optical hybrid **250** in FIG. 38. In FIG. 39, the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** are directed to a horizontal-polarization optical receiver **200h** that includes a 90-degree optical hybrid **250h** and four detectors **220h-I+**, **220h-I-**, **220h-Q+**, and **220h-Q-**. The vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-v** are directed to a vertical-polarization optical receiver **200v** that includes a 90-degree optical hybrid **250v** and four detectors **220v-I+**, **220v-I-**, **220v-Q+**, and **220v-Q-**. The h-polarization optical receiver **200h** may be used to determine the relative size of the horizontal-polarization Raman signal **160-h** as well as the in-phase and quadrature components of the horizontal-polarization Raman signal **160-h**. Similarly, the v-polarization optical receiver **200v** may be used to determine the relative size of the vertical-polarization Raman signal **160-v** as well as the in-phase and quadrature components of the vertical-polarization Raman signal **160-v**.

[0175] The 90-degree optical hybrid **250h** in FIG. 39 combines the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** to produce four horizontally polarized combined output beams:

two in-phase combined beams **210h-I+** and **210h-I-**, and two quadrature combined beams **210h-Q+** and **210h-Q-**. Each of the four horizontally polarized combined beams includes a portion of the horizontal-polarization Raman signal **160-h** and a portion of the horizontal-polarization probe beam **120pr-h**. The four combined beams are directed to four respective detectors (**220h-I+**, **220h-I-**, **220h-Q+**, **220h-Q-**), and each detector produces a respective photocurrent signal (i_{h-I+} , i_{h-I-} , i_{h-Q+} , i_{h-Q-}) that corresponds to the coherent mixing of a portion of the horizontal Raman signal **160-h** and a portion of the horizontal probe beam **120pr-h**. Each of the four photocurrents i_{h-I+} , i_{h-I-} , i_{h-Q+} , and i_{h-Q-} may be referred to as a horizontal-polarization electronic signal. The photocurrents from the detectors are subtracted to produce a subtracted horizontal in-phase photocurrent signal i_{h-I} which is equal to $i_{h-I+} - i_{h-I-}$ and a subtracted horizontal quadrature photocurrent signal i_{h-Q} which is equal to $i_{h-Q+} - i_{h-Q-}$.

[0176] The 90-degree optical hybrid **250v** in FIG. 39 combines the vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-v** to produce four vertically polarized combined output beams: two in-phase combined beams **210v-I+** and **210v-I-**, and two quadrature combined beams **210v-Q+** and **210v-Q-**. Each of the four vertically polarized combined beams includes a portion of the vertical-polarization Raman signal **160-v** and a portion of the vertical-polarization probe beam **120pr-v**. The four combined beams are directed to four respective detectors (**220v-I+**, **220v-I-**, **220v-Q+**, **220v-Q-**), and each detector produces a respective photocurrent signal (i_{v-I+} , i_{v-I-} , i_{v-Q+} , i_{v-Q-}) that corresponds to the coherent mixing of a portion of the vertical Raman signal **160-v** and a portion of the vertical probe beam **120pr-v**. Each of the four photocurrents i_{v-I+} , i_{v-I-} , i_{v-Q+} , and i_{v-Q-} may be referred to as a vertical-polarization electronic signal. The photocurrents from the detectors are subtracted to produce a subtracted vertical in-phase photocurrent signal i_{v-I} which is equal to $i_{v-I+} - i_{v-I-}$ and a subtracted vertical quadrature photocurrent signal i_{v-Q} which is equal to $i_{v-Q+} - i_{v-Q-}$.

[0177] Each of the subtracted photocurrent signals i_{h-I} , i_{h-Q} , i_{v-I} , and i_{v-Q} may be sent to detection electronics **230** that produces voltage signals and digital output signals corresponding to the subtracted photocurrent signals. Based on the digital output signals (which are determined from the four horizontal-polarization electronic signals and the four vertical-polarization electronic signals), a processor may determine (i) the polarization of the Raman signal **160** and (ii) a phase associated with the Raman signal (e.g., a phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr**). Determining the polarization of a Raman signal **160** may include determining a relative size or ratio of the horizontal and vertical polarization components of the Raman signal. For example, the relative size of the horizontal polarization component of the Raman signal **160** may be determined by adding characteristics (e.g., areas or amplitudes) associated with the two horizontal photocurrent signals i_{h-I} and i_{h-Q} . Similarly, the relative size of the vertical polarization component of the Raman signal **160** may be determined by adding characteristics associated with the two vertical photocurrent signals i_{v-I} and i_{v-Q} . As an example, if the relative size of the horizontal polarization component of the Raman signal **160** is 1 and the relative size of the vertical

polarization component of the Raman signal **160** is 100, then the Raman signal **160** may be determined to be substantially vertically polarized.

[0178] Based on the digital output signals, a processor may determine a phase associated with the Raman signal. For example, the processor may determine (i) a phase difference $\Delta\phi_h$ between the horizontal Raman signal **160-h** and the horizontal probe beam **120pr-h** and (ii) a phase difference $\Delta\phi_v$ between the vertical Raman signal **160-v** and the vertical probe beam **120pr-v**. Based on the digital output signals, a processor may determine in-phase and quadrature portions associated with each of the horizontal Raman signal **160-h** and vertical Raman signal **160-v**. The phase difference $\Delta\phi_h$ between the horizontal Raman signal **160-h** and the horizontal probe beam **120pr-h** may be determined from the expression $\Delta\phi_h = \arctan(Q_h/IP_h)$, where Q_h and IP_h are the quadrature and in-phase portions associated with the horizontal Raman signal. The phase difference $\Delta\phi_v$ between the vertical Raman signal **160-v** and the vertical probe beam **120pr-v** may be determined from the expression $\Delta\phi_v = \arctan(Q_v/IP_v)$, where Q_v and IP_v are the quadrature and in-phase portions associated with the vertical Raman signal.

[0179] An optical receiver **200** may include one or more detectors **220**. An optical receiver **200** may include one detector **220** (e.g., as illustrated in FIGS. 1, 2, and 18), or an optical receiver **200** may include multiple detectors **220** (e.g., as illustrated in FIGS. 17, 36, 38, and 39). An optical receiver **200** with multiple detectors **220** may include 2, 3, 4, 8, 16, or any other suitable number of detectors. For example, an optical receiver **200** may include two detectors **220** arranged so that their respective photocurrents are subtracted (e.g., as illustrated in FIG. 36). As another example, an optical receiver **200** may include four detectors **220** (e.g., as illustrated in FIG. 38) or eight detectors **220** (e.g., as illustrated in FIG. 39). In an optical receiver **200** with multiple detectors **220**, portions of a probe beam **120pr** and a Raman signal **160** may be coherently mixed together at one or more of the multiple detectors **220**, and each of these one or more detectors may produce a photocurrent signal i corresponding to the coherent mixing of the probe beam and the Raman signal. Any of the optical receivers **200** described herein as having a single detector **220** may also be configured to have two or more detectors. For example, the optical receiver in FIG. 1 (which includes one detector **220**) may include a second detector (not illustrated in FIG. 1), and the detection electronics **230** may be configured to receive and process photocurrent signals from each of the two detectors.

[0180] FIGS. 40-45 each illustrate an example Raman spectroscopy system **100** that includes one or more optical fibers **116**. Each of the optical fibers **116** in FIGS. 40-45 may be referred to as an optical-fiber extension or a fiber-optic extension, and each of the Raman spectroscopy systems may be referred to as a Raman spectroscopy system with optical-fiber extension or a Raman spectroscopy system with fiber-optic extension. An optical-fiber extension refers to one or more optical fibers **116** that transmit light to or from an enclosure **101** of a Raman spectroscopy system **100**. The optical fibers **116** in each of FIGS. 40-45 direct the combined pump-Stokes beam **140** from the enclosure **101** to a sample **150** and direct the resulting Raman signal **160** back to the enclosure. An optical-fiber extension allows the Raman signal **160** of a sample **150** located external to a

Raman spectroscopy system **100** and some distance away from the system to be measured.

[0181] Each of the Raman spectroscopy systems **100** in FIGS. **40-45** includes an enclosure **101** and one or more optical fibers **116**, where at least a portion of the optical fibers is located external to the enclosure. The enclosure **101** of a Raman spectroscopy system **100** may be referred to as a chassis or housing and may be made from metal (e.g., aluminum), plastic, or any other suitable substantially rigid material. An enclosure **101** may substantially enclose or contain one or more parts of a Raman spectroscopy system **100** and may include a feedthrough or a fiber-optic adapter that allows one or more optical fibers to exit from or connect to the enclosure. In each of FIGS. **42-45**, the enclosure **101** contains a pump light source **110_{pu}**, Stokes light source **110_S**, and optical receiver **200**. An enclosure **101** may also contain all or part of a processor. For example, an enclosure **101** may contain a processor that receives a digital output signal **240** from detection electronics **230**, and the processor may analyze the digital output signal to determine a characteristic of a corresponding photocurrent signal *i*. In some embodiments, a processor or a portion of a processor may be located in the detection electronics **230** of a Raman spectroscopy system.

[0182] An optical fiber **116** (which may be referred to as a fiber-optic cable, fiber optic, or fiber) refers to a flexible glass or plastic fiber that transmits light with relatively low optical loss (e.g., less than 1 dB of optical-power loss per kilometer of fiber length). Light that propagates in an optical fiber **116** may travel primarily through a fiber-optic core that is surrounded by a cladding. The core may have a higher refractive index than the cladding, which provides optical confinement and guidance for light that propagates within an optical fiber. The optical fiber **116** in each of FIGS. **40-45** may include any suitable type of optical fiber. For example, an optical fiber **116** may be a single-mode (SM) optical fiber (e.g., with a core diameter of approximately 4 to 14 μm) or a multi-mode (MM) optical fiber (e.g., with a core diameter of approximately 50 to 100 μm). As another example, an optical fiber **116** may be a polarization-maintaining (PM) optical fiber, which is a type of SM fiber having two propagation axes that each allow linearly polarized light to propagate along the fiber and substantially maintain the linear polarization. As another example, an optical fiber **116** may be a hollow-core optical fiber where light propagates primarily along a hollow region of the fiber. A hollow-core optical fiber may provide lower optical loss, reduced optical nonlinearities, a higher optical damage threshold, or a larger optical bandwidth as compared to an optical fiber having a solid core made from glass or plastic. As another example, an optical fiber **116** may be a multi-core optical fiber having two or more cores along which light may propagate (e.g., the input and output optical fibers in FIG. **41** may be replaced by a single dual-core optical fiber in which the pump-Stokes beam **140** propagates along one core and the Raman signal propagates along the other core). The optical fiber **116** in each of FIGS. **40-45** may have any suitable length, such as for example a length of approximately 1 m, 2 m, 5 m, 10 m, 100 m, 1 kilometer (km), or 10 km. For example, a Raman spectroscopy system with optical-fiber extension that is used in a medical clinic or hospital may have an optical fiber **116** with a length of less than 10 m, while a Raman spectroscopy

system with optical-fiber extension that is used to investigate an oil well may have an optical fiber with a length of 1-10 km.

[0183] In each of FIGS. **40-41**, a combined pump-Stokes beam **140** (which includes a pump beam **120_{pu}** and a Stokes beam **120_S**) is directed by an optical fiber **116** from the enclosure **101** of a Raman spectroscopy system **100** to a sample **150**. A Raman signal **160** is produced by coherent Raman scattering of the pump and Stokes beams of light at the sample **150**, and the Raman signal is directed by an optical fiber **116** from the sample to the enclosure **101**. A Raman spectroscopy system with optical-fiber extension may include 1, 2, 3, 4, 5, 10, or any other suitable number of optical fibers **116**. In FIG. **40**, the pump-Stokes beam **140** and the Raman signal **160** are directed to and from the sample **150** by a single optical fiber **116**, and in FIG. **41**, the pump-Stokes beam **140** and the Raman signal **160** are directed to and from the sample **150** separately by two optical fibers **116**. The Raman spectroscopy system **100** in FIG. **40** includes one optical fiber **116** that (i) directs the combined pump-Stokes beam **140** to the sample **150** and (ii) directs the Raman signal **160** back to the enclosure **101** of the Raman spectroscopy system. The Raman spectroscopy system **100** in FIG. **41** includes two optical fibers: an output optical fiber **116_a** that directs the combined pump-Stokes beam **140** to the sample **150**, and an input optical fiber **116_b** that directs the Raman signal **160** back to the enclosure **101** of the Raman spectroscopy system. In other embodiments, a Raman spectroscopy system **100** may include two or more optical fibers that direct the pump and Stokes beams to a sample or two or more optical fibers that direct the Raman signal back to the enclosure of the system. For example, one output optical fiber that directs a pump-Stokes beam to a sample may have two or more input optical fibers positioned around the output optical fiber, where each of the input optical fibers is configured to direct a portion of a Raman signal back to a Raman spectroscopy system.

[0184] In FIGS. **40-45**, the pump-Stokes beam **140** may be emitted as a free-space optical beam directly from the output end of an optical fiber **116**, and the resulting free-space pump-Stokes beam **140** may be a diverging beam. Alternatively, a lens may be located near the output end of an optical fiber **116**, and the lens may produce a free-space pump-Stokes beam **140** that is focused or collimated. The sample **150** in FIGS. **40-41** is located a distance *D* from the end of the optical fiber **116**. The distance *D* may have any suitable value, such as for example a value of approximately 1 mm, 2 mm, 5 mm, 10 mm, 100 mm, 1 m, 5 m, 10 m, 100 m, or 1 km. For example, the distance *D* may be between 0 mm and 10 mm for a Raman spectroscopy system with optical-fiber extension that is used to measure a person's skin for indications of skin cancer. As another example, the distance *D* may be between 1 m and 1 km for a Raman spectroscopy system used for remote measurement of a package, a person, a vehicle, or a manufacturing process.

[0185] In each of FIGS. **42-45**, the Raman spectroscopy system **100** includes an enclosure **101** that contains a pump light source **110_{pu}**, a Stokes light source **110_S**, and an optical receiver **200** (which includes a probe light source **110_{pr}**, detector **220**, and detection electronics **230**). The pump light source **110_{pu}** produces a pump beam of light **120_{pu}** at a pump frequency (which may be referred to as a first frequency and may be represented by ν_{pu} , ν_1 , ω_{pu} , or ω_1), and the Stokes light source **110_S** produces a Stokes

beam of light **120S** at a Stokes frequency (which may be referred to as a second frequency and may be represented by ν_S , ω_S , or ω_2). The pump and Stokes frequencies may be offset by a frequency offset Ω , where Ω equals $\nu_{pu} - \nu_S$ (or equivalently, $\Omega = \omega_1 - \omega_2$). The pump light source **110pu** may be referred to as a first light source, and the pump beam of light **120pu** may be referred to as a first beam of light. The Stokes light source **110S** may be referred to as a second light source, and the Stokes beam of light **120S** may be referred to as a second beam of light.

[0186] In each of FIGS. 42-45, the pump and Stokes beams are directed to a sample **150** by an optical fiber **116**, and the sample produces a Raman signal **160** in response to the pump and Stokes beams. The Raman signal **160** may be produced by coherent Raman scattering of the pump and Stokes beams at the sample **150**. For example, the Raman signal **160** may be produced by coherent Raman scattering that occurs within the sample **150** or at the surface of the sample. At least a portion of the Raman signal **160** produced by the sample **150** is coupled into an optical fiber **116** and directed to an optical receiver **200** that detects the Raman signal. The optical receiver **200** detects the Raman signal **160** using an optical heterodyne technique in which the Raman signal **160** is coherently mixed with a probe beam of light **120pr** at an optical detector **220**. An optical receiver **200** may include one or more optical detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with at least a portion of a probe beam **120pr** to produce an electronic signal. The optical receiver **200** in each of FIGS. 42-45 includes one optical detector **220** that coherently mixes the Raman signal **160** with the probe beam **120pr** to produce a photocurrent signal *i*. In other embodiments, an optical receiver **200** may include two or more optical detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with a portion of a probe beam **120pr** to produce a corresponding photocurrent signal.

[0187] In each of FIGS. 42-45, the probe light source **110pr** produces a probe beam of light **120pr** at a probe frequency (which may be referred to as a third frequency and may be represented by ν_{pr} , ω_{pr} , or ω_3). The probe beam **120pr** is combined with the Raman signal **160**, and the combined probe-Raman signal **210** is directed to an optical detector **220**. The Raman signal **160** and the probe beam **120pr** are coherently mixed at the detector **220** to produce a corresponding photocurrent signal *i*. Coherent mixing of the Raman signal **160** and the probe beam **120pr** may refer to coherently mixing at least a portion of the Raman signal **160** with at least a portion of the probe beam **120pr**. For example, an optical receiver **200** may include multiple detectors **220**, and each detector may coherently mix a portion of a Raman signal with a portion of a probe beam **120pr**. As another example, a detector **220** may coherently mix at least a portion of a probe beam **120pr** with a spectral portion of a Raman signal **160**, where the spectral portion of the Raman signal that is coherently mixed includes optical frequency components of the Raman signal that are located within a particular frequency range of the frequency ν_{pr} of the probe beam (e.g., the particular frequency range may depend on the electronic bandwidth of the detector **220**).

[0188] In each of FIGS. 42-45, the detection electronics **230** (which may include or may be referred to as an electronic circuit) produces a digital output signal **240** corresponding to the photocurrent signal *i*. An electronic

circuit **230** may include an electronic amplifier **232** and a digitizer **236** (e.g., like that illustrated in FIG. 2). The electronic amplifier **232** may amplify the photocurrent signal *i* to produce a voltage signal **234** that corresponds to the photocurrent signal, and the digitizer **236** may produce a digital representation of the voltage signal. The digital representation of the voltage signal **234** may include digital values that approximate the shape of the voltage signal. The digital representation of the voltage signal **234** may be sent to a processor as part of the digital output signal **240**. The digital output signal **240** may be referred to as corresponding to the photocurrent signal *i*, since the digital output signal **240** includes a digital representation of the voltage signal **234**, and the voltage signal corresponds to the photocurrent signal *i*.

[0189] The digital output signal **240** in each of FIGS. 42-45 may be sent to a processor, and the processor may determine a characteristic **162** of the photocurrent signal *i* based on the digital output signal. For example, the digital output signal may include a digital representation of a voltage signal **234**. The processor may determine a characteristic of the digital representation of the voltage signal **234**, and that characteristic may be referred to as being a characteristic of the corresponding photocurrent signal *i*. The characteristic **162** of a photocurrent signal may include one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization. Additionally, a processor may associate a Raman frequency shift with a determined characteristic **162** of a photocurrent signal *i*. The Raman frequency shift may equal $\nu_{pu} - \nu_{pr}$, where ν_{pu} is the pump frequency, and ν_{pr} is the probe frequency.

[0190] The probe light source **110pr** in each of FIGS. 42-45 may include a wavelength-tunable laser configured to sequentially change the probe-beam frequency ν_{pr} to multiple different frequencies. At each of the different probe-beam frequencies, the optical detector **220** may coherently mix the probe beam **120pr** and the Raman signal **160** to produce a corresponding photocurrent signal, and a processor may determine a characteristic of each of the photocurrent signals. In some embodiments, the frequency offset Ω between the pump and Stokes frequencies may be approximately equal to a vibrational frequency of a particular material, and a processor may determine, based on a determined characteristic **162** of a photocurrent signal *i*, (i) whether the particular material is present in a sample **150** or (ii) an amount or a concentration of the particular material in the sample. A processor of a Raman spectroscopy system **100** may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. A processor may be similar to the computer system **5500** illustrated in FIG. 55 and described herein. In some embodiments, a processor or a portion of a processor may be located in the detection electronics **230** of a Raman spectroscopy system.

[0191] The Raman spectroscopy system **100** in each of FIGS. 42 and 43 includes one optical fiber **116** that (i) directs a pump beam **120pu** and a Stokes beam **120S** to a sample **150** and (ii) directs a Raman signal **160** back to the enclosure **101** of the system. The combined pump-Stokes beam **140** and the Raman signal **160** propagate in opposite directions within the same optical fiber **116**. The Raman spectroscopy

system **100** in each of FIGS. **44** and **45** includes two optical fibers: an output optical fiber **116a** that directs a pump beam **120_{pu}** and a Stokes beam **120_S** to a sample **150**, and an input optical fiber **116b** that directs a Raman signal **160** back to the enclosure **101** of the system.

[0192] The optical beams within the enclosure **101** of a Raman spectroscopy system **100** may be free-space optical beams, fiber-coupled optical beams, or waveguide-coupled optical beams (e.g., a beam that propagates in an optical waveguide of a PIC), or any combination thereof. In FIG. **42**, the optical beams within the enclosure **101** are primarily free-space optical beams, while in FIG. **43**, the optical beams within the enclosure **101** are primarily fiber-coupled optical beams. In FIG. **42**, the pump beam **120_{pu}** and the Stokes beam **120_S** are combined at a free-space optical combiner **130a** to produce a free-space combined pump-Stokes beam **140** that is coupled into the optical fiber **116** by a lens **114**. The Raman signal **160** produced by a sample is coupled into the opposite end of the optical fiber **116** and propagates along the optical fiber to the enclosure **101** of the Raman spectroscopy system **100**. The Raman signal **160** is emitted from the optical fiber **116** and collimated by the lens **114**. The optical combiner **130e** (which may be a dichroic or non-dichroic beamsplitter or a polarization beamsplitter) splits off at least a portion of the Raman signal **160**, which is directed to the optical receiver **200**. The free-space optical combiner **130b** combines the probe beam **120_{pr}** with the Raman signal **160** to produce a free-space probe-Raman signal **210** that is directed to a detector **220**.

[0193] In FIG. **43**, the fiber-coupled pump beam **120_{pu}** and Stokes beam **120_S** are combined at a fiber-optic combiner **130a** to produce a fiber-coupled pump-Stokes beam **140** that is directed to an optical circulator **131** by an optical fiber. The optical circulator **131** is a three-port fiber-optic component that receives input light at one port and directs that light to exit the circulator from another port. For example, light that enters the optical circulator **131** at port **1** is directed to exit the circulator from port **2**, and light that enters the circulator at port **2** is directed to exit the circulator from port **3**. The optical circulator **131** in FIG. **43** receives the combined pump-Stokes beam **140** at port **1** and directs the combined pump-Stokes beam to the optical fiber **116**, which is coupled to port **2**. Additionally, the optical circulator **131** receives the Raman signal **160** from the optical fiber **116** at port **2** and directs the Raman signal to the optical receiver **200** via an optical fiber coupled to port **3**. The fiber-optic combiner **130b** combines the fiber-coupled probe beam **120_{pr}** with the fiber-coupled Raman signal **160** to produce a fiber-coupled probe-Raman signal **210** that is directed to a detector **220**. In other embodiments, instead of using a fiber-optic circulator **131** (as illustrated in FIG. **43**), a Raman spectroscopy system may use a fiber-optic combiner as an optical splitter that splits off at least a portion of a Raman signal and directs the split-off Raman signal to an optical receiver.

[0194] In FIG. **44**, the optical beams within the enclosure **101** are primarily free-space optical beams, while in FIG. **45**, the optical beams within the enclosure **101** are primarily fiber-coupled optical beams. In FIG. **44**, the pump beam **120_{pu}** and the Stokes beam **120_S** are combined at a free-space optical combiner **130a** to produce a free-space combined pump-Stokes beam **140** that is coupled into an output optical fiber **116a** by a lens **114a**. The Raman signal **160** produced by a sample is coupled into an input optical fiber

116b and propagates along the input fiber to the enclosure **101** of the Raman spectroscopy system **100**. The Raman signal **160** is emitted from the optical fiber **116** and collimated by a lens **114b** to produce a free-space Raman signal **160** that is directed to an optical receiver **200**. The free-space optical combiner **130b** combines the probe beam **120_{pr}** with the Raman signal **160** to produce a free-space probe-Raman signal **210** that is directed to a detector **220**.

[0195] In FIG. **45**, the fiber-coupled pump beam **120_{pu}** and Stokes beam **120_S** are combined at a fiber-optic combiner **130a** to produce a fiber-coupled pump-Stokes beam **140** that propagates along an output optical fiber **116a** and is directed to a sample. The fiber-optic combiner **130a** in each of FIGS. **43** and **45** is similar to the fiber-optic combiner **130** illustrated in FIG. **25**. In FIG. **45**, the Raman signal **160** produced by the sample is coupled into an input optical fiber **116b** and propagates along the input fiber into the enclosure **101** of the Raman spectroscopy system **100** and to a fiber-optic combiner **130b**. The fiber-optic combiner **130b** combines the fiber-coupled probe beam **120_{pr}** with the fiber-coupled Raman signal **160** to produce a fiber-coupled probe-Raman signal **210** that is directed to a detector **220**. The fiber-optic combiner **130b** in each of FIGS. **43** and **45** is similar to the fiber-optic combiner **130** illustrated in FIG. **28**.

[0196] A technical advantage of a Raman spectroscopy system **100** with optical-fiber extension is the ability to measure a sample **150** located outside of the system. A Raman spectroscopy system with optical-fiber extension allows a sample **150** to be measured in place instead of having to first collect a sample that is then put inside a system for measurement. Additionally, since an optical fiber **116** is typically flexible and relatively lightweight, an operator of the system may direct the optical fiber to a particular location with relative ease while the enclosure **101** and its contents remain fixed in place. The flexible optical fiber **116** may allow an operator to make measurements by positioning the terminal end of the optical fiber to direct a combined pump-Stokes beam **140** at an object of interest (e.g., a person's skin, water, wastewater, or a manufacturing process). For example, a Raman spectroscopy system with optical-fiber extension may be used during a medical procedure, such as for example, to identify tumor margins during surgery, diagnose skin cancer, identify issues during a colonoscopy, or measure interstitial fluid located under the skin. The relatively low optical loss of an optical fiber **116** may allow the measurement of objects that are located a relatively long distance from the enclosure **101** of a Raman spectroscopy system **100**. Additionally, the relatively small size of an optical fiber **116** may allow the measurement of objects located in relatively hard-to-reach places. For example, an optical fiber **116** inserted into an oil well may allow the measurement of materials located deep underground.

[0197] FIGS. **46-47** each illustrate a cross-section of a portion of an enclosure **101** with an example fiber-optic feedthrough. A fiber-optic feedthrough allows light propagating in an optical fiber **116** to be delivered (i) from inside the enclosure **101** of a Raman spectroscopy system **100** to outside the enclosure or (ii) from outside the enclosure to inside the enclosure. In FIG. **46**, a fiber-optic feedthrough for an optical fiber **116** is provided by an opening in the enclosure **101** and an O-ring **310**. The optical fiber **116** passes through a hole in the enclosure **101** and into the interior of the enclosure (located on the left side of FIG. **46**).

The O-ring 310 provides for a feedthrough that mechanically secures the optical fiber 116 and prevents movement or damage to the optical fiber. In FIG. 47, a fiber-optic feedthrough is provided by a fiber-optic adapter 330. An internal optical fiber 116i located in the interior of the enclosure 101 is connected to an external optical fiber 116e by a fiber-optic adapter 330. The internal and external optical fibers each have a fiber-optic connector 320 that is secured to the fiber-optic adapter 330 (e.g., by screwing the connector to mating threads on the adapter). The dashed-line inset in FIG. 47 illustrates the fiber-optic connectors 320 connected to the fiber-optic adapter 330 so that the ends of the optical fibers 116i and 116e are in contact and light may be coupled from one fiber to the other. The fiber-optic adapter 330 provides for an optical connection between the ends of the two optical fibers so that the combined pump-Stokes beam 140 and the Raman signal 160 are transmitted between the fibers with low optical loss (e.g., less than 0.5 dB of optical loss). The internal optical fiber 116i couples the combined pump-Stokes beam 140 to the external optical fiber 116e, which directs the pump-Stokes beam to a sample 150. Additionally, the external optical fiber 116e directs a Raman signal 160 from the sample 150 back to the enclosure 101 and couples the Raman signal to the internal optical fiber 116i. Any of the example Raman spectroscopy systems 100 in FIGS. 40-45 may include (i) an optical fiber 116 that enters the interior of the enclosure 101 through a hole in the enclosure (e.g., as illustrated in FIG. 46) or (ii) an external optical fiber 116e that is optically connected to an internal optical fiber 116i by a fiber-optic adapter 330 (e.g., as illustrated in FIG. 47).

[0198] Each of the optical fibers 116 in FIGS. 40-47 may be a passive optical fiber that transmits light with relatively low optical loss and does not provide optical amplification. In other embodiments, an optical fiber 116 may include a fiber-optic amplifier 126 that provides optical amplification to light traveling along the optical fiber. The fiber-optic amplifier may include an optical gain fiber doped with rare-earth materials (e.g., neodymium, erbium, or ytterbium) or bismuth. For example, a portion of the output optical fiber 116a in FIG. 45 that is located within the enclosure 101 may include a fiber-optic amplifier that amplifies the pump beam 120pu or the Stokes beam 120S. As another example, a portion of the input optical fiber 116b in FIG. 45 that is located within the enclosure 101 may include a fiber-optic amplifier that amplifies the Raman signal 160. As another example, the internal optical fiber 116i in FIG. 47 may include a fiber-optic amplifier that amplifies the pump or Stokes beam before exiting the enclosure 101 or amplifies the Raman signal 160 before the Raman signal is directed to an optical receiver 200. As another example, the pump light source 110pu or the Stokes light source 110S in any of FIGS. 42-45 may include a seed laser diode followed by an optical amplifier (e.g., a SOA 124 or a fiber-optic amplifier 126).

[0199] FIG. 48 illustrates the terminal end 117 of an example optical fiber 116 along with a lens 114. The terminal end 117 of an optical fiber 116 refers to an end of the optical fiber where light is coupled into the fiber or light is emitted from the fiber. The terminal end 117 may have a polished surface that provides an optical interface with low optical scattering. Additionally, an anti-reflection coating may be deposited onto the surface of the terminal end 117 to reduce the optical-reflection loss of light that is coupled into or out of the optical fiber 116. A combined pump-Stokes beam 140 emitted from the terminal end 117 of an optical fiber 116

may propagate as a free-space optical beam that includes the pump and Stokes beams. The free-space optical beam may be a diverging beam, or a lens 114 may be positioned near the terminal end 117 to produce a collimated or focused free-space beam.

[0200] The optical fiber 116 in FIG. 48 (which may correspond to the optical-fiber extension 116 in FIG. 40, 42, 43, 46, or 47) directs a combined pump-Stokes beam 140 to a sample 150 and directs the associated Raman signal 160 back to the enclosure 101 of a Raman spectroscopy system 100. The combined pump-Stokes beam 140 and the Raman signal 160 may propagate along the optical fiber 116 primarily through the fiber-optic core 190 which is surrounded by a fiber-optic cladding 192. In FIG. 48, the combined pump-Stokes beam 140 is emitted from the terminal end 117 of the optical fiber 116 as a free-space optical beam that includes the pump and Stokes beams. The lens 114, which is located near the terminal end 117, receives the pump-Stokes beam 140 emitted from the terminal end and produces a free-space pump-Stokes beam 140 that is directed to the sample 150. Additionally, the lens 114 in FIG. 48 receives the Raman signal 160 produced by the sample 150 in response to the pump and Stokes beams. The lens 114 may focus the Raman signal 160 to couple the Raman signal into the optical fiber 116 via the terminal end 117.

[0201] In FIG. 48, the free-space pump-Stokes beam 140 emitted from the terminal end 117 may be a diverging free-space optical beam, and the lens 114 may be configured to produce a free-space pump-Stokes beam that is collimated or focused. For example, the lens 114 may produce a collimated pump-Stokes beam 140, and the distance D from the lens to the sample 150 may be any suitable distance from 0 mm to 1 km (e.g., the distance D may be approximately 0 mm, 1 mm, 2 mm, 5 mm, 10 mm, 0.1 m, 1 m, 10 m, 100 m, or 1 km). As another example, the lens 114 may be configured to focus the pump-Stokes beam 140 onto the sample 150. In this case, the distance D from the lens to the sample 150 may be between approximately f and 4f, where f is the focal length of the lens (e.g., for a lens with a 10-mm focal length, the distance to the sample may be 10-40 mm).

[0202] In other embodiments, the terminal end 117 of an optical fiber 116 may directly emit a combined pump-Stokes beam 140 without having a lens positioned near the terminal end, and the pump-Stokes beam may be emitted as a diverging free-space optical beam that is directed to a sample 150. Additionally, a Raman signal 160 produced by the sample 150 may be directly coupled into the optical fiber 116 without first being focused by a lens. For example, an optical fiber 116 without a lens may be used to investigate a sample 150 located between 0 mm and 10 mm from the terminal end 117.

[0203] A lens 114 may be attached to or integrated into the terminal end 117 of an optical fiber 116, or a lens may be located some distance from the terminal end 117. For example, the terminal end 117 of an optical fiber 116 may include a gradient refractive index or a lensed tip that is integrated into the terminal end and that acts as a lens. As another example, a lens 114 may be a spherical lens, an aspheric lens, or a gradient-index (GRIN) lens, and the lens may be located some distance from the terminal end 117 or may be attached to the terminal end. The distance from the terminal end 117 to the lens 114 may be less than approximately 4f, where f is the focal length of the lens. For example, the distance from the terminal end 117 to the lens

117 may be approximately equal to for 2f. For a lens 114 that is attached to a terminal end 117, the distance from the terminal end to the lens may be referred to as being 0 mm.

[0204] FIG. 49 illustrates example input and output optical fibers 116b and 116a along with a lens 114. The input and output optical fibers 116b and 116a in FIG. 49 may correspond to the input and output optical fibers 116b and 116a in FIG. 41, 44, or 45. The output optical fiber 116a in FIG. 49 directs the combined pump-Stokes beam 140 to a sample 150, and the input optical fiber 116b directs the associated Raman signal 160 back to the enclosure 101 of a Raman spectroscopy system 100. The combined pump-Stokes beam 140 is emitted from the terminal end 117a of the output optical fiber 116a as a free-space optical beam that includes the pump and Stokes beams. The lens 114 receives the pump-Stokes beam 140 emitted from the terminal end 117a and produces a free-space pump-Stokes beam 140 that is directed to the sample 150. The free-space pump-Stokes beam 140 produced by the lens 114 may be a collimated or focused free-space beam. Additionally, the lens 114 in FIG. 49 receives the Raman signal 160 produced by the sample 150 in response to the pump and Stokes beams and couples the Raman signal into the input optical fiber 116b. The lens 114 may focus the Raman signal 160 to couple the signal into the input optical fiber 116b via the terminal end 117b.

[0205] In FIG. 49, a single lens 114 is used to (i) receive the pump-Stokes beam 140 from the output optical fiber 116a and direct the pump-Stokes beam to the sample 150 and (ii) receive the Raman signal 160 from the sample and couple the Raman signal into the input optical fiber 116b. In other embodiments, a Raman spectroscopy system 100 that includes multiple input or output optical fibers may include multiple lenses 114. For example, a Raman spectroscopy system 110 that includes one input optical fiber and one output optical fiber may include two lenses 114: an output lens located near the terminal end 117a of the output optical fiber 116a, and an input lens located near the terminal end 117b of the input optical fiber 116b. The output lens may receive the pump-Stokes beam 140 emitted from the terminal end 117a and produce a free-space pump-Stokes beam 140 that is directed to the sample 150. The free-space pump-Stokes beam 140 produced by the output lens may be a collimated or focused free-space beam. The input lens may receive the Raman signal 160 produced by the sample 150 in response to the pump and Stokes beams and couple the Raman signal into the input optical fiber 116b via the terminal end 117b. The input lens may focus the Raman signal 160 to couple the Raman signal into the input optical fiber 116b.

[0206] FIG. 50 illustrates example input and output optical fibers 116b and 116a along with a parabolic mirror 194. The input and output optical fibers 116b and 116a in FIG. 50 may correspond to the input and output optical fibers 116b and 116a in FIG. 41, 44, or 45. The output optical fiber 116a in FIG. 50 directs a combined pump-Stokes beam 140 to a sample 150, and the input optical fiber 116b directs the associated Raman signal 160 back to the enclosure 101 of a Raman spectroscopy system 100. The combined pump-Stokes beam 140 is emitted from the terminal end 117a of the output optical fiber 116a as a free-space optical beam that includes the pump and Stokes beams. The output lens 114 receives the pump-Stokes beam 140 emitted from the terminal end 117a and produces a combined pump-Stokes beam 140 that is directed to the sample 150. The free-space

pump-Stokes beam 140 produced by the lens 114 may be a collimated or focused free-space beam.

[0207] The mirror 194 includes a through hole 196 that the free-space pump-Stokes beam 140 propagates through while traveling to the sample 150. The through hole 194 may be a circular or conical hole having a diameter greater than the beam diameter of the pump-Stokes beam 140. For example, the lens 114 may produce a collimated pump-Stokes beam 140 having a 2-mm beam diameter, and the through hole 194 may have a diameter of approximately 4 mm. The mirror 194 has a reflective surface 198 that receives the Raman signal 160 produced by the sample 150 in response to the pump and Stokes beams and reflects the Raman signal to direct the signal to the terminal end 117b of the input optical fiber 116b. The reflective surface 198 may be substantially flat, and a lens (not illustrated in FIG. 50) located between the reflective surface and the terminal end 117b may focus the Raman signal 160 into the optical fiber 116b. Alternatively, the reflective surface 198 may have a curved shape that focuses the Raman signal 160. For example, as illustrated in FIG. 50, the mirror 194 may be an off-axis parabolic mirror where the reflective surface 198 has a parabolic shape. The parabolic reflective surface 198 may focus the Raman signal 160 so that it is coupled into the input optical fiber 116b via the terminal end 117b.

[0208] FIG. 51 illustrates an example Raman spectroscopy system 100 that includes a visible light source 110v. The visible light source 110v produces a visible beam of light 120v that is combined at an optical combiner 130v with the combined pump-Stokes beam 140 to produce a combined pump-Stokes-visible beam 142. The combined pump-Stokes-visible beam 142 (which includes the pump beam 120pu, the Stokes beam 120S, and the visible beam 120v) travels through an optical fiber 116 and is emitted as a free-space optical beam that is directed to a sample 150. The sample produces a Raman signal 160 in response to the pump and Stokes beams, and at least a portion of the Raman signal is coupled into the optical fiber 116 and propagates along the fiber to the enclosure 101 of the Raman spectroscopy system 100. The optical combiner 130e splits off at least a portion of the Raman signal 160, which is directed to the optical receiver 200 for detection. In other embodiments, an optical circulator 131 (e.g., as illustrated in FIG. 43) may be used in place of the optical combiner 130e in FIG. 51.

[0209] The pump beam 120pu and the Stokes beam 120S may each have wavelengths that are not visible to the human eye (e.g., the pump and Stokes wavelengths may be greater than approximately 900 nm), and the pump and Stokes beams may not produce a visible spot of light at the sample 150. The visible beam 120v may have a wavelength that is visible to the human eye. For example, the visible light source 110v may include a laser that produces a visible beam of light 120v having a wavelength between approximately 380 nm (blue) and approximately 780 nm (red). As another example, the visible light source 110v may include (i) a laser diode with an operating wavelength of approximately 450-490 nm or 635 nm or (ii) a solid-state laser with an operating wavelength of approximately 532 nm.

[0210] In FIG. 51, the visible beam 120v is directed to the sample 150 along with the pump and Stokes beams. The visible beam 120v produces a visible alignment spot 144 at a location where the combined pump-Stokes-visible beam 142 is incident on the sample 150. The visible alignment spot 144 includes light from the visible beam 120v that is

scattered or reflected from the sample **150**. The visible alignment spot **144** indicates the location of the pump and Stokes beams at the sample **150** and may be used by an operator of the system **100** to aim the combined pump-Stokes beam **140** to the sample. For example, an operator may position the terminal end **117** of the optical fiber **116** to direct the combined pump-Stokes-visible beam **142** to the sample **150** using the location of the visible alignment spot **144** as an alignment aid. Any of the Raman spectroscopy systems **100** illustrated in FIGS. **40-45** may include a visible light source **110v** that produces a visible beam of light **120v**, similar to that illustrated in FIG. **51**.

[0211] In FIG. **51**, the visible beam **120v** propagates to the sample **150** in an optical fiber **116** along with the pump and Stokes beams. In other embodiments, a visible beam **120v** produced by a visible light source **110v** may propagate in a different optical fiber **116** from the pump and Stokes beams. For example, in FIG. **45**, a visible beam **120v** may be coupled into the input optical fiber **116b** and directed to propagate to the sample (and in the opposite direction of the Raman signal **160**). As another example, a Raman spectroscopy system **100** may include a separate optical fiber configured to direct a visible beam **120v** to a sample **150**. The pump-Stokes beam **140** may propagate in a first output optical fiber, and the visible beam **120v** may propagate in a second output fiber. The first and second output optical fibers may be aligned so that the free-space pump-Stokes beam **140** and the free-space visible beam **120v** propagate together and the visible alignment spot **144** produced at the sample **150** is substantially overlapped with or directly adjacent to the location of the pump and Stokes beams at the sample. For example, the free-space visible beam **120v** may propagate along an optical axis that is approximately parallel to and directly adjacent to the optical axis along which the combined pump-Stokes beam **140** propagates.

[0212] FIG. **52** illustrates an example Raman spectroscopy system **100** with a moveable end holder **350**. The end holder **350** may include a mechanical housing that contains the terminal end **117** of one or more optical fibers **116** along with one or more lenses **114** or a mirror **194** with a through hole **196**. For example, the terminal end **117** and lens **114** in FIG. **48** may be contained within an end holder, or the terminal ends **117a** and **117b** and the lens **114** in FIG. **49** may be contained within an end holder. The end holder **350** may allow an operator of the Raman spectroscopy system **100** to position the terminal end **117** of an optical fiber **116** to direct a free-space pump-Stokes beam **140** to a sample **150**. For example, an operator may grasp the end holder **350** in their hand **351**, as illustrated in FIG. **52**, and move or rotate the end holder to aim the pump-Stokes beam **140** to a particular location. Additionally, the Raman spectroscopy system **100** may include a visible light source **110v** that produces a visible beam **120v**. The visible beam **120v** may propagate to the sample **150** along with the pump and Stokes beams, and the operator may use the visible alignment spot **144** as an alignment aid for directing the pump and Stokes beams.

[0213] The Raman spectroscopy system **100** in FIG. **51** or **52** may operate so that initially the visible light source **110v** is turned on and the pump and Stokes light sources are turned off. When the pump and Stokes light sources are turned off, the pump and Stokes beams may include little to no light. With the visible light source **110v** turned on, an operator may position the end holder **350** in FIG. **52** to direct the visible beam **120v** to a desired location (as indicated by

the location of the visible alignment spot **144**). Then, the pump and Stokes light sources may be turned on to produce the combined pump-Stokes beam **140** and the resulting Raman signal **160** may be measured by the system. The end holder **350** may include a switch or a button that allows an operator to turn on or off the pump and Stokes light sources.

[0214] FIG. **53** illustrates an example Raman spectroscopy system **100** with a balanced-detection optical receiver **200**. The optical receiver **200** includes two detectors (signal detector **220-sig** and reference detector **220-ref**) arranged in a balanced-detection configuration. The Raman spectroscopy system **100** in FIG. **53** may detect a Raman signal **160** by coherently mixing the Raman signal **160** with the probe beam of light **120pr** at the signal detector **220-sig**. Additionally, the reference detector **220-ref** may be used to reduce or remove common-mode noise that is present in both the signal beam **210-sig** and the reference beam **210-ref**. The Raman spectroscopy system **100** in FIG. **53** is similar to the Raman spectroscopy systems in FIGS. **42-45**, except the optical receiver **200** in FIG. **53** includes two detectors arranged for balanced detection. Additionally, the system in FIG. **53** is configured to produce a Stokes reference beam **120S-ref** and a probe reference beam **120pr-ref** that are detected by the reference detector **220-ref**. The light directed to and from a sample in FIG. **53** may be delivered via one or more optical fibers **116** (e.g., similar to the systems illustrated in FIGS. **40-45**), and the Raman spectroscopy system **100** in FIG. **53** may be referred to as a Raman spectroscopy system with optical-fiber extension and balanced detection.

[0215] In FIG. **53**, the combined pump-Stokes beam **140** is directed to a sample **150** by an output optical fiber **116a**. The light that returns from the sample **150** (which includes a Raman signal **160** as well as residual light from the pump beam **120pu** and Stokes beam **120S**) is directed back to the system by an input optical fiber **116b**. The residual pump light **120pu'** and the residual Stokes light **120S'** refers to light that is “leftover” after the pump and Stokes beams have interacted with the sample **150** to produce the Raman signal **160**. The residual pump beam of light **120pu'** may be produced by light from the pump beam **120pu** that is reflected from, transmitted through, or scattered by the sample **150**. Similarly, the residual Stokes beam of light **120S'** may be produced by light from the Stokes beam **120S** that is reflected from, transmitted through, or scattered by the sample **150**. The optical filter **134** blocks the light from the residual pump beam **120pu'** and transmits the Raman signal **160** and the residual Stokes light **120S'**. The Raman signal **160** and the residual Stokes beam of light **120S'** are combined with the probe beam **120pr** at the optical combiner **130b** to produce the combined signal beam **210-sig**, which is directed to the signal detector **220-sig**.

[0216] In addition to producing the Stokes beam of light **120S**, the Stokes light source **110S** in FIG. **53** produces a Stokes reference beam of light **120S-ref**. Similarly, in addition to producing the probe beam of light **120pr**, the probe light source **110pr** produces a probe reference beam of light **120pr-ref**. The probe and Stokes reference beams are combined at the optical combiner **130-ref** to produce the reference beam **210-ref**, which is directed to the reference detector **220-ref**. The probe and Stokes reference beams may each be produced by splitting off a portion of a primary beam. For example, the Stokes light source **110S** may include a laser that produces a primary output beam of light,

and a small portion (e.g., between 1% and 10%) of light from the primary output beam may be split off to produce the Stokes reference beam of light **120S-ref**. The Stokes beam of light **120S** may be produced from the remaining portion (e.g., between 90% and 99%) of light from the primary output beam that is not split off.

[0217] The signal detector **220-sig** in FIG. **53** receives the signal beam **210-sig** (which includes the Raman signal **160**, probe beam **120pr**, and residual Stokes beam **120S'**) and produces a signal photocurrent i_{sig} corresponding to the Raman signal, probe beam, and residual Stokes beam, where a portion of the signal photocurrent corresponds to coherent mixing between the Raman signal and the probe beam. The reference detector **220-ref** receives the reference beam **210-ref** (which includes the probe reference beam **120pr-ref** and the Stokes reference beam **120S-ref**) and produces a reference photocurrent i_{ref} corresponding to the probe and Stokes reference beams.

[0218] The detection electronics **230** in FIG. **53** receives the signal photocurrent i_{sig} and the reference photocurrent i_{ref} and produces a digital output signal **240** that corresponds to the two photocurrents. For example, the detection electronics **230** may include a subtraction module that determines a subtraction signal that corresponds to or that equals a difference between (i) a signal corresponding to the signal photocurrent i_{sig} and (ii) a signal corresponding to the reference photocurrent i_{ref} . The detection electronics **230** may include two electronic amplifiers **232**, where each electronic amplifier is configured to produce a voltage signal corresponding to one of the photocurrents. The subtraction signal may be determined as the difference between a first voltage signal corresponding to the signal photocurrent i_{sig} and a second voltage signal corresponding to the reference photocurrent i_{ref} . The detection electronics **230** may include a digitizer that produces a digital representation of the subtraction signal, and the digital output signal **240** may include the digital representation of the subtraction signal. The digital output signal **240** may be sent to a processor, and the processor may determine a characteristic **162** of the subtraction signal based on the digital output signal **240**. In other embodiments, the detection electronics **230** may include: two electronic amplifiers **232**; a first digitizer that produces a first digital signal corresponding to the signal photocurrent i_{sig} ; and a second digitizer that produces a second digital signal corresponding to the reference photocurrent i_{ref} . For example, the first digital signal may include a digital representation of a first voltage signal that corresponds to the signal photocurrent i_{sig} , and the second digital signal may include a digital representation of a second voltage signal that corresponds to the reference photocurrent i_{ref} . The digital output signal **240** may include the two digital signals corresponding to the two photocurrents. A processor may determine a digital subtraction signal from the two digital signals, and a characteristic **162** of a subtraction signal may be determined from the digital subtraction signal.

[0219] The balanced-detection configuration in FIG. **53** uses two detectors to substantially reduce or remove intensity noise that may be present in the probe beam **120pr** or the Stokes beam **120S**. The signal beam **210-sig** detected by the signal detector **220-sig** includes the Raman signal **160** along with the probe beam **120pr** and the residual Stokes beam **120S'**. The reference beam **210-ref** detected by the reference detector **220-ref** includes the probe reference beam **120pr-ref** and the Stokes reference beam **120S-ref**. Since the probe

beam **120pr** and probe reference beam **120pr-ref** may be derived from the same light source, the two beams may each include correlated (or, common-mode) noise signals. Similarly, the residual Stokes beam **120S'** and the Stokes reference beam **120S-ref** may each include correlated noise signals. In a balanced-detection optical receiver **200**, a signal corresponding to the reference photocurrent i_{ref} may be subtracted from a signal corresponding to the signal photocurrent i_{sig} to produce a subtraction signal. The subtraction signal may include a signal associated with the Raman signal **160**, while the common-mode noise present in the probe beam and the Stokes beam may be substantially removed by the subtraction operation that produces the subtraction signal. As a result, the subtraction signal may have a reduced noise compared to each of the photocurrent signals i_{sig} and i_{ref} alone.

[0220] FIG. **54** illustrates an example method **5400** for measuring a Raman signal **160**. The method may be performed by a Raman spectroscopy system as described herein. For example, the method may be performed by the Raman spectroscopy system **100** illustrated in FIG. **1** or **2** and may be used to measure a Raman signal **160**. The method may begin at step **5410** by producing a first beam of light at a first frequency (ν_1). At step **5420**, a second beam of light at a second frequency (ν_2) is produced. The first beam of light may be referred to as a pump beam of light **120pu**, and the second beam of light may be referred to as a Stokes beam of light **120S**. The first and second frequencies may be offset by a frequency offset Ω , where $\Omega = \nu_1 - \nu_2$. For example, the first and second frequencies may each correspond to a wavelength between approximately 300 nm and approximately 5,000 nm, and the frequency offset Ω may be between approximately 5 THz and approximately 100 THz. The first and second beams of light may each be produced by a light source that includes one or more laser diodes, where each laser diode is a fixed-wavelength laser diode or a wavelength-tunable laser diode.

[0221] At step **5430**, the first and second beams of light are directed to a sample **150**, and at step **5440**, a Raman signal **160** is collected, where the Raman signal is produced by the sample in response to the first and second beams of light. For example, the Raman signal may be produced by coherent Raman scattering of the first and second beams of light within the sample. The first and second beams of light may be directed to the sample by one or more optical elements that include a free-space optical element, an optical fiber, or an optical waveguide. Similarly, one or more optical elements (which may include a free-space optic, an optical fiber, or an optical waveguide) may collect the Raman signal and direct the Raman signal to an optical receiver. For example, the sample **150** may be located external to the Raman spectroscopy system **100**, and one or more optical fibers **116** may (i) direct the first and second beams of light to the sample and (ii) direct the Raman signal back to the system.

[0222] At step **5450**, the Raman signal is detected. Detection of the Raman signal **160**, which may be performed by an optical receiver **200**, includes steps **5452** and **5454**. At step **5452**, a third beam of light at a third frequency ν_3 is produced. The third beam of light (which may be referred to as a probe beam of light **120pr**) may be produced by a wavelength-tunable laser, where the third frequency is adjustable by changing a wavelength of light produced by the wavelength-tunable laser. At step **5454**, a portion of the

Raman signal is coherently mixed with at least a portion of the third beam of light to produce an electronic signal (e.g., a photocurrent signal i). The Raman signal and the third beam of light may be coherently mixed at a detector of an optical receiver. The portion of the Raman signal that is coherently mixed with the third beam of light may include optical frequency components of the Raman signal that are within a particular frequency range of the third frequency, and the particular frequency range may depend on the electronic bandwidth of the detector. For example, the particular frequency range may extend from approximately $\nu_3 - \Delta f$ to approximately $\nu_3 + \Delta f$, where ν_3 is the third frequency, and Δf is the electronic bandwidth of the detector.

[0223] At step 5460, a characteristic of the electronic signal is determined, at which point the method may end. For example, a processor may receive a digital signal corresponding to the electronic signal, and the processor may determine the characteristic of the electronic signal based on the digital signal. The characteristic of the electronic signal may be associated with the Raman signal 160 and may include a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, or a polarization. Additionally, the characteristic of the electronic signal may be associated with a Raman shift at a frequency $\nu_1 - \nu_3$, where ν_1 is the first frequency, and ν_3 is the third frequency. Based on one or more determined signal characteristics 162, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0224] Various example aspects directed to a Raman spectroscopy system are described below.

[0225] Aspect 1. A system comprising: a first light source configured to produce a first beam of light at a first frequency; a second light source configured to produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; one or more optical elements configured to: direct the first and second beams of light to a sample; and collect a Raman signal produced by the sample in response to the first and second beams of light; an optical receiver configured to detect the Raman signal, the optical receiver comprising: a third light source configured to produce a third beam of light at a third frequency, wherein the third light source comprises a wavelength-tunable laser, wherein the third frequency is adjustable by changing a wavelength of light produced by the wavelength-tunable laser; and an optical detector configured to coherently mix a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and a processor configured to determine a characteristic of the electronic signal.

[0226] Aspect 2. The system of aspect 1, wherein: the third frequency is ν_3 ; the third light source is further configured to change the frequency of the third beam of light by a frequency change Δf to a frequency $\nu_3 + \Delta f$; the detector is further configured to coherently mix another portion of the Raman signal with at least a portion of the third beam of light at the frequency $\nu_3 + \Delta f$ to produce another electronic signal; and the processor is further configured to determine a characteristic of the another electronic signal.

[0227] Aspect 3. The system of aspect 2, wherein: the characteristic of the electronic signal is associated with a

Raman shift at a frequency $\nu_1 - \nu_3$, wherein ν_1 is the first frequency; and the characteristic of the another electronic signal is associated with another Raman shift at a frequency $\nu_1 - (\nu_3 + \Delta f)$.

[0228] Aspect 4. The system of aspect 2, wherein the frequency change Δf is between approximately 10 megahertz (MHz) and approximately 10 gigahertz (GHz).

[0229] Aspect 5. The system of aspect 1, wherein: the frequency offset Ω is approximately equal to a vibrational frequency of a particular material; and the processor is further configured to determine an amount or a concentration of the particular material in the sample based on the characteristic of the electronic signal.

[0230] Aspect 6. The system of aspect 1, wherein the frequency offset Ω is between approximately 5 terahertz (THz) and approximately 100 THz.

[0231] Aspect 7. The system of aspect 1, wherein the first light source or the second light source comprises a wavelength-tunable laser, wherein the frequency offset Ω is adjustable by changing a wavelength of the wavelength-tunable laser.

[0232] Aspect 8. The system of aspect 1, wherein the first light source or the second light source comprises two or more fixed-wavelength lasers, each of the fixed-wavelength lasers having a different operating wavelength, wherein the frequency offset Ω is adjustable by selecting one of the fixed-wavelength lasers for operation.

[0233] Aspect 9. The system of aspect 1, wherein: the frequency offset Ω is a first frequency offset Ω_1 , and the Raman signal is a first Raman signal; subsequent to the optical receiver detecting the first Raman signal: the first light source is further configured to change the first frequency to produce a second frequency offset Ω_2 different from the first frequency offset Ω_1 ; and the optical receiver is further configured to detect a second Raman signal produced by the sample in response to the first and second beams of light with the second frequency offset Ω_2 , wherein the detector is configured to coherently mix a portion of the second Raman signal with at least a portion of the third beam of light to produce a second electronic signal; and the processor is further configured to determine a characteristic of the second electronic signal.

[0234] Aspect 10. The system of aspect 1, wherein: the frequency offset Ω is a first frequency offset Ω_1 , and the Raman signal is a first Raman signal; subsequent to the optical receiver detecting the first Raman signal: the second light source is further configured to change the second frequency to a new second frequency to produce a second frequency offset Ω_2 different from the first frequency offset Ω_1 ; the third light source is further configured to change the third frequency to a new third frequency, wherein the new third frequency is within 200 GHz of the new second frequency; and the optical receiver is further configured to detect a second Raman signal produced by the sample in response to the first and second beams of light with the second frequency offset Ω_2 , wherein the detector is configured to coherently mix a portion of the second Raman signal with at least a portion of the third beam of light to produce a second electronic signal; and the processor is further configured to determine a characteristic of the second electronic signal.

[0235] Aspect 11. The system of aspect 1, wherein the wavelength of light produced by the wavelength-tunable

laser is adjustable over a wavelength range having a width between approximately 10 nanometers (nm) and approximately 100 nm.

[0236] Aspect 12. The system of aspect 1, wherein the wavelength-tunable laser comprises a wavelength-tunable laser diode having a wavelength-tuning range with a width between approximately 10 nanometers (nm) and approximately 100 nm.

[0237] Aspect 13. The system of aspect 1, wherein the wavelength-tunable laser comprises a sampled-grating distributed Bragg reflector (SG-DBR) laser.

[0238] Aspect 14. The system of aspect 1, wherein the wavelength of light produced by the wavelength-tunable laser is between approximately 1490 nanometers (nm) and approximately 1570 nm.

[0239] Aspect 15. The system of aspect 1, wherein the wavelength of light produced by the wavelength-tunable laser is between approximately 1000 nanometers (nm) and approximately 1100 nm.

[0240] Aspect 16. The system of aspect 1, wherein the third light source comprises: two or more laser diodes, wherein each of the laser diodes is a fixed-wavelength laser diode or a wavelength-tunable laser diode; and an optical multiplexer configured to combine light produced by each of the laser diodes into a single output beam of light.

[0241] Aspect 17. The system of aspect 16, wherein the third light source is configured to operate only one laser diode at a time, wherein: the third light source comprises N laser diodes, wherein N is an integer greater than or equal to 2; during a first measurement period, a first one of the N laser diodes is configured to produce light, and N-1 of the laser diodes, excluding the first one, are configured to not produce light; and during a second measurement period, a second one of the N laser diodes is configured to produce light, and N-1 of the laser diodes, excluding the second one, are configured to not produce light.

[0242] Aspect 18. The system of aspect 1, wherein the third light source comprises N wavelength-tunable laser diodes, wherein N is an integer greater than or equal to 2, and the third light source is configured to tune over one or more wavelength ranges having a total width between $(0.7)N \cdot \Delta\lambda_{av}$ and $N \cdot \Delta\lambda_{av}$, wherein $\Delta\lambda_{av}$ is an average wavelength-tuning range of the N laser diodes.

[0243] Aspect 19. The system of aspect 1, wherein the second frequency and the third frequency differ by greater than 100 megahertz (MHZ) and less than 200 gigahertz (GHz).

[0244] Aspect 20. The system of aspect 1, wherein the first or second light source is a wavelength-tunable light source, wherein the respective first or second frequency is adjustable over a frequency range corresponding to a wavelength range having a width between approximately 10 nanometers (nm) and approximately 100 nm.

[0245] Aspect 21. The system of aspect 1, wherein the first frequency corresponds to a wavelength between approximately 1220 nanometers (nm) and approximately 1450 nm, and the second frequency corresponds to a wavelength between approximately 1490 nm and approximately 1570 nm.

[0246] Aspect 22. The system of aspect 1, wherein each of the first, second, and third beams of light has a spectral linewidth of less than 200 megahertz (MHZ).

[0247] Aspect 23. The system of aspect 1, wherein one or more of the first, second, and third beams of light have a spectral linewidth of less than 1 megahertz (MHZ).

[0248] Aspect 24. The system of aspect 1, wherein each of the first light source, the second light source, and the third light source comprises one or more laser diodes, wherein each laser diode is a fixed-wavelength laser diode or a wavelength-tunable laser diode.

[0249] Aspect 25. The system of aspect 1, wherein the first light source, the second light source, or the third light source comprises a seed laser configured to produce seed light and an optical amplifier configured to amplify the seed light to produce an output beam of light, wherein the optical amplifier comprises a semiconductor optical amplifier (SOA) or a fiber-optic amplifier.

[0250] Aspect 26. The system of aspect 1, wherein the portion of the Raman signal that is coherently mixed with the third beam of light comprises optical frequency components of the Raman signal within a particular frequency range of the third frequency, wherein the particular frequency range is based on an electronic bandwidth of the optical detector.

[0251] Aspect 27. The system of aspect 1, wherein the detector has an electronic bandwidth of approximately Δf , and the electronic signal produced by the detector comprises one or more electronic frequency components, each electronic frequency component having a frequency less than or equal to approximately Δf .

[0252] Aspect 28. The system of aspect 1, wherein the characteristic of the electronic signal comprises one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization.

[0253] Aspect 29. The system of aspect 1, wherein the frequency offset Ω equals $\nu_1 - \nu_2$, wherein ν_1 is the first frequency, and ν_2 is the second frequency.

[0254] Aspect 30. The system of aspect 1, wherein each of the first, second, and third frequencies corresponds to a wavelength between approximately 300 nanometers (nm) and approximately 5,000 nm.

[0255] Aspect 31. The system of aspect 1, wherein the first frequency corresponds to a wavelength between approximately 1300 nanometers (nm) and approximately 1400 nm.

[0256] Aspect 32. The system of aspect 1, wherein the first frequency corresponds to a wavelength between approximately 890 nanometers (nm) and approximately 920 nm.

[0257] Aspect 33. The system of aspect 1, wherein the first frequency corresponds to a wavelength between approximately 700 nanometers (nm) and approximately 850 nm.

[0258] Aspect 34. The system of aspect 1, wherein the third frequency corresponds to a wavelength between approximately 1500 nanometers (nm) and approximately 1600 nm.

[0259] Aspect 35. The system of aspect 1, wherein the detector has an electronic bandwidth between approximately 100 megahertz (MHz) and approximately 50 gigahertz (GHz).

[0260] Aspect 36. The system of aspect 1, wherein the first light source, the second light source, or the third light source comprises a distributed feedback (DFB) laser diode.

[0261] Aspect 37. The system of aspect 1, wherein the first light source, the second light source, or the third light source

comprises a wavelength-tunable laser diode comprising an external-cavity laser diode, a thermally tuned laser diode, or a sampled-grating distributed Bragg reflector (SG-DBR) laser.

[0262] Aspect 38. The system of aspect 1, wherein the first light source, the second light source, or the third light source comprises a light-emitting diode (LED), super-luminescent light source, short-pulse laser, broadband light source, fiber laser, or solid-state laser.

[0263] Aspect 39. The system of aspect 1, wherein the Raman signal is produced by coherent Raman scattering of the first and second beams of light within the sample.

[0264] Aspect 40. The system of aspect 1, wherein the sample comprises a biological material.

[0265] Aspect 41. The system of aspect 1, wherein the sample comprises an inorganic material.

[0266] Aspect 42. The system of aspect 1, wherein the sample comprises a crystalline material.

[0267] Aspect 43. The system of aspect 1, further comprising a half-wave plate configured to rotate a polarization of the first or second beam of light prior to being directed to the sample.

[0268] Aspect 44. The system of aspect 1, further comprising a quarter-wave plate configured to convert a polarization of the first or second beam of light to a circular or elliptical polarization prior to being directed to the sample.

[0269] Aspect 45. The system of aspect 1, further comprising a half-wave plate configured to rotate a polarization of the third beam of light.

[0270] Aspect 46. The system of aspect 1, further comprising an optical filter located between the sample and the optical receiver, the optical filter configured to transmit one or more wavelengths associated with the Raman signal and block one or more wavelengths associated with the first or second beam of light.

[0271] Aspect 47. The system of aspect 1, further comprising an optical polarizer located between the sample and the optical receiver, wherein the optical polarizer is oriented to transmit light with a polarization associated with the Raman signal.

[0272] Aspect 48. A method for measuring a Raman signal, the method comprising: producing a first beam of light at a first frequency; producing a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; directing the first and second beams of light to a sample; collecting a Raman signal produced by the sample in response to the first and second beams of light; detecting the Raman signal, comprising: producing a third beam of light at a third frequency, wherein the third beam of light is produced by a wavelength-tunable laser, wherein the third frequency is adjustable by changing a wavelength of light produced by the wavelength-tunable laser; and coherently mixing a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and determining a characteristic of the electronic signal.

[0273] Aspect 50. One or more computer-readable non-transitory storage media embodying software that is operable when executed to: produce a first beam of light at a first frequency; produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; direct the first and second beams of light to a sample; collect a Raman signal produced by the sample in response to the first and second beams of

light; detect the Raman signal, comprising: produce a third beam of light at a third frequency, wherein the third beam of light is produced by a wavelength-tunable laser, wherein the third frequency is adjustable by changing a wavelength of light produced by the wavelength-tunable laser; and coherently mix a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and determine a characteristic of the electronic signal.

[0274] Various example aspects directed to another Raman spectroscopy system are described below.

[0275] Aspect 1. A system comprising: a first light source configured to produce a first beam of light at a first frequency; a second light source configured to produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; one or more optical elements configured to: direct the first and second beams of light to a sample; and collect a Raman signal produced by coherent Raman scattering of the first and second beams of light within the sample; an optical receiver configured to detect the Raman signal, the optical receiver comprising: a third light source configured to produce a third beam of light at a third frequency; and an optical detector configured to coherently mix a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal, wherein the portion of the Raman signal that is coherently mixed with the third beam of light comprises optical frequency components of the Raman signal within a particular frequency range of the third frequency, wherein the particular frequency range is based on an electronic bandwidth of the detector; and a processor configured to determine a characteristic of the electronic signal.

[0276] Aspect 2. The system of aspect 1, wherein the particular frequency range extends from approximately $\nu_3 - \Delta f$ to approximately $\nu_3 + \Delta f$, wherein ν_3 is the third frequency, and Δf is the electronic bandwidth of the detector.

[0277] Aspect 3. The system of aspect 1, wherein the electronic signal produced by the detector comprises one or more electronic frequency components, each electronic frequency component having a frequency less than or equal to Δf , wherein Δf is the electronic bandwidth of the detector.

[0278] Aspect 4. The system of aspect 1, wherein the detector has an electronic bandwidth between approximately 100 megahertz (MHz) and approximately 10 gigahertz (GHz).

[0279] Aspect 5. The system of aspect 1, wherein the electronic signal comprises one or more electronic frequency components less than or equal to approximately 10 gigahertz (GHz).

[0280] Aspect 6. The system of aspect 1, wherein the detector comprises a PN photodiode, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), silicon photomultiplier (SiPM), or photomultiplier tube (PMT).

[0281] Aspect 7. The system of aspect 1, wherein: the electronic signal comprises a photocurrent signal produced by the detector; and the optical receiver further comprises: an electronic amplifier configured to amplify the photocurrent signal to produce a voltage signal corresponding to the photocurrent signal; and a digitizer configured to produce a digital representation of the voltage signal.

[0282] Aspect 8. The system of aspect 7, wherein the processor is configured to determine the characteristic of the

electronic signal based on the digital representation of the voltage signal, wherein the characteristic of the electronic signal comprises one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization.

[0283] Aspect 9. The system of aspect 7, wherein the voltage signal is a time-domain signal, and the processor is further configured to determine a Fourier transform of the digital representation of the voltage signal to determine a frequency-domain representation of the voltage signal.

[0284] Aspect 10. The system of aspect 1, wherein the processor is further configured to associate a Raman frequency shift with the determined characteristic of the electronic signal, wherein the Raman frequency shift equals $\nu_1 - \nu_3$, wherein ν_1 is the first frequency, and ν_3 is the third frequency.

[0285] Aspect 11. The system of aspect 1, wherein the electronic signal comprises a photocurrent signal corresponding to the coherent mixing of the portion of the Raman signal and the third beam of light.

[0286] Aspect 12. The system of aspect 1, wherein the electronic signal comprises a coherent-mixing term that is proportional to a product of (i) an amplitude of an electric field of the Raman signal and (ii) an amplitude of an electric field of the third beam of light.

[0287] Aspect 13. The system of aspect 12, wherein the coherent-mixing term is proportional to $E_R \cdot E_3 \cdot \cos[2\pi(\nu_R - \nu_3)t + \Delta\phi]$, wherein: E_R is the amplitude of the electric field of the Raman signal; E_3 is the amplitude of the electric field of the third beam of light; ν_R is a frequency of the electric field of the Raman signal; ν_3 is a frequency of the electric field of the third beam of light; and $\Delta\phi$ is a phase difference between the electric field of the Raman signal and the electric field of the third beam of light.

[0288] Aspect 14. The system of aspect 1, wherein the first, second, or third light source is a wavelength-tunable light source, wherein the first, second, or third frequency is adjustable over a frequency range corresponding to a wavelength range having a width between approximately 10 nanometers (nm) and approximately 100 nm.

[0289] Aspect 15. The system of aspect 1, wherein the second frequency and the third frequency differ by greater than 100 megahertz (MHz) and less than 200 gigahertz (GHz).

[0290] Aspect 16. The system of aspect 1, wherein: the third frequency is ν_3 ; the third light source is further configured to change the frequency of the third beam of light by a frequency change ΔF to a frequency $\nu_3 + \Delta F$; the detector is further configured to coherently mix another portion of the Raman signal with at least a portion of the third beam of light at the frequency $\nu_3 + \Delta F$ to produce another electronic signal; and the processor is further configured to determine a characteristic of the another electronic signal.

[0291] Aspect 17. The system of aspect 1, wherein: the third light source comprises a wavelength-tunable laser, wherein the third light source is further configured to sequentially change the frequency of the third beam of light to a plurality of different frequencies; the detector is further configured to coherently mix another portion of the Raman signal with at least a portion of one of the different frequencies of the third beam of light to produce a corresponding one of a plurality of electronic signals; and the processor is

further configured to determine a characteristic of each of the plurality of electronic signals.

[0292] Aspect 18. The system of aspect 17, wherein the processor is further configured to determine, based on the determined characteristics of the electronic signals, (i) whether a particular material is present in the sample or (ii) an amount or a concentration of the particular material in the sample.

[0293] Aspect 19. The system of aspect 17, wherein each of the different frequencies to which the third light source is changed is offset from an adjacent one of the different frequencies by between approximately 10 megahertz (MHz) and approximately 10 gigahertz (GHz).

[0294] Aspect 20. The system of aspect 17, wherein the processor is further configured to determine a Raman spectrum based on the determined characteristics of the electronic signals.

[0295] Aspect 21. The system of aspect 20, wherein: the frequency offset Ω is approximately equal to a vibrational frequency of a particular material; and the processor is further configured to determine an amount or a concentration of the particular material in the sample based on the determined Raman spectrum.

[0296] Aspect 22. The system of aspect 1, wherein the first light source or the second light source comprises a wavelength-tunable laser, wherein the frequency offset Ω is adjustable by changing a wavelength of the wavelength-tunable laser.

[0297] Aspect 23. The system of aspect 1, wherein the frequency offset Ω is between approximately 5 terahertz (THz) and approximately 100 THz.

[0298] Aspect 24. The system of aspect 1, wherein the frequency offset Ω is approximately equal to a vibrational frequency of a particular material.

[0299] Aspect 25. The system of aspect 24, wherein the processor is further configured to determine whether the particular material is present in the sample based on the characteristic of the electronic signal.

[0300] Aspect 26. The system of aspect 24, wherein the processor is further configured to determine an amount or a concentration of the particular material in the sample based on the characteristic of the electronic signal.

[0301] Aspect 27. The system of aspect 1, wherein the Raman signal is an optical signal having a spectral linewidth between approximately 30 gigahertz (GHz) and approximately 300 GHz.

[0302] Aspect 28. The system of aspect 1, wherein the Raman signal is an optical signal having a center frequency approximately equal to $2\nu_1 - \nu_2$, wherein ν_1 is the first frequency, and ν_2 is the second frequency.

[0303] Aspect 29. The system of aspect 1, wherein the Raman signal is an optical signal having a center frequency within 200 gigahertz (GHz) of the first frequency or the second frequency.

[0304] Aspect 30. The system of aspect 1, wherein: the optical receiver is further configured to detect residual light from the first beam of light after the first beam of light has interacted with the sample, wherein the optical receiver further comprises: a fourth light source configured to produce a fourth beam of light at a fourth frequency, wherein the fourth frequency is within 50 GHz of the first frequency; and an additional optical detector configured to coherently mix at least a portion of the residual light with at least a portion of the fourth beam of light to produce an additional

electronic signal; and the processor is further configured to determine a characteristic of the additional electronic signal.

[0305] Aspect 31. A method for measuring a Raman signal, the method comprising: producing a first beam of light at a first frequency; producing a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; directing the first and second beams of light to a sample; collecting a Raman signal produced by coherent Raman scattering of the first and second beams of light within the sample; detecting the Raman signal, comprising: producing a third beam of light at a third frequency; and coherently mixing, by an optical detector, a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal, wherein the portion of the Raman signal that is coherently mixed with the third beam of light comprises optical frequency components of the Raman signal within a particular frequency range of the third frequency, wherein the particular frequency range is based on an electronic bandwidth of the detector; and determining a characteristic of the electronic signal.

[0306] Aspect 32. One or more computer-readable non-transitory storage media embodying software that is operable when executed to: produce a first beam of light at a first frequency; produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; direct the first and second beams of light to a sample; collect a Raman signal produced by coherent Raman scattering of the first and second beams of light within the sample; detect the Raman signal, comprising: produce a third beam of light at a third frequency; and coherently mix, by an optical detector, a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal, wherein the portion of the Raman signal that is coherently mixed with the third beam of light comprises optical frequency components of the Raman signal within a particular frequency range of the third frequency, wherein the particular frequency range is based on an electronic bandwidth of the detector; and determine a characteristic of the electronic signal.

[0307] Various example aspects directed to another Raman spectroscopy system are described below.

[0308] Aspect 1. A system comprising: a first light source configured to produce a first beam of light at a first frequency; a second light source configured to produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; one or more optical elements configured to: direct the first and second beams of light to a sample; and collect a Raman signal produced by the sample in response to the first and second beams of light; an optical receiver configured to detect the Raman signal, the optical receiver comprising: a third light source configured to produce a third beam of light at a third frequency; and one or more optical detectors, wherein each detector is configured to coherently mix a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and a processor configured to determine a characteristic of the electronic signal.

[0309] Aspect 2. The system of aspect 1, wherein the processor is further configured to determine a phase difference between the Raman signal and the third beam of light.

[0310] Aspect 3. The system of aspect 1, wherein the processor is further configured to determine an in-phase portion and a quadrature portion associated with the Raman signal.

[0311] Aspect 4. The system of aspect 1, wherein the optical receiver further comprises a 90-degree optical hybrid configured to: combine the Raman signal with the third beam of light to produce four combined beams, the four combined beams comprising two in-phase combined beams and two quadrature combined beams, wherein each combined beam comprises a portion of the Raman signal and a portion of the third beam of light; and direct each of the combined beams to one of four detectors of the optical receiver.

[0312] Aspect 5. The system of aspect 4, wherein: prior to combining the Raman signal with the third beam of light, the 90-degree optical hybrid is configured to split the Raman signal or the third beam of light into a first part and a second part; and the 90-degree optical hybrid comprises a phase shifter configured to impart a 90-degree phase change to the first part with respect to the second part. Aspect 6. The system of aspect 4, wherein: the four detectors are each configured to coherently mix the portion of the Raman signal and the portion of the third beam of light to produce one of four electronic signals; and the processor is further configured to determine a phase difference between the Raman signal and the third beam of light.

[0313] Aspect 7. The system of aspect 1, wherein the processor is further configured to determine a polarization of the Raman signal.

[0314] Aspect 8. The system of aspect 1, wherein the optical receiver further comprises: a Raman-signal polarization beamsplitter configured to split the Raman signal into a horizontal-polarization Raman signal and a vertical-polarization Raman signal; a third-beam polarization beamsplitter configured to split the third beam into a horizontal-polarization third beam and a vertical-polarization third beam; a horizontal-polarization optical receiver comprising one or more of the optical detectors, wherein each detector is configured to coherently mix at least a portion of the horizontal-polarization Raman signal and horizontal-polarization third beam to produce a horizontal-polarization electronic signal; and a vertical-polarization optical receiver comprising another one or more of the optical detectors, wherein each detector is configured to coherently mix at least a portion of the vertical-polarization Raman signal and vertical-polarization third beam to produce a vertical-polarization electronic signal.

[0315] Aspect 9. The system of aspect 8, wherein determining the characteristic of the electronic signal comprises determining one or more characteristics of the horizontal-polarization and vertical-polarization electronic signals.

[0316] Aspect 10. The system of aspect 9, wherein the processor is further configured to determine a polarization of the Raman signal based on the characteristics of the horizontal-polarization and vertical-polarization electronic signals.

[0317] Aspect 11. The system of aspect 8, wherein: the horizontal-polarization optical receiver comprises: four optical detectors; and a first 90-degree optical hybrid configured to: combine the horizontal-polarization Raman signal with the horizontal-polarization third beam to produce four combined horizontal-polarization beams, each combined horizontal-polarization beam comprising a portion of the hori-

horizontal-polarization Raman signal and a portion of the horizontal-polarization third beam; and direct each of the combined horizontal-polarization beams to one of the four detectors; and the vertical-polarization optical receiver comprises: another four optical detectors; and a second 90-degree optical hybrid configured to: combine the vertical-polarization Raman signal with the vertical-polarization third beam to produce four combined vertical-polarization beams, each combined vertical-polarization beam comprising a portion of the vertical-polarization Raman signal and a portion of the vertical-polarization third beam; and direct each of the combined vertical-polarization beams to one of the another four optical detectors.

[0318] Aspect 12. The system of aspect 11, wherein: the four detectors of the horizontal-polarization optical receiver are each configured to coherently mix the portion of the horizontal-polarization Raman signal and the portion of the horizontal-polarization third beam to produce one of four horizontal-polarization electronic signals; the four detectors of the vertical-polarization optical receiver are each configured to coherently mix the portion of the vertical-polarization Raman signal and the portion of the vertical-polarization third beam to produce one of four vertical-polarization electronic signals; and based on the four horizontal-polarization electronic signals and the four vertical-polarization electronic signals, the processor is further configured to determine (i) a polarization of the Raman signal and (ii) a phase difference between the Raman signal and the third beam of light.

[0319] Aspect 13. The system of aspect 1, wherein the optical elements comprise an optical combiner configured to combine the first and second beams of light to produce a combined beam that is directed to the sample.

[0320] Aspect 14. The system of aspect 13, wherein the optical combiner is part of a photonic integrated circuit (PIC), wherein the first and second beams of light are combined into an optical waveguide of the PIC.

[0321] Aspect 15. The system of aspect 13, wherein the optical combiner is a fiber-optic combiner, wherein the first and second beams of light are combined into an optical fiber.

[0322] Aspect 16. The system of aspect 1, wherein the optical receiver further comprises an optical combiner configured to combine the Raman signal and the third beam of light to produce one or more combined beams that are each directed to one of the optical detectors.

[0323] Aspect 17. The system of aspect 16, wherein the optical combiner is part of a photonic integrated circuit (PIC), wherein the portion of the Raman signal and the portion of the third beam of light are combined into an optical waveguide of the PIC.

[0324] Aspect 18. The system of aspect 16, wherein the optical combiner is a fiber-optic combiner, wherein the portion of the Raman signal and the portion of the third beam of light are combined into an optical fiber.

[0325] Aspect 19. The system of aspect 1, wherein the optical elements comprise a photonic integrated circuit (PIC) comprising one or more optical waveguides, wherein: one or more optical waveguides are configured to direct the first and second beams of light to the sample; and one or more other optical waveguides are configured to direct the Raman signal and the third beam of light to the one or more of the detectors.

[0326] Aspect 20. The system of aspect 1, wherein the first, second, or third light source is a wavelength-tunable

light source, wherein the first, second, or third frequency is adjustable over a frequency range corresponding to a wavelength range having a width between approximately 10 nanometers (nm) and approximately 100 nm.

[0327] Aspect 21. The system of aspect 1, wherein the frequency offset Ω is between approximately 5 terahertz (THz) and approximately 100 THz.

[0328] Aspect 22. The system of aspect 1, wherein: the frequency offset Ω is approximately equal to a vibrational frequency of a particular material; and the processor is further configured to determine, based on the characteristic of the electronic signal, (i) whether the particular material is present in the sample or (ii) an amount or a concentration of the particular material in the sample.

[0329] Aspect 23. The system of aspect 1, wherein: the electronic signal comprises a photocurrent signal produced by the detector; and the optical receiver further comprises: an electronic amplifier configured to amplify the photocurrent signal to produce a voltage signal corresponding to the photocurrent signal; and a digitizer configured to produce a digital representation of the voltage signal.

[0330] Aspect 24. The system of aspect 23, wherein the processor is configured to determine the characteristic of the electronic signal based on the digital representation of the voltage signal, wherein the characteristic of the electronic signal comprises one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization.

[0331] Aspect 25. The system of aspect 1, wherein each detector has an electronic bandwidth of approximately Δf , and the electronic signal produced by each detector comprises one or more electronic frequency components, each electronic frequency component having a frequency less than or equal to Δf .

[0332] Aspect 26. The system of aspect 1, wherein the one or more detectors comprise a PN photodiode, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), silicon photomultiplier (SiPM), or photomultiplier tube (PMT).

[0333] Aspect 27. A method for measuring a Raman signal, the method comprising: producing a first beam of light at a first frequency; producing a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; directing the first and second beams of light to a sample; collecting a Raman signal produced by the sample in response to the first and second beams of light; detecting the Raman signal, comprising: producing a third beam of light at a third frequency; and coherently mixing a portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and determining a characteristic of the electronic signal.

[0334] Aspect 28. One or more computer-readable non-transitory storage media embodying software that is operable when executed to: produce a first beam of light at a first frequency; produce a second beam of light at a second frequency, wherein the first and second frequencies are offset by a frequency offset Ω ; direct the first and second beams of light to a sample; collect a Raman signal produced by the sample in response to the first and second beams of light; detect the Raman signal, comprising: produce a third beam of light at a third frequency; and coherently mix a

portion of the Raman signal with at least a portion of the third beam of light to produce an electronic signal; and determine a characteristic of the electronic signal.

[0335] Various example aspects directed to another Raman spectroscopy system are described below.

[0336] Aspect 1. A system comprising: a pump light source configured to produce a pump beam of light at a pump frequency; one or more optical elements configured to: direct the pump beam to a sample; and collect a Raman signal produced by the sample in response to the pump beam, wherein the Raman signal is produced by spontaneous Raman scattering of the pump beam within the sample; an optical receiver configured to detect the Raman signal, the optical receiver comprising: a probe light source configured to produce a probe beam of light at a probe frequency; and one or more optical detectors, wherein each detector is configured to coherently mix a portion of the Raman signal with at least a portion of the probe beam of light to produce an electronic signal; and a processor configured to determine a characteristic of the electronic signal.

[0337] Aspect 2. The system of aspect 1, wherein the probe light source comprises a wavelength-tunable laser configured to tune the probe frequency of the probe beam of light to a plurality of frequencies across at least a portion of the Raman signal.

[0338] FIG. 55 illustrates an example computer system 5500. One or more computer systems 5500 may perform one or more steps of one or more methods described or illustrated herein. One or more computer systems 5500 may provide functionality described or illustrated herein. Software running on one or more computer systems 5500 may perform one or more steps of one or more methods described or illustrated herein or may provide functionality described or illustrated herein. A computer system 5500 may include or may be referred to as a processor, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. Herein, reference to a computer system may encompass one or more computer systems, where appropriate. A processor of a Raman spectroscopy system 100 may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus.

[0339] Computer system 5500 may take any suitable physical form. As an example, computer system 5500 may be an embedded computer system, a system-on-chip (SOC), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a single-board computer system (SBC), a desktop computer system, a laptop or notebook computer system, a graphics processing unit (GPU), a mainframe, a mesh of computer systems, a server, a tablet computer system, or any suitable combination of two or more of these. As another example, all or part of computer system 5500 may be combined with, coupled to, or integrated into a variety of devices, including, but not limited to, a medical device, wearable device, camera, camcorder, personal digital assistant (PDA), mobile telephone, smartphone, electronic reading device (e.g., an e-reader), game console, smart watch, clock, calculator, television monitor, flat-panel display, computer monitor, eyewear, or head-mounted display. Where appropriate, computer system 5500 may include one or more computer systems 5500; be unitary or distributed; span multiple locations; span multiple machines; span multiple data centers; or reside in a cloud, which may include one or more cloud components in one or

more networks. Where appropriate, one or more computer systems 5500 may perform without substantial spatial or temporal limitation one or more steps of one or more methods described or illustrated herein. As an example, one or more computer systems 5500 may perform in real time or in batch mode one or more steps of one or more methods described or illustrated herein. One or more computer systems 5500 may perform at different times or at different locations one or more steps of one or more methods described or illustrated herein, where appropriate.

[0340] As illustrated in the example of FIG. 55, computer system 5500 may include a processor 5510, memory 5520, storage 5530, an input/output (I/O) interface 5540, a communication interface 5550, or a bus 5560. Computer system 5500 may include any suitable number of any suitable components in any suitable arrangement.

[0341] Processor 5510 may include hardware for executing instructions, such as those making up a computer program. As an example, to execute instructions, processor 5510 may retrieve (or fetch) the instructions from an internal register, an internal cache, memory 5520, or storage 5530; decode and execute them; and then write one or more results to an internal register, an internal cache, memory 5520, or storage 5530. A processor 5510 may include one or more internal caches for data, instructions, or addresses. A processor 5510 may include one or more internal registers for data, instructions, or addresses. Processor 5510 may include any suitable number of any suitable internal registers, where appropriate. Where appropriate, processor 5510 may include one or more arithmetic logic units (ALUs); may be a multi-core processor; or may include one or more processors 5510.

[0342] Memory 5520 may include main memory for storing instructions for processor 5510 to execute or data for processor 5510 to operate on. As an example, computer system 5500 may load instructions from storage 5530 or another source (such as, for example, another computer system 5500) to memory 5520. Processor 5510 may then load the instructions from memory 5520 to an internal register or internal cache. To execute the instructions, processor 5510 may retrieve the instructions from the internal register or internal cache and decode them. During or after execution of the instructions, processor 5510 may write one or more results (which may be intermediate or final results) to the internal register or internal cache. Processor 5510 may then write one or more of those results to memory 5520. One or more memory buses (which may each include an address bus and a data bus) may couple processor 5510 to memory 5520. Bus 5560 may include one or more memory buses. Memory 5520 may include random access memory (RAM). This RAM may be volatile memory, where appropriate. Where appropriate, this RAM may be dynamic RAM (DRAM) or static RAM (SRAM). Memory 5520 may include one or more memories 5520, where appropriate.

[0343] Storage 5530 may include mass storage for data or instructions. As an example, storage 5530 may include a hard disk drive (HDD), a floppy disk drive, flash memory, an optical disc, a magneto-optical disc, magnetic tape, or a Universal Serial Bus (USB) drive or a combination of two or more of these. Storage 5530 may include removable or non-removable (or fixed) media, where appropriate. Storage 5530 may be internal or external to computer system 5500, where appropriate. Storage 5530 may be non-volatile, solid-state memory. Storage 5530 may include read-only memory

(ROM). Where appropriate, this ROM may be mask ROM (MROM), programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), flash memory, or a combination of two or more of these. Storage **5530** may include one or more storage control units facilitating communication between processor **5510** and storage **5530**, where appropriate. Where appropriate, storage **5530** may include one or more storages **5530**.

[0344] I/O interface **5540** may include hardware, software, or both, providing one or more interfaces for communication between computer system **5500** and one or more I/O devices. Computer system **5500** may include one or more of these I/O devices, where appropriate. One or more of these I/O devices may enable communication between a person and computer system **5500**. As an example, an I/O device may include a keyboard, keypad, microphone, monitor, mouse, printer, scanner, speaker, camera, stylus, tablet, touch screen, trackball, another suitable I/O device, or any suitable combination of two or more of these. An I/O device may include one or more sensors. Where appropriate, I/O interface **5540** may include one or more device or software drivers enabling processor **5510** to drive one or more of these I/O devices. I/O interface **5540** may include one or more I/O interfaces **5540**, where appropriate.

[0345] Communication interface **5550** may include hardware, software, or both providing one or more interfaces for communication (such as, for example, packet-based communication) between computer system **5500** and one or more other computer systems **5500** or one or more networks. As an example, communication interface **5550** may include a network interface controller (NIC) or network adapter for communicating with an Ethernet or other wire-based network or a wireless NIC (WNIC); a wireless adapter for communicating with a wireless network, such as a WI-FI network; or an optical transmitter (e.g., a laser or a light-emitting diode) or an optical receiver (e.g., a photodetector) for communicating using fiber-optic communication or free-space optical communication. Computer system **5500** may communicate with an ad hoc network, a personal area network (PAN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), or one or more portions of the Internet or a combination of two or more of these. One or more portions of one or more of these networks may be wired or wireless. As an example, computer system **5500** may communicate with a wireless PAN (WPAN) (such as, for example, a BLUETOOTH WPAN), a WI-FI network, a Worldwide Interoperability for Microwave Access (WiMAX) network, a cellular telephone network (such as, for example, a Global System for Mobile Communications (GSM) network), or other suitable wireless network or a combination of two or more of these. As another example, computer system **5500** may communicate using fiber-optic communication based on 100 Gigabit Ethernet (100 GbE), 10 Gigabit Ethernet (10 GbE), or Synchronous Optical Networking (SONET). Computer system **5500** may include any suitable communication interface **5550** for any of these networks, where appropriate. Communication interface **5550** may include one or more communication interfaces **5550**, where appropriate.

[0346] Bus **5560** may include hardware, software, or both coupling components of computer system **5500** to each other. As an example, bus **5560** may include an Accelerated Graphics Port (AGP) or other graphics bus, an Enhanced Industry Standard Architecture (EISA) bus, a front-side bus

(FSB), a HYPERTRANSPORT (HT) interconnect, an Industry Standard Architecture (ISA) bus, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCIe) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local bus (VLB), or another suitable bus or a combination of two or more of these. Bus **5560** may include one or more buses **5560**, where appropriate.

[0347] Various modules, circuits, systems, methods, or algorithm steps described in connection with the implementations disclosed herein may be implemented as electronic hardware, computer software, or any suitable combination of hardware and software. Computer software (which may be referred to as software, computer-executable code, computer code, a computer program, computer instructions, or instructions) may be used to perform various functions described or illustrated herein, and computer software may be configured to be executed by or to control the operation of computer system **5500**. As an example, computer software may include instructions configured to be executed by processor **5510**. Owing to the interchangeability of hardware and software, the various illustrative logical blocks, modules, circuits, or algorithm steps have been described generally in terms of functionality. Whether such functionality is implemented in hardware, software, or a combination of hardware and software may depend upon the particular application or design constraints imposed on the overall system.

[0348] A computing device may be used to implement various modules, circuits, systems, methods, or algorithm steps disclosed herein. As an example, all or part of a module, circuit, system, method, or algorithm disclosed herein may be implemented or performed by a general-purpose single- or multi-chip processor, a digital signal processor (DSP), an ASIC, a FPGA, a GPU, any other suitable programmable-logic device, discrete gate or transistor logic, discrete hardware components, or any suitable combination thereof. A general-purpose processor may be a microprocessor, or any conventional processor, controller, microcontroller, GPU, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0349] One or more implementations of the subject matter described herein may be implemented as one or more computer programs (e.g., one or more modules of computer-program instructions encoded or stored on a computer-readable non-transitory storage medium). As an example, the steps of a method or algorithm disclosed herein may be implemented in a processor-executable software module which may reside on a computer-readable non-transitory storage medium. A computer-readable non-transitory storage medium may include any suitable storage medium that may be used to store or transfer computer software and that may be accessed by a computer system. Herein, a computer-readable non-transitory storage medium or media may include one or more semiconductor-based or other integrated circuits (ICs) (such, as for example, field-programmable gate arrays (FPGAs) or application-specific ICs (ASICs)), hard disk drives (HDDs), hybrid hard drives

(HHDs), optical discs (e.g., compact discs (CDs), CD-ROM, digital versatile discs (DVDs), Blu-ray discs, or laser discs), optical disc drives (ODDs), magneto-optical discs, magneto-optical drives, floppy diskettes, floppy disk drives (FDDs), magnetic tapes, flash memories, solid-state drives (SSDs), RAM, RAM-drives, ROM, SECURE DIGITAL cards or drives, any other suitable computer-readable non-transitory storage media, or any suitable combination of two or more of these, where appropriate. A computer-readable non-transitory storage medium may be volatile, non-volatile, or a combination of volatile and non-volatile, where appropriate.

[0350] Certain features described herein in the context of separate implementations may also be combined and implemented in a single implementation. Conversely, various features that are described in the context of a single implementation may also be implemented in multiple implementations separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

[0351] While operations may be depicted in the drawings as occurring in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all operations be performed. Further, the drawings may schematically depict one more example processes or methods in the form of a flow diagram or a sequence diagram. However, other operations that are not depicted may be incorporated in the example processes or methods that are schematically illustrated. For example, one or more additional operations may be performed before, after, simultaneously with, or between any of the illustrated operations. Moreover, one or more operations depicted in a diagram may be repeated, where appropriate. Additionally, operations depicted in a diagram may be performed in any suitable order. Furthermore, although particular components, devices, or systems are described herein as carrying out particular operations, any suitable combination of any suitable components, devices, or systems may be used to carry out any suitable operation or combination of operations. In certain circumstances, multitasking or parallel processing operations may be performed. Moreover, the separation of various system components in the implementations described herein should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems may be integrated together in a single software product or packaged into multiple software products.

[0352] Various embodiments have been described in connection with the accompanying drawings. However, it should be understood that the figures may not necessarily be drawn to scale. As an example, distances or angles depicted in the figures are illustrative and may not necessarily bear an exact relationship to actual dimensions or layout of the devices illustrated.

[0353] One or more of the figures described herein may include example data that is prophetic. For example, the graphs illustrated FIGS. 3-16 and 19 may include or may be referred to as prophetic examples.

[0354] The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifica-

tions to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes or illustrates respective embodiments herein as including particular components, elements, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend.

[0355] The term “or” as used herein is to be interpreted as an inclusive or meaning any one or any combination, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, the expression “A or B” means “A, B, or both A and B.” As another example, herein, “A, B, or C” means at least one of the following: A; B; C; A and B; A and C; B and C; A, B, and C. An exception to this definition will occur if a combination of elements, devices, steps, or operations is in some way inherently mutually exclusive.

[0356] As used herein, words of approximation such as, without limitation, “approximately,” “substantially,” or “about” refer to a condition that when so modified is understood to not necessarily be absolute or perfect but would be considered close enough to those of ordinary skill in the art to warrant designating the condition as being present. The extent to which the description may vary will depend on how great a change can be instituted and still have one of ordinary skill in the art recognize the modified feature as having the required characteristics or capabilities of the unmodified feature. In general, but subject to the preceding discussion, a numerical value herein that is modified by a word of approximation such as “approximately” may vary from the stated value by $\pm 0.5\%$, $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$, $\pm 10\%$, $\pm 12\%$, or $\pm 15\%$. The term “substantially constant” refers to a value that varies by less than a particular amount over any suitable time interval. For example, a value that is substantially constant may vary by less than or equal to 20%, 10%, 1%, 0.5%, or 0.1% over a time interval of approximately 104 s, 103 s, 102 s, 10 s, 1 s, 100 ms, 10 ms, 1 ms, 100 μ s, 10 μ s, or 1 μ s. The term “substantially constant” may be applied to any suitable value, such as for example, an optical power, an electrical current, a wavelength, an optical or electrical frequency, or an optical or electrical phase.

[0357] As used herein, the terms “first,” “second,” “third,” etc. may be used as labels for nouns that they precede, and these terms may not necessarily imply a particular ordering (e.g., a particular spatial, temporal, or logical ordering). As an example, a system may be described as determining a “first result” and a “second result,” and the terms “first” and “second” may not necessarily imply that the first result is determined before the second result.

[0358] As used herein, the terms “based on” and “based at least in part on” may be used to describe or present one or more factors that affect a determination, and these terms may not exclude additional factors that may affect a determination. A determination may be based solely on those factors which are presented or may be based at least in part on those factors. The phrase “determine A based on B” indicates that B is a factor that affects the determination of A. In some

instances, other factors may also contribute to the determination of A. In other instances, A may be determined based solely on B.

[0359] Various example aspects included in this disclosure may be presented in a range format. It should be understood that a description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of a disclosed aspect. Accordingly, the description of a range should be considered to have specifically disclosed all suitable sub-ranges as well as individual numerical values within that range, unless expressly indicated otherwise or indicated otherwise by context. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed sub-ranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6, etc., as well as individual values within that range, for example, 1.1, 2, 2.3, 5, and 5.9.

[0360] Although the foregoing embodiments have been described in some detail for purposes of clarity of understanding, the invention is not limited to the details provided. There may be numerous alternative ways of implementing the invention. The disclosed embodiments are illustrative and not restrictive.

1. A Raman spectroscopy system comprising:
 - a pump light source configured to produce a pump beam of light at a pump frequency;
 - a Stokes light source configured to produce a Stokes beam of light at a Stokes frequency, wherein the pump and Stokes frequencies are offset by a frequency offset Ω ;
 - one or more optical fibers configured to (i) direct the pump and Stokes beams to a sample located external to the system and (ii) direct a Raman signal to the system, wherein the Raman signal is produced by coherent Raman scattering of the pump and Stokes beams of light at the sample;
 - an optical receiver configured to detect the Raman signal, the optical receiver comprising:
 - a probe light source configured to produce a probe beam of light at a probe frequency;
 - an optical detector configured to coherently mix the Raman signal with the probe beam of light to produce a corresponding photocurrent signal; and
 - an electronic circuit configured to produce a digital output signal corresponding to the photocurrent signal; and
 - a processor configured to determine a characteristic of the photocurrent signal based on the digital output signal.
2. The Raman spectroscopy system of claim 1, wherein the one or more optical fibers comprise one optical fiber, wherein the one optical fiber is configured to direct the pump and Stokes beams to the sample and direct the Raman signal to the system.
3. The Raman spectroscopy system of claim 2, further comprising a lens located near a terminal end of the optical fiber, wherein the lens is configured to:
 - receive the pump and Stokes beams from the terminal end of the optical fiber and produce a free-space beam that is directed to the sample; and
 - receive the Raman signal from the sample and couple the Raman signal into the optical fiber via the terminal end.
4. The Raman spectroscopy system of claim 2, further comprising an optical circulator configured to:
 - receive the pump and Stokes beams and direct the pump and Stokes beams to the optical fiber; and

- receive the Raman signal from the optical fiber and direct the Raman signal to the optical receiver.

5. The Raman spectroscopy system of claim 1, wherein the one or more optical fibers comprise:

- an output optical fiber configured to direct the pump and Stokes beams to the sample; and
- an input optical fiber configured to direct the Raman signal to the optical receiver.

6. The Raman spectroscopy system of claim 5, further comprising a lens configured to:

- receive the pump and Stokes beams from a terminal end of the output optical fiber and produce a free-space beam that is directed to the sample; and
- receive the Raman signal from the sample and couple the Raman signal into the input optical fiber via a terminal end of the input optical fiber.

7. The Raman spectroscopy system of claim 5, further comprising:

- an output lens configured to receive the pump and Stokes beams from the terminal end of the output optical fiber and produce a free-space beam that is directed to the sample; and
- an input lens configured to receive the Raman signal from the sample and couple the Raman signal into the input optical fiber via the terminal end of the input optical fiber.

8. The Raman spectroscopy system of claim 5, further comprising:

- an output lens configured to receive the pump and Stokes beams from the terminal end of the output optical fiber and produce a free-space beam that is directed to the sample; and
- a mirror comprising:
 - a through hole that the free-space beam propagates through while traveling to the sample; and
 - a reflective surface configured to receive the Raman signal from the sample and direct the Raman signal to a terminal end of the input optical fiber.

9. The Raman spectroscopy system of claim 8, wherein the mirror is an off-axis parabolic mirror wherein the reflective surface has a parabolic shape configured to focus the Raman signal into the input optical fiber via the terminal end of the input optical fiber.

10. The Raman spectroscopy system of claim 1, wherein the one or more optical fibers comprise an optical fiber having a terminal end that is configured to produce a free-space beam comprising the pump and Stokes beams.

11. The Raman spectroscopy system of claim 10, further comprising a lens configured to collimate the free-space beam or focus the free-space beam.

12. The Raman spectroscopy system of claim 10, wherein the terminal end of the optical fiber is configured to be positioned by an operator of the system to direct the free-space beam to the sample.

13. The Raman spectroscopy system of claim 1, further comprising a visible light source configured to produce a visible beam of light, wherein the visible beam of light is combined with the pump and Stokes beams, and the visible beam of light is configured to produce a visible alignment spot at the sample where the pump and Stokes beams are incident on the sample.

14. The Raman spectroscopy system of claim 1, further comprising a fiber-optic combiner configured to combine the pump beam of light and the Stokes beam of light to produce

a combined pump-Stokes beam of light that is directed to one of the one or more optical fibers.

15. The Raman spectroscopy system of claim 1, further comprising a fiber-optic combiner configured to (i) combine the Raman signal and the probe beam of light to produce a combined probe-Raman signal and (ii) direct the combined probe-Raman signal to the optical detector.

16. The Raman spectroscopy system of claim 1, wherein: the Stokes light source is further configured to produce a Stokes reference beam of light;

the one or more optical fibers are further configured to direct residual Stokes light to the system, the residual Stokes light comprising light from the Stokes beam of light after the Stokes beam of light has interacted with the sample;

the probe light source is further configured to produce a probe reference beam of light;

the optical detector is a signal detector, wherein the photocurrent signal produced by the signal detector is a signal photocurrent that corresponds to the Raman signal, the probe beam of light, and the residual Stokes beam of light, wherein a portion of the signal photocurrent corresponds to the coherent mixing between the Raman signal and the probe beam of light;

the optical receiver further comprises a reference detector configured to produce a reference photocurrent corresponding to the probe reference beam of light and the Stokes reference beam of light;

the electronic circuit is further configured to determine a subtraction signal that equals a difference between a signal corresponding to the signal photocurrent and a signal corresponding to the reference photocurrent; and the processor is further configured to determine a characteristic of the subtraction signal.

17. The Raman spectroscopy system of claim 1, further comprising an enclosure, wherein the pump light source, Stokes light source, and optical receiver are contained within the enclosure, and at least a portion of the one or more optical fibers is located external to the enclosure.

18. The Raman spectroscopy system of claim 17, further comprising a fiber-optic adapter, wherein the one or more optical fibers comprise an external optical fiber located external to the enclosure, and the fiber-optic adapter is configured to connect the external optical fiber to an internal optical fiber located inside the enclosure.

19. The Raman spectroscopy system of claim 1, wherein the processor is further configured to associate a Raman frequency shift with the determined characteristic of the photocurrent signal, wherein the Raman frequency shift equals $\nu_{pu} - \nu_{pr}$, wherein ν_{pu} is the pump frequency, and ν_{pr} is the probe frequency.

20. The Raman spectroscopy system of claim 1, wherein the characteristic of the photocurrent signal comprises one or more of: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization.

21. The Raman spectroscopy system of claim 1, wherein: the probe light source comprises a wavelength-tunable laser, wherein the probe light source is further configured to sequentially change the probe frequency of the probe beam of light to a plurality of different frequencies;

the optical detector is further configured to coherently mix the Raman signal with the probe beam of light at each of the different frequencies to produce a plurality of corresponding photocurrent signals; and

the processor is further configured to determine a characteristic of each of the plurality of corresponding photocurrent signals.

22. The Raman spectroscopy system of claim 1, wherein: the frequency offset Ω is approximately equal to a vibrational frequency of a particular material; and

the processor is further configured to determine, based on the characteristic of the photocurrent signal, (i) whether the particular material is present in the sample or (ii) an amount or a concentration of the particular material in the sample.

23. The Raman spectroscopy system of claim 1, wherein the pump light source or the Stokes light source comprises a wavelength-tunable laser, wherein the frequency offset Ω is adjustable by changing a wavelength of the wavelength-tunable laser.

24. The Raman spectroscopy system of claim 1, wherein the pump light source or the Stokes light source comprises a seed laser configured to produce seed light and an optical amplifier configured to amplify the seed light to produce an output beam of light, wherein the optical amplifier comprises a semiconductor optical amplifier (SOA) or a fiber-optic amplifier.

25. The Raman spectroscopy system of claim 1, wherein the electronic circuit comprises:

an electronic amplifier configured to amplify the photocurrent signal to produce a voltage signal corresponding to the photocurrent signal; and

a digitizer configured to produce a digital representation of the voltage signal, wherein the digital representation of the voltage signal is part of the digital output signal.

26. The Raman spectroscopy system of claim 1, wherein coherently mixing the Raman signal with the probe beam of light to produce the corresponding photocurrent signal comprises coherently mixing a portion of the Raman signal with the probe beam of light, wherein the portion of the Raman signal that is coherently mixed comprises optical frequency components of the Raman signal located within a particular frequency range of the probe frequency, wherein the particular frequency range depends on an electronic bandwidth of the detector.

27. The Raman spectroscopy system of claim 1, wherein the photocurrent signal comprises a coherent-mixing term that is proportional to a product of (i) an amplitude of an electric field of the Raman signal and (ii) an amplitude of an electric field of the probe beam of light.

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